

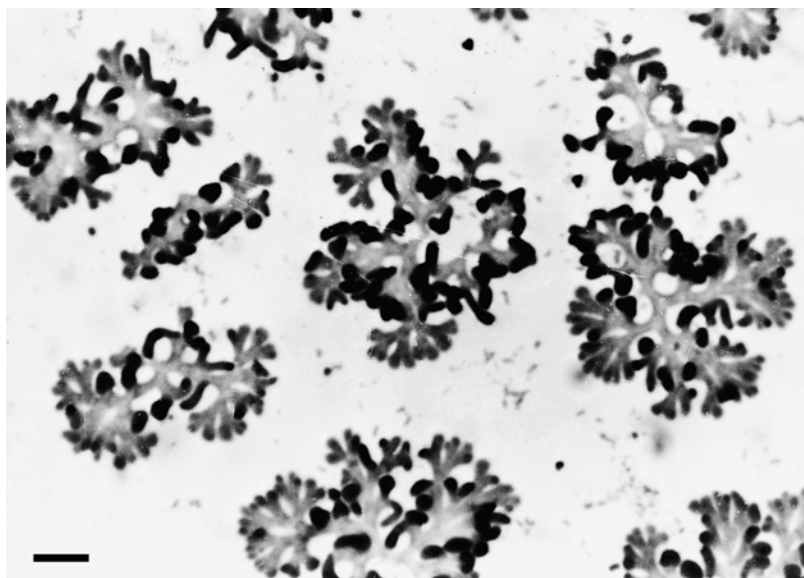


FIGURE BXII.γ.32. Habitat form of *N. ramosa*, showing the typical bush-like microcolony structure. Formalin-carbol-fuchsin. Bar = 20 μm.

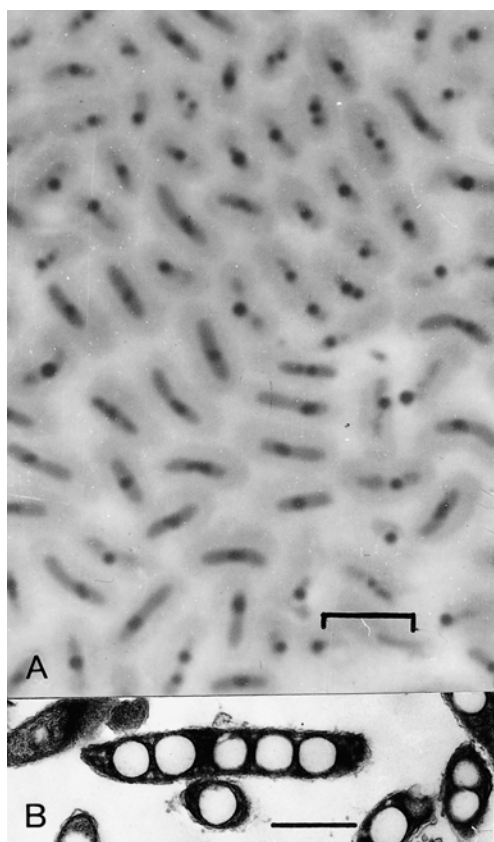


FIGURE BXII.γ.33. *Nevskia ramosa*, strain Soe 1 (DSM 11499). A, without distinct stalks in this growth phase. Phase-contrast micrograph. Bar = 10 μm. B, ultrastructure of organisms grown with Na-lactate. Cells filled with poly-β-hydroxybutyrate granules (white areas). Bar = 1 μm.

slime all around the cells. Cells filled with numerous refractile globules (Fig. BXII.γ.33B) which consist mainly of polyhydroxyalkanoates. Resting stages are not known. **Gram negative**. In young cultures, some cells are **motile** by means of a polar fla-

gellum: they are set free from the microcolonies and can start the development of new colonies (Fig. BXII.γ.34). **Strictly aerobic**. Optimum growth temperature: 20–25°C. **Chemoorganotrophic**, with a broad range of organic compounds, having an oxidative type of metabolism. **Colorless colonies** on nutrient agar. Weak catalase activity, no cytochrome oxidase activity. Cytochromes of the *b* and *c* types are present. Originally found in the surface pellicle of an aquarium. Occur mainly on the surface of freshwater environments (neuston community).

Phylogenetic analysis of 16S rDNA sequences revealed that *N. ramosa* is a member of the phylum *Proteobacteria* and represents a deep branch of the *Gammaproteobacteria* with no known close relatives (Fig. BXII.γ.35).

The original strain described by Famintzin (1892) as the type strain was not isolated in pure culture.

The mol% *G* + *C* of the DNA is: 67.8–69.0 (HPLC), neotype strain Soe 1 (DSM 11499); 69.0, strain OL 1 (DSM 11500).

Type species: *Nevskia ramosa* Famintzin 1892, 484.

FURTHER DESCRIPTIVE INFORMATION

The two 16S rDNA sequences obtained have been deposited with EMBL (accession No. AJ001010 and AJ001011 for strains Soe 1 and OL 1, respectively). The 16S rDNA sequences of the two strains revealed 99.2% similarity; genomic fingerprinting by ERIC-PCR showed different banding patterns on agarose gel (Stürmeyer et al., 1998).

For *in situ* identification by hybridization, two oligonucleotide probes targeting positions 656–674 (NEV 656, 5'-CGCCTCCCTCTACCGTTT-3') and 177–195 (NEV 177, 5'-GCTCTTGCGAGATCATGC-3') of the 16S rRNA gene are available (Glöckner et al., 1998).

The type species was described by Famintzin (1892) as being 2–6 × 12 μm in size, but he also mentioned organisms identical to *N. ramosa* of “much smaller sizes”. Henrici and Johnson (1935) described an organism practically identical to *N. ramosa* in morphology but of smaller dimensions, i.e., 1 × 3–4 μm. Babenzien (1965, 1967) isolated some strains from the surface of swamp ditches and ponds and the average size was 0.7 × 2.4–2.7 μm.

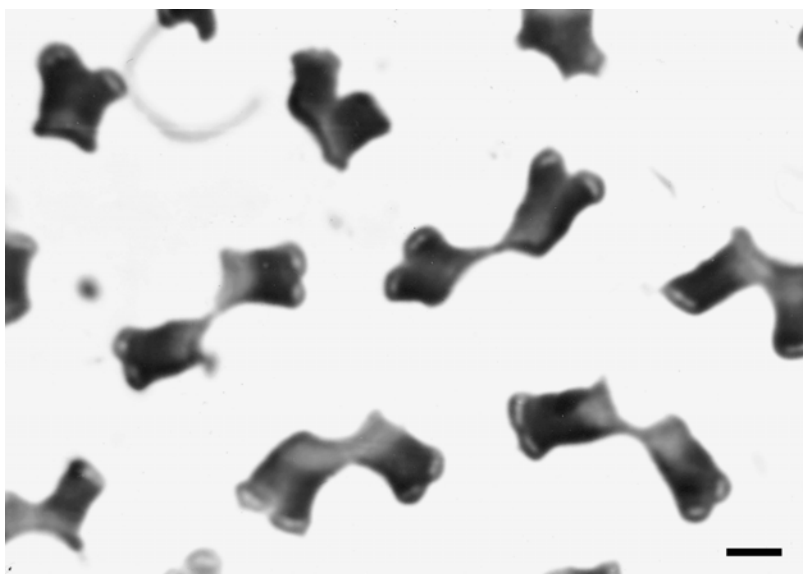


FIGURE BXII.γ.34. *Nevskia ramosa*, the beginning of colony growth, cells at the tip of the slime stalks. Formalin-toluidine blue. Bar = 10 μm.

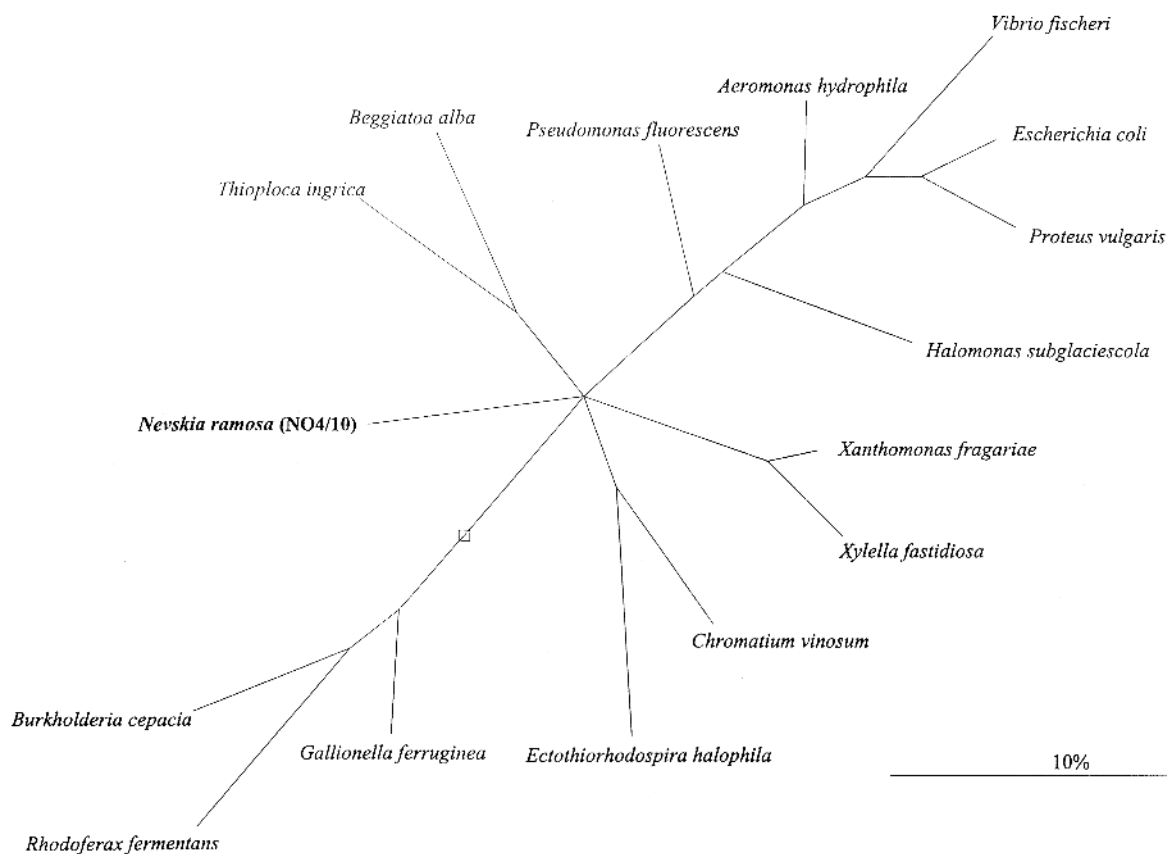


FIGURE BXII.γ.35. Phylogenetic tree, showing phylogenetic relationships between *Nevskia ramosa*, *Nevskia* clones NO4 and NO10, and closest relatives among the *Gammaproteobacteria*, based on comparative analysis of 16S rDNA. Bar = 10% estimated divergence. The open box indicates the root. (Reproduced with permission from Glöckner et al., *Applied and Environmental Microbiology* 64: 1895–1901, 1998, ©American Society for Microbiology, Washington, D.C..)

Hirsch (1992) reported other *Nevskia*-like organisms also in this size range. The neotype strain DSM 11499 has a size of 0.7–1.1 × 1.4–2.3 μm during the logarithmic phase of growth and a tendency to form more elongated cells within the lag phase.

The globular bodies were thought to be composed of “etherical oils” and could be dissolved with 70% ethanol (Famintzin, 1892), whereas it was assumed by Henrici and Johnson (1935) that the vacuoles contained sulfur. According to Babenzien and

Hirsch (1974), globules contain mainly poly-β-hydroxybutyrate. Cells grown at the air-water interface typically contain 2–5 refractile globules.

In the enrichment cultures of Henrici and Johnson (1935), the bacteria were distinctly favored by the addition of sulfide, which indicated that *Nevskia* was either a sulfur bacterium or a microaerophile. As reported by Babenzien (1965) and Hirsch (1981), enrichment cultures were definitely favored by addition of 0.1% sodium lactate or acetate to lake water.

Pure cultures of *N. ramosa* have a strictly aerobic metabolism and grow with numerous substrates (Table BXII.γ.26) added to lake water or simple synthetic medium (Stürmeyer et al., 1998). No anaerobic growth with nitrate, thiosulfate, and sulfate as electron donors.

The development of a pellicle and stalk formation is primarily dependent on the availability of combined nitrogen. If ammonia, nitrate, or amino acids are supplied to the nutrient broth, the cultures grow submersed. It is assumed that the *Nevskia* biofilm in nature traps ammonia from the air before it is dissolved in the water.

By cultivating *N. ramosa* as a surface film, the fluid medium remains clear.

The slime stalks consist mainly of polysaccharides with rhamnose and small amounts of glucose and mannose, but other hydrophobic components may be present.

Nitrogenase activity was not detected.

According to numerous reports (Babenzien, 1966; Babenzien and Hirsch, 1974; Hirsch, 1981; Heldal and Tumyr, 1986; Kangatharalingam and Priscu, 1993; Stürmeyer et al., 1998), *Nevskia ramosa* is widely distributed in shallow aquatic habitats; it has never been reported from soils. Most findings have been from the water surface and, in fact, *Nevskia* appears to be a typical neuston microorganism. As indicated by the dull appearance and hydrophobicity of the biofilm, the flat rosettes live essentially outside of the water phase. While *Nevskia* cells are sensitive to UV radiation if kept in the dark after UV exposure, the isolates overcome UV damage by a very effective photorepair mechanism if incubated subsequently under light.

ENRICHMENT AND ISOLATION PROCEDURES

Surface films of swamp ditches, ponds, greenhouse pools, etc. are collected with sterile glass slides, needle loops, paper strips, or foils. The samples are transferred to Petri dishes with sterile water from the same sites, supplemented with 5 mM sodium lactate, and incubated in the dark at room temperature for about 10 days. Under microscopic control, bacterial microcolonies from the water surface are then picked up with a loop and transferred repeatedly into sterile Erlenmeyer flasks containing the same medium. For dilution purposes and to obtain pure cultures, multiwell chambers are also suitable. Pure cultures were obtained by repeated streaking of diluted samples on agar plates (1% Difco

agar) containing lake water medium. To obtain submersed cultures, the medium was supplemented with 5 mM NH₄Cl.

Pure cultures can also be cultivated as surface films in a synthetic medium of the following composition: sodium lactate, 5 mM; MgSO₄·7H₂O, 0.2 mM; CaCl₂·2H₂O, 0.1 mM; KH₂PO₄, 25 mM; trace element solution SL 9, 0.5 ml/l; and vitamin solution, 0.5 ml/l. The pH is adjusted to 7.0. Incubation temperature 20–25°C (Stürmeyer et al., 1998).

MAINTENANCE PROCEDURES

Stock cultures can be maintained in Erlenmeyer flasks containing liquid medium, covered with cotton plugs, with transfers every three months. The strains in the DSM (Braunschweig, Germany) are stored at –70°C and are available in lyophilized form.

TAXONOMIC COMMENTS

Since the taxonomy of *N. ramosa* was based solely on morphological data, it was previously found in the section on “Phylogenetically unaffiliated bacteria” (*The Prokaryotes*, 2nd Ed.) and Section 21 “Budding and/or appendaged bacteria” (*Bergey’s Manual of Systematic Bacteriology*, 1st Ed.).

It is now clear that *N. ramosa* should not be affiliated with *Caulobacterales*, *Gallionella*, *Thiobacterium*, or *Beggiatoa*.

TABLE BXII.γ.26. Utilization of organic substrates by <i>Nevskia ramosa</i>			
Component	Concentration (mM)	Growth	Rosette formation
D-Ribose	5	+	+
D-Glucose	5	+	+
D-Fructose	5	–	–
Sucrose	5	+	+
Starch	0.20%	+	–
Cellulose	0.20%	+	–
Formate	5	+	+
Acetate	5	+	+
Lactate	5	+	+
Lactate/NH ₄ Cl	5	+	–
Butyrate	8	–	–
Palmitate	2	–	–
L-Malate	10	–	–
Pyruvate	10	+	+
Citrate	5	–	–
Fumarate	10	+	+
Succinate	10	–	–
Ethanol	5	+	+
Glycerol	5	+	+
L-Alanine	5	–	–
L-Arginine	5	+	–
L-Cysteine	5	–	–
L-Glutamate	5	+	–
Benzoate	2	+	–
Tween 20	0.001%	+	–
Tween 80	0.001%	+	–

List of species of the genus *Nevskia*

1. *Nevskia ramosa* Famintzin 1892, 484^{AL}
ra.mo'sa. L. adj. *ramosus* branched.

The characteristics are the same as those described for the genus. A spectrum of substrates and the mode of growth are shown in Table BXII.γ.26. The neotype strain Soe 1 (DSM 11499) was isolated from a small bog lake (L. Soel-

kensee, Germany) and formerly designated as strain IMET 10965.
The mol% G + C of the DNA is: 67.8 (HPLC), strain Soe 1 (DSM 11499); 69.0 (HPLC), strain OL 1 (DSM 11500).
Type strain: Soe 1, DSM 11499.
GenBank accession number (16S rRNA): AJ001010.

Genus VI. Pseudoxanthomonas Finkmann, Altendorf, Stackebrandt and Lipski 2000, 280^{VP}

ANDRÉ LIPSKI AND ERKO S. STACKEBRANDT

Pseu.do.xan.tho.mo' nas. Gr. *pseudēs* false; M.L. n. *Xanthomonas* a bacterial generic name; M.L. fem. n. *Pseudoxanthomonas* false *Xanthomonas*.

Rod-shaped cells. Gram negative. **Aerobic. Heterotrophic.** Colonies are yellow. Possess a ubiquinone with eight isoprene units. **The predominant fatty acids in the iso-/anteiso-type fatty acid profile are C_{15:0} iso and C_{15:0} anteiso***

Type species: Pseudoxanthomonas broegbernensis Finkmann, Altendorf, Stackebrandt and Lipski 2000, 280.

DIFFERENTIATION OF THE GENUS *PSEUDOXANTHOMONAS* FROM OTHER GENERA

The genus can be separated from the genera *Xanthomonas* and *Stenotrophomonas* by the absence of the fatty acid 3-hydroxy-iso-tridecanoic acid and from the species *Xylella fastidiosa* by the presence of branched-chain fatty acids.

TAXONOMIC COMMENTS

Pseudoxanthomonas broegbernensis is clearly a member of the family *Xanthomonadaceae*. The phylogenetic position within this family is not yet settled. Depending on the selection and number of reference sequences, the genus *Xanthomonas* or *Xylella* is the most closely related genus. The more precise phylogenetic relatedness of these genera must await analyses using more sequences of strains of the genus *Pseudoxanthomonas*, which is presently represented by a single sequence only.

At present, the 16S rDNA sequence of *P. broegbernensis* shares the closest relationship (97% similarity) to the sequence of an undescribed ultramicrobacterium strain 12-3 (Iizuka et al., 1998) whose accession number is AB008507.

List of species of the genus *Pseudoxanthomonas*

1. ***Pseudoxanthomonas broegbernensis*** Finkmann, Altendorf, Stackebrandt and Lipski 2000, 280^{VP}
broeg.ber'nen.sis. M.L. fem. adj. *broegbernensis* pertaining to Brögbern (location near Lingen/Germany, from where the organism was isolated).

The characteristics are as described for the genus and as listed in Table BXII.γ.27. Nitrate is not reduced, but

nitrite is reduced to nitrous oxide (Finkmann et al., 2000). Predominant compounds of the fatty acid profile are C_{15:0} iso, C_{15:0} anteiso, C_{16:0}, C_{16:1 ω7c}, and C_{17:1 iso ω8c}. The major hydroxy fatty acid is C_{11:0 iso 3OH}. The fatty acids C_{17:0 cyclo} and C_{13:0 iso 3OH} are absent. No pathogenic potential for humans, animals, or plants known.

TABLE BXII.γ.27. Characteristics of *Pseudoxanthomonas broegbernensis*^{a,b}

Characteristic	Reaction
Nitrate reduction to nitrite or N ₂	—
Nitrite reduction to N ₂ O	+
Tween 80 hydrolysis	+
Urease, arginine dihydrolase	—
Alkalinization of citrate and malonate media	—
Lecithinase	—
Growth at 37°C	—
<i>Acid from:</i>	
Glucose, xylose, fructose, lactose, maltose	+
Rhamnose, sucrose	—
<i>Hydrolysis of:</i> ^c	
pNP-N-acetyl-β-glucosaminide, pNP-β-galactopyranoside	+
pNP-β-glucopyranoside, pNP-α-D-maltoside, pNP-β-xyloside	+
Bis-pNP-phosphate, pNP-phosphocholine, thymidine-5'-mono-phosphate-pNP-ester	—
<i>Hydrolysis of:</i> ^c	
L-Alanine-pNA, L-arginine-pNA, γ-L-glutamate-pNA, L-leucine-pNA, L-lysine-pNA	+
Glycine-pNA, L-proline-pNA, L-valine-pNA	—
<i>Growth with:</i>	
N-Acetylglucosamine, D-cellobiose, D-galactose, D-maltose, D-mannose, D-sucrose, D-trehalose	+
L-Arabinose, D-fructose, gluconate, D-ribose, D-xylose, D-mannite, sorbitol, propionate, Taconitate, adipate, DL-3-hydroxybutyrate, DL-lactate, L-malate, pyruvate, L-histidine, L-hydroxyproline, L-ornithine, L-proline, putrescine	—
<i>Growth in the presence of:</i>	
NaCl, 3%	—
Cadmium acetate, 0.001%; ZnCl ₂ , 0.01%; lead acetate, 0.01%	—
<i>Growth in presence of antimicrobial agents (μg or U per disk):</i>	
Erythromycin (10 μg); streptomycin (10 μg); nalidixic acid (30 μg); kanamycin (30 μg); ampicillin (10 μg); penicillin G (10 U); gentamicin (10 μg); fucidin (10 μg); tetracycline (30 μg); novobiocin (30 μg)	+
Neomycin, 30 μg	—

^a Symbols: see standard definitions.

^b Data from Lipski et al. (1992); Finkmann et al. (2000).

^c pNP, *para*-nitrophenyl; pNA: *para*-nitroanilide.

The only strain of this genus was isolated from an experimental biofilter supplied with the waste gas of an animal rendering plant.

The mol% G + C of the DNA is: not known.

Type strain: B1616/1, DSM 12573.

GenBank accession number (16S rRNA): AJ012231.

Genus VII. *Rhodanobacter* Nalin, Simonet, Vogel and Normand 1999, 22^{VP}

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Rho.da'no.bac.ter. L. masc. n. *Rhodanus* River Rhône; M.L. masc. n. *bacter* equivalent of *bacterium* a small rod; M.L. masc. n. *Rhodanobacter* rod isolated close to the River Rhône.

Gram-negative rods. Nonmotile. Colonies on SMA agar plates are yellow with clean edges, 0.2–1 mm in diameter after 3 d at 30°C. Do not form spores. Branching does not occur. Aerobic; chemoorganotrophic. **Catalase and oxidase positive.** Optimum growth temperature 30°C. Based on analysis of 16S rDNA sequences, *Rhodanobacter* belongs in the class *Gamma proteobacteria*, order *Xanthomonadales*, and family *Xanthomonadaceae*.

The mol% G + C of the DNA is: 63.

Type species: ***Rhodanobacter lindaniclasticus*** Nalin, Simonet, Vogel and Normand 1999, 22.

FURTHER DESCRIPTIVE INFORMATION

Rhodanobacter lindaniclasticus can degrade γ -hexachlorocyclohexane (lindane; γ -HCH) under aerobic conditions. The first gene in the lindane degradation pathway, *linA*, is highly conserved (99.4%) between *R. lindaniclasticus* and the other known aerobic lindane degrading bacterium, *Sphingomonas paucimobilis* (Thomas et al., 1996b).

ENRICHMENT AND ISOLATION PROCEDURES

Thomas et al. (1996b) collected γ -HCH-contaminated soil from several wood treatment facilities and added 5 g to a mixture of 10 g each Biolyte CX80 and CX85 in 100 ml of Ringer (1/4) solution. Following 1 h on a rotary shaker and 10 min of ultrasonication, 1 ml of the resulting supernatant was used as an inoculum for the enrichment culture and added to 3 l of a mineral medium containing γ -HCH as the sole carbon and energy source. Over 18 months, γ -HCH was added and the pH adjusted as necessary, using 1 N NaOH. Aliquots were periodically plated onto SMA agar and tested for ability to utilize γ -HCH.

DIFFERENTIATION OF THE GENUS *RHODANOBACTER* FROM OTHER GENERA

Rhodanobacter can be distinguished from other members of the *Xanthomonadaceae* by its ability to utilize γ -HCH as a growth substrate. Other distinguishing characteristics are listed in Table BXII.γ.17 in the chapter on *Xanthomonas*.

List of species of the genus *Rhodanobacter*

1. *Rhodanobacter lindaniclasticus* Nalin, Simonet, Vogel and Normand 1999, 22^{VP}

lin.da'ni.clas.ti.cus. Fr. masc. n. *lindane* commercial name of γ -HCH; Gr. adj. *clasticus* breaking, from G. part. perf. *klastos* broken; M.L. masc. adj. *lindaniclasticus* lindane-breaking.

Colonies on nutrient agar are yellow (nothoxanthin), circular, and convex, 0.2–1 mm in diameter. Growth on Milieu Base Salin (MBS) with γ -HCH as sole carbon source. Catalase, oxidase, and Tween 80 esterase positive. Urease,

protease, tryptophanase, nitrate reductase, amylase, and DNase negative. Utilizes the following carbon sources: D-glucose, esculin, cellobiose, trehalose, D-xylose, lactate, caprate, malate, and citrate. Optimal growth at 30°C; also grows at 41°C; no growth at 4°C.

The type strain was isolated from soil at a wood-treatment site.

The mol% G + C of the DNA is: 63 (HPLC).

Type strain: RP5557, LMG 18385.

GenBank accession number (16S rRNA): AF039167.

Genus VIII. *Schineria* Tóth, Kovács, Schumann, Kovács, Steiner, Halbritter and Máriageti 2001, 406^{VP}

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Schĩ ner.i.a. N.L. fem. n. *Schineria* pertaining to Schiner who first described the fly *Wohlfahrtia magnifica* in 1862.

Gram-negative nonmotile aerobic rods. Catalase positive. Major fatty acids, C_{14:0}, C_{16:0}, and C_{18:1}. Major polar lipids include phosphatidylethanolamine, phosphatidylglycerol, and phosphatidylserine.

The mol% G + C of the DNA is: 42 (HPLC).

Type species: ***Schineria larvae*** Tóth, Kovács, Schumann, Kovács, Steiner, Halbritter and Máriageti 2001, 406.

FURTHER DESCRIPTIVE INFORMATION

The inner cell membrane forms invaginations that protrude into the cytoplasm (Tóth et al., 2001).

The *Schineria* strains studied were isolated from larvae of *Wohlfahrtia magnifica*, a dipteran that is an obligate parasite and that infects many vertebrates, including domestic livestock; it is possible that the strong chitin-degrading capability of *Schineria* strains plays a role in molting of the host larvae (Tóth et al., 2001).

Analysis of 16S rDNA sequences placed *Schineria larvae* in the *Gamma*proteobacteria (Tóth et al., 2001).

ENRICHMENT AND ISOLATION PROCEDURES

Schineria larvae strains were isolated on King's B medium from homogenates of washed *Wohlfahrtia magnifica* stage 1 and stage 2 larvae (Tóth et al., 2001).

MAINTENANCE PROCEDURES

Schineria larvae strains were maintained on King's B medium (Tóth et al., 2001).

List of species of the genus *Schineria*

1. ***Schineria larvae*** Tóth, Kovács, Schumann, Kovács, Steiner, Halbritter and Máriageti 2001, 406^{VP}
lar'vae. L. gen. fem. *larvae* pertaining to the origin of the type strain isolated from *Wohlfahrtia magnifica* maggots.

Single straight encapsulated rods (0.8–0.9 × 2.0–3.0 µm). Produce H₂S. Hydrolyze chitin and urea; do not hy-

drolyze casein, esculin, or gelatin. Nonhemolytic. Do not utilize citrate. Voges–Proskauer test negative. No acid produced from carbohydrates in bioMérieux API CH50 test strips.

The mol% G + C of the DNA is: 42 (HPLC).

Type strain: L1/68, CIP 107108, DSM 13226.

GenBank accession number (16S rRNA): AF252143.

Genus IX. *Stenotrophomonas* Palleroni and Bradbury 1993, 608^{VP*}

NORBERTO J. PALLERONI

Ste.no.tro.pho.mo' nas. Gr. adj. *stenus* narrow; Gr. n. *trophus* one who feeds; Gr. n. *monas* a unit, monad; M.L. fem. n. *Stenotrophomonas* a unit feeding on few substrates.

Straight or curved rods but not helical, 0.5 × 1.5 µm, occur singly or in pairs. Do not accumulate poly-β-hydroxybutyrate granules as intracellular reserve material. No resting stages are known. Gram negative. **Motile by two or more polar flagella.** **Aerobic, having a strictly respiratory type of metabolism with oxygen as electron acceptor. Nitrate is reduced but it is not used as a nitrogen source for growth.** The colonies are yellow, greenish, or gray. The yellow color is not due to carotenoids or xanthomonadins. The colonies may turn a dark brown color with age. **The major polyamines are spermidine and cadaverine.** Aside from fatty acids in common with *Xanthomonas* species, one of the species (*S. maltophilia*) is characterized by the presence of C_{17:0} cyclopropane fatty acid and ubiquinone Q8. No growth occurs at 4 or 41°C; the optimum growth temperature is ~35°C. **Oxidase negative.** Gelatinase and catalase positive. Strong lipolytic activity, as judged by the hydrolysis of Tween 80. **Chemoorganotrophic.** One of the species (*S. maltophilia*) is widely distributed in nature and is commonly isolated from clinical materials and nosocomial infections.

The mol% G + C of the DNA is: 66.9 ± 0.8.

Type species: ***Stenotrophomonas maltophilia*** (Hugh 1981) Palleroni and Bradbury 1993, 608 (*Pseudomonas maltophilia* Hugh 1981, 195; *Xanthomonas maltophilia* Swings, De Vos, Van Den Mooter and De Ley 1983, 412.)

FURTHER DESCRIPTIVE INFORMATION

Morphological characteristics *Stenotrophomonas* cells are morphologically uniform and appear as slender straight or curved rods, single or in pairs (Fig. BXII.γ.36). No granules of reserve materials are visible under the phase contrast microscope in cells grown under various conditions. No studies on fine structure of thin sections have been reported in the literature.

The cells have two or more polar flagella (Fig. BXII.γ.37),

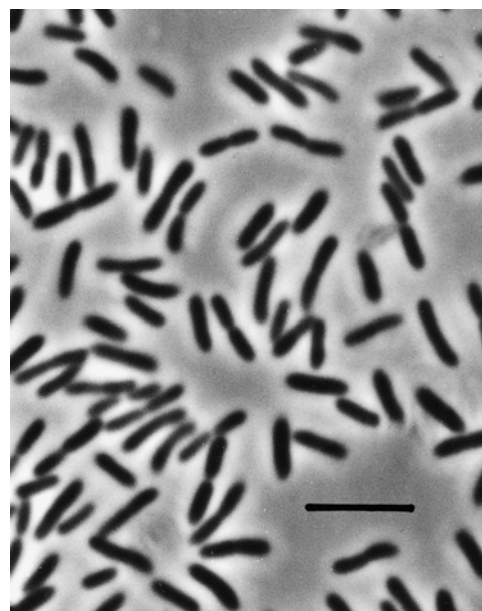


FIGURE BXII.γ.36. Cells of *Stenotrophomonas maltophilia* strain ATCC 13637^T grown in nutrient broth. Phase contrast. Bar = 5 µm.

*Editorial Note: The literature search for the chapter on *Stenotrophomonas* was completed in January, 2000. During the course of unavoidable publication delays, a number of new species were described or reclassified after the chapter was completed. It was not possible to include these species in the text or to include their characteristics in the comparative tables. The reader is encouraged to consult the studies listed in the Further Reading section.

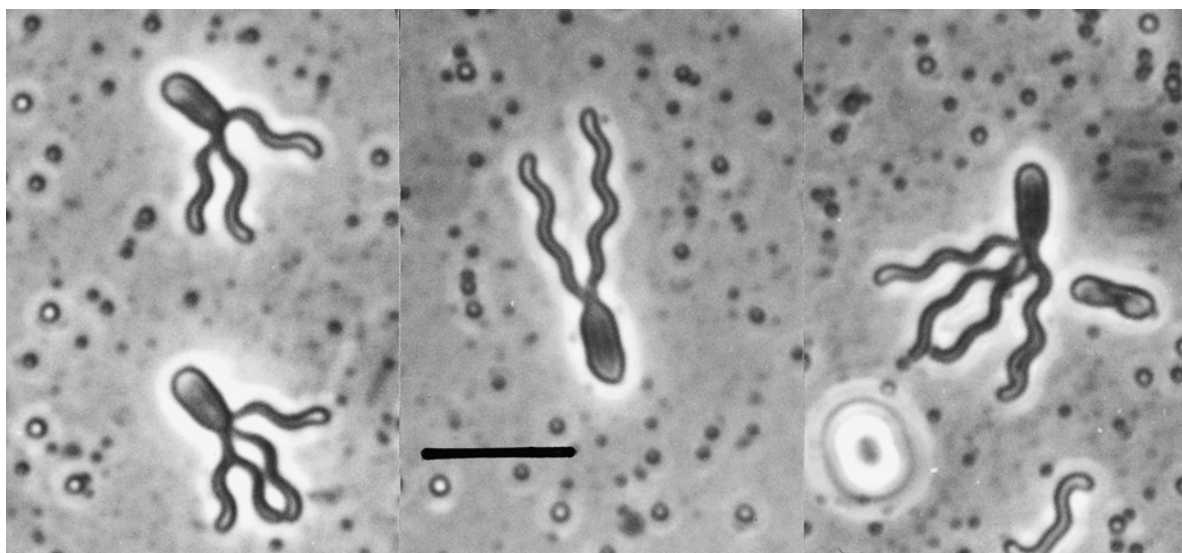


FIGURE BXII-γ.37. *Stenotrophomonas maltophilia* strain ATCC 13637^T. Leifson flagella stain. Bar = 5 μm.

whereas *Xanthomonas* species typically have only one polar flagellum per cell. A flagellin of molecular weight 33 kDa has been isolated (Montie and Stover, 1983).

Polar fimbriae occur in *S. maltophilia*, as well as in several other pseudomonads (Fuerst and Hayward, 1969). These elements may be necessary for attachment to host cells and other solid surfaces, and may also be involved in the twitching motility of the *Stenotrophomonas* cells on solid surfaces (Henrichsen, 1975).

Chemical composition Most studies on cell composition in this genus refer to *S. maltophilia*. This species is resistant to EDTA treatment, whereas *P. aeruginosa* and other species of this genus were found to be particularly sensitive, due to removal of components (metal ions, proteins, lipopolysaccharides, lipids) from the outer membrane of these organisms by the chelator (Wilkinson, 1970; Wilkinson et al., 1973).

An early report on the lipid composition of several species of *Xanthomonas* (Rietschel et al., 1975) shows a marked similarity to the lipid composition of whole cells of *S. maltophilia*, and clear differences from the composition of pseudomonads of other phenotypic groups (Moss et al., 1972, 1973), in agreement with the taxonomic relationships shown by rRNA-DNA hybridization experiments (Palleroni et al., 1973). Both *S. maltophilia* and *Xanthomonas* have a lipid A composition based on glucosamine (Hase and Rietschel, 1976), and similarities in the LPS composition include a high content of rhamnose, the presence of D-galacturonic acid, the absence of a heptose, and a low KDO content (Neal and Wilkinson, 1979, 1982). The major fatty acids of the outer membrane of *S. maltophilia* include 9-methyl-decanoic, 2-hydroxy-9-methyldecanoic, 3-hydroxy-9-methyldecanoic, 3-hydroxydodecanoic, and 3-hydroxy-11-methyldecanoic acids (Wilkinson et al., 1983b).

An interesting feature of the LPS of *S. maltophilia* is the presence of a pentose derivative identified as 3-O-methyl-L-xylose, which is also found in unrelated species such as *Rhodopseudomonas viridis* (Weckesser et al., 1974). Other observations on the LPS structure and a possible basis for the serological cross-reaction between *Stenotrophomonas* and *Brucella* have been reported (Di Fabio et al., 1987), as well as details of the synthesis of the tetrasaccharide repeating unit of *Stenotrophomonas* LPS (Liptak et

al., 1986). A specific extraction procedure has been described for the study of outer membrane proteins (Chin and Dai, 1990), which participate in the antibiotic susceptibility of *S. maltophilia* strains isolated from clinical species (Cullmann, 1991; Lecso-Bornet et al., 1992; Wilcox et al., 1994).

By transfer to a medium containing carbenicillin and polyvinyl pyrrolidone, Pease et al. (1989) were able to induce the formation of cell-wall-deficient forms in various stages, which were demonstrated using scanning electron microscopy. Induction of rapid reversion to normal bacterial forms has been attributed to plasmids by the authors. This rapid reversion makes it difficult (although not impossible) to maintain cell-wall-deficient strains in media containing carbenicillin.

The efficiency of solute diffusion through the outer membrane of *S. maltophilia* is roughly 3–5% of that of *E. coli*. This low value in *S. maltophilia* is attributed to the low copy number of porins rather than to differences in the size of the channels (Yamazaki et al., 1989).

No hopanes have been found in *S. maltophilia* cells (Rohmer et al., 1979). *Stenotrophomonas maltophilia* contains spermidine and cadaverine as the major polyamine components (Auling et al., 1991; Yang et al., 1993b).

The whole cell fatty acid composition of *Stenotrophomonas* will be discussed under Taxonomic Comments.

Pigmentation The colonies of *S. maltophilia* strains on various complex and chemically defined media are often yellowish. The color, which does not diffuse into the medium, is not due to carotenoid pigments but perhaps to flavins (Stanier et al., 1966; Palleroni and Bradbury, 1993). Although characterization of the pigment is still tentative, it is probably not a xanthomonadin (the pigment characteristic of *Xanthomonas*) or a closely related compound (Starr et al., 1977).

According to a published description (Hugh and Gilardi, 1980), the colonies of *S. maltophilia* characteristically appear lavender-green in color on blood agar media. A greenish discoloration of erythrocytes develops in the area of confluent growth, but there are no signs of hemolysis around isolated colonies. *S. africana* forms faint fluorescent green colonies on Trypticase soy agar and gray colonies on sheep blood agar and on MacConkey

agar at both 30 and 37°C. (Drancourt et al., 1997). No information on the chemical nature of the pigment(s) is available.

Some strains of *S. maltophilia* develop a brown discoloration in agar media—in fact this was the basis for the name “*Pseudomonas melanogena*” (Iizuka and Komagata, 1963a; Komagata et al., 1974)—which may be due not to water-soluble melanin but to a secondary chemical reaction on some extracellular products (Hugh and Gilardi, 1980). The intensity of this brown color is enhanced after growth at 42°C.

Nutrition and growth conditions Many strains of *Stenotrophomonas* require the addition of growth factors (methionine or cystine plus glycine) for growth on chemically defined media (Iizuka and Komagata, 1964b; Hugh and Gilardi, 1980). Methionine-independent strains, however, have been known for many years (Ikemoto et al., 1980). As suggested elsewhere in this chapter, they may represent a different species of *Stenotrophomonas*.

Even though *Stenotrophomonas* species are able to reduce nitrate, this anion is not a nitrogen source for growth. This lack of growth is a useful characteristic for the differentiation of this genus from most other groups of aerobic pseudomonads (Stanier et al., 1966).

A strong ammonia odor develops after growth on blood agar medium (Hugh and Gilardi, 1980). This property is absent from *S. africana* (Drancourt et al., 1997).

Studies on the nutrition of *S. maltophilia* have shown that this species is not as nutritionally versatile as other species of aerobic pseudomonads of the genus *Pseudomonas* and other species formerly assigned to this genus (Stanier et al., 1966). This restricted nutritional capacity was the basis for the proposed new generic name (*Stenotrophomonas*) (Palleroni and Bradbury, 1993). Typically, strains of the two species of this genus utilize some disaccharides (maltose, cellobiose, lactose) as sole carbon and energy sources (Hugh and Ryschenkow, 1961; Stanier et al., 1966; Drancourt et al., 1997). These substrates are rarely utilized by other pseudomonads. Growth on lactose is poor, perhaps because of the fact that the cells do not use galactose.

The strong lipolytic activity of *S. maltophilia* is due, at least in part, to the presence of an esterase in the outer membrane. The esterase may play a role in the utilization of esters as substrates for growth (Debette and Prensier, 1989). The membrane-bound non-specific esterases of many strains have been analyzed by polyacrylamide gel electrophoresis and isoelectric focusing, and the relationship of these observations to the distribution of selected phenotypic properties of the strains supports the idea that species names such as *Pseudomonas beteli* and *P. hibiscicola* should be considered synonyms of *S. maltophilia* (Singer et al., 1994), as suggested by Van Den Mooter and Swings (1990).

A micromethod for the nutritional screening of strains of *S. maltophilia* has been described (Frenay et al., 1983). In the earlier study by Stanier et al. (1966), 23 strains of *S. maltophilia* were subjected to an extensive nutritional screening (see the description of this species later in this chapter). No simple alcohols, polyalcohols or glycols, aromatic compounds, amines, various nitrogenous compounds, or paraffin hydrocarbons were utilized by the great majority of strains. Of the 145 substrates tested, only 23 supported growth of all or most strains. The strains tested in this study—provided by Dr. P. Thibault of the Pasteur Institute—had been isolated from various materials of clinical origin. However, later isolations from different natural habitats have resulted in the identification of strains of *S. maltophilia* capable of degrading sulfonic-aromatic compounds (Lee and Clark, 1993), the

herbicide Dicamba (3,6-dichloro-2-methoxybenzoic acid) (Yang et al., 1994; Wang et al., 1997), toluene and xylenes (Su and Kafkewitz, 1994), benzoate (Liu et al., 1996b), or chitin (Andreeva et al., 1996; Wiwat et al., 1996). *S. maltophilia* has been isolated from enrichments with kerosene and crude oil, inoculated with soil from petroleum regions in Japan. Although the isolates used hydrocarbons, they rapidly lost this capacity after subculture under laboratory conditions (Iizuka and Komagata, 1964b).

Some of these properties place *Stenotrophomonas* species among those of interest for applications in biotechnology and for bioremediation of contaminated environments. However, it is important to keep in mind that organisms reputed to be opportunistic human pathogens should be used under strictly controlled conditions in large-scale biotechnological projects.

Metabolism Studies on many basic aspects of metabolism and metabolic pathways in this group of bacteria are rather fragmentary. The strains of *Stenotrophomonas* species give a negative oxidase reaction, which is correlated with the lack of cytochrome *c* (Stanier et al., 1966). Neither an α -band at 551–554 nm nor a β -band at 522–524 nm was seen in the spectra. The oxidase reaction was mistakenly recorded as positive by Palleroni and Bradbury (1993), following the report of Van Den Mooter and Swings (1990) who recorded a positive reaction for the majority of their strains. Some strains gave at most a faint positive oxidase reaction.

The metabolism of fructose involves the participation of a phosphoenol-pyruvate (PEP) phosphotransferase system (Sawyer et al., 1977b). *S. maltophilia* is able to use lactose for growth, a property very rare in other aerobic pseudomonads. As mentioned above when dealing with the nutritional aspects of this species, growth on this disaccharide, however, is rather poor, which may be related to the fact that the cells do not use galactose as a carbon source.

In the tyrosine biosynthetic pathway, both *Pseudomonas* and *Stenotrophomonas* strains have a prephenate activity linked to NAD; this activity is sensitive to feedback inhibition in *Stenotrophomonas* but not in *Pseudomonas*. The latter genus also has an arogenate dehydrogenase linked to NAD, an enzyme that is lacking in *Stenotrophomonas* (Byng et al., 1980). Studies on the regulatory pattern of prephenate and arogenate dehydratase activities in the phenylalanine biosynthetic pathways have suggested that *Stenotrophomonas* be classified in the same group as the fluorescent species of *Pseudomonas* (Whitaker et al., 1981b).

In spite of the fragmentary biochemical information that is available, various reports on the isolation and properties of a number of enzymes are found in the literature. These refer to proteases (Debette, 1991; Margesin and Schinner, 1991; Singer and Debette, 1993), peptidases (Stevens et al., 1990; Suga et al., 1995; Kabashima et al., 1996), cocaine esterase (Britt et al., 1992a), a secondary alcohol dehydrogenase that has little activity on primary and aromatic alcohols (Britt et al., 1992b), deoxyribonuclease (Scully and Winder, 1991), and enzymes involved in ADP ribosylation (Edmonds et al., 1989), a mechanism of action found in important bacterial toxins. The peptidyl dipeptidase activity of *S. maltophilia* appeared to be the most active among those of 24 different bacterial species examined (Stevens et al., 1990).

Studies on the *S. maltophilia* β -lactamases will be mentioned in the section on antibiotic susceptibility.

As stated elsewhere, *S. maltophilia* is unable to use nitrate as a nitrogen source, but it can reduce this anion to nitrite (Iizuka

and Komagata, 1964b; Stanier et al., 1966). The nitrate reductase of some strains has been characterized (Woodward et al., 1990; Ketchum and Payne, 1992). A membrane-bound methyl viologen nitrate reductase and a soluble molybdopterin have been described in a study on 23 strains of *S. maltophilia* (Woodward et al., 1990).

The above studies point to the fact that in the reduction of nitrate to nitrite, the former can act as an electron sink when the organisms are growing under semiaerobic conditions. In a study of utilization of toluene and xylenes by a strain of *S. maltophilia*, these aromatic substrates were used under low oxygen and anoxic conditions in the presence of nitrate (Su and Kafkewitz, 1994). For strains with nitrite and nitrate reductase activity, see the note at the end of the list of species.

Phages and plasmids The information available on these subjects is scanty and fragmentary. The sensitivity of *Xanthomonas* showed a major overlap with the sensitivity of *S. maltophilia* among the various species then assigned to the genus *Pseudomonas* (Starr, 1981). This finding is a further confirmation of the phylogenetic relationship between *Xanthomonas* and *Stenotrophomonas* as members of the *Gamma*proteobacteria.

It has to be admitted, however, that the practical application of phage sensitivity as a diagnostic tool frequently fails to fulfill expectations. As stated by Billing and Garrett (1980) with reference to problems of identification of plant pathogenic bacteria, "phages potentially useful in diagnosis have been found in some cases, but all too often in other cases they have proved non-specific or too specific." However, there are instances in which phage sensitivity may become an important piece of evidence pointing to basic phylogenetic relationships. For example, a phage lytic for *Pseudomonas* species, both fluorescent and non-fluorescent, but not for *S. maltophilia*, was isolated from sewage and described in detail (Kelln and Warren, 1971). On the other hand, the isolation of a temperate phage capable of plaque formation on both *P. aeruginosa* and *S. maltophilia* has been described (Moillo, 1973).

Some factors required for phage nucleic acid replication may be conserved through evolution in some groups of Gram-negative bacteria. One of the elements required for *in vitro* replication of sex-specific single stranded RNA coliphage Q β is the so-called host factor (HF), which is a heat-resistant RNA-binding protein of molecular weight 12,000, usually present in *E. coli* as a hexamer (Franze de Fernández et al., 1972). A component able to cross-react with an *E. coli* HF antiserum was detected in the extracts of a number of species of aerobic pseudomonads, among them, *S. maltophilia* (DuBow and Ryan, 1977). Cell extracts of members of rRNA similarity groups I and V (*Pseudomonas*, *Xanthomonas*, and *Stenotrophomonas*) had HF activity after heat treatment. Heat-treated extracts of species of pseudomonads belonging to different branches of *Proteobacteria* did not stimulate replication of Q β RNA and, in addition, they inhibited the activity of systems containing saturating amounts of *E. coli* HF, in other words, they contained a heat-resistant inhibitor.

The promiscuous IncP-1 plasmid R18 was analyzed by means of Tn7 insertion mutations in the *Tral* region, involved in conjugational transmissibility from *P. aeruginosa* to other species, including *S. maltophilia* (Schilf and Krishnapillai, 1986).

From a collection of 18 strains of *S. maltophilia*, 5 were shown to harbor plasmids, of which one was studied in some detail. Its molecular weight is 4.4×10^6 daltons, and the presence of several single restriction sites makes it promising as a cloning vector (Shen et al., 1992).

Antigenic properties A 30 kDa immunoglobulin binding protein was isolated from *S. maltophilia* (Grover et al., 1991a) and later partially characterized (Grover and Odell, 1992). The protein binds IgGs of various subclasses and IgA at their Fc regions. A substantial number of reports on the composition of several antigens in the outer membrane of *S. maltophilia* by Prof. Wilkinson and collaborators can be found in the literature (Winn et al., 1995, 1996; Winn and Wilkinson, 1995, 1996, 1997).

A serological classification of *S. maltophilia* strains based on the heat-stable O antigens has been proposed, extending the 15 serotypes defined by antisera in previous investigations (Hugh and Ryschenkow, 1961) to a total of 26 (Schable et al., 1989).

The isolation of a membrane protein capable of cross-reacting with the β subunit and carboxyl tail of human pregnancy chorionic gonadotropin (hCG), was described (Grover et al., 1991b, 1993b; Carrell et al., 1993). The partial sequence of the receptor soon followed (Grover et al., 1993a). A complete sequence of the gene encoding the *S. maltophilia* gonadotropin-like protein is now available (Grover et al., 1995). Exposed sites on *S. maltophilia* cells bind both hCG and the native hCG-like ligand with equal affinity (Carrell and Odell, 1992).

Antibiotic sensitivity The high frequency of detection of *S. maltophilia* as an opportunistic pathogen and isolation from materials of clinical origin, has prompted numerous studies on the susceptibility of the strains to various antibiotics, as well as the intrinsic and exogenous conditions that affect such sensitivity (Cullmann, 1991; Lecso-Bornet et al., 1992; Sader et al., 1994; Wilcox et al., 1994; Poulos et al., 1995; Rahmati-Bahram et al., 1995, 1996; Vanhoof et al., 1995; Alonso and Martínez, 1997). The conditions for laboratory tests have also been examined (Yao et al., 1995).

S. maltophilia is resistant to most β -lactam antibiotics, and the aminoglycosides and quinolones have a modest activity (García-Rodríguez et al., 1991). However, some quinolones have been found to be quite active (Furet and Pechere, 1991), in particular, ciprofloxacin (Visalli et al., 1997). In a Japanese study on *S. maltophilia* strains isolated from clinical materials, high sensitivity to minocycline was observed. The sensitivity did not show the yearly changes recorded for *P. aeruginosa* (Tabe and Igari, 1994). *S. africana* has an antibiotic susceptibility restricted to netilmicin, ciprofloxacin, trimethoprim-sulfamethoxazole, and colimycin (Drancourt et al., 1997).

A report on the transmission of antibiotic resistance from *S. maltophilia* to *Pseudomonas aeruginosa* calls attention to the combined danger that these two species represent as the most common Gram-negative opportunistic pathogens in the hospital environment (Babalova et al., 1995). Laboratory experiments with a strain of *S. maltophilia* have shown the possibility of transferring to other species its multiple resistance to drugs (which included amikacin and cephalosporins) coded for by genes in an R plasmid (Krcmery et al., 1985). Interestingly, antibiotic and heavy metal resistance genes of Gram-positive origin, also detected in *Staphylococcus aureus*, have been identified in a *S. maltophilia* strain of clinical origin (Alonso et al., 2000).

Among the resistance factors of *S. maltophilia*, β -lactamases have received much attention (Paton et al., 1994; Felici and Amicosante, 1995; Bonfiglio et al., 1995; Kelly and Mortensen, 1995). Sequence studies on the L1 metallo- β -lactamase have discovered significant differences from similar enzymes present in *Aeromonas hydrophila* and *Bacillus cereus*, which resulted in the addition of a new subclass of this type of enzymes to the ones already known (Walsh et al., 1994). The L1 penicillinase *bla*S gene of *S. malto-*

philia was cloned in *E. coli*, where it could express its capacity for imipem hydrolysis (Dufresne et al., 1988).

A recent study of the L2 serine β -lactamase from *S. maltophilia* has demonstrated that this enzyme is a clavulanic acid-sensitive cephalosporinase related to a β -lactamase from *Yersinia enterocolitica* (Walsh et al., 1997a).

A number of antibiotics have been tested against *S. africana* using the disk diffusion technique (Drancourt et al., 1997). This species appeared to be susceptible to netilmicin, ciprofloxacin, trimethoprim-sulfamethoxazole, and colimycin.

An antifungal compound, maltophilin, is produced by *S. maltophilia*. It is active against human and plant pathogenic fungi (Jakobi et al., 1996).

Relationships to plants A predominance of *S. maltophilia* over other pseudomonads has been observed in the rhizosphere of several cultivated plants, such as cabbage, rape, mustard, corn, and beet (Debette and Blondeau, 1980). Some of these plants have proteins of high sulfur content, and the abundance of *S. maltophilia*, which requires S-containing amino acids like methionine, may be due to excretion of such compounds by the roots. However, methionine could not be detected in significant amounts in the rhizosphere.

In the rhizosphere of oilseed rape, *S. maltophilia* is able to interfere with growth of plant pathogenic fungi (Berg et al., 1996). The antifungal compound maltophilin (Jakobi et al., 1996) may be involved in this activity. Use of chitinolytic strains of *S. maltophilia* has been proposed for biological control of summer patch on turf grass (Kobayashi et al., 1995).

S. maltophilia has been found on the surface of leaves of the grass *Lolium perenne*, although at a low frequency in comparison to other aerobic pseudomonads (Austin et al., 1978).

Pathogenicity for humans *S. maltophilia* has been implicated as a direct agent of disease and as the cause of secondary infections that aggravate various human pathological conditions. It has been identified in cystic fibrosis (Gladman et al., 1992; Denton et al., 1996), cancer (Khardori et al., 1990; Elsner et al., 1997), meningitis (Nguyen and Muder, 1994; Papadakis et al., 1997), soft tissue infections (Vartivarian et al., 1994), septic bursitis (Papadakis et al., 1996), urinary tract infections (Vartivarian et al., 1996), bacteremia (Muder et al., 1996), endophthalmitis following surgical procedures (Kaiser et al., 1997), and pneumonia (Elsner et al., 1997). The literature records concern for the emergence of the species as a dangerous opportunistic pathogen (Marshall et al., 1989), and for the role of anti-pseudomonadal antibiotics in this process.

A method based on the PCR technique for the detection of *Pseudomonas aeruginosa*, *Burkholderia cepacia*, and *S. maltophilia* in sputum of cystic fibrosis patients has been described, but the specificity is low for *S. maltophilia*, which may be due to improper PCR conditions (Karpatis and Jonasson, 1996). It is interesting to mention here that strains of *S. maltophilia* do not produce alginate, and that genetic studies have demonstrated failure to transfer a plasmid carrying the gene *algT* from *P. aeruginosa* to *S. maltophilia*. The studies also failed to identify sequences homologous to this gene in samples examined by Southern analysis (Goldberg et al., 1993).

The type strain of *S. africana* was isolated from an HIV patient who exhibited primary meningoencephalitis (Drancourt et al., 1997).

Ecology The variety of materials from which *S. maltophilia* can be isolated is truly remarkable. A list compiled by Hugh and

Gilardi includes raw and pasteurized milk; well, stagnant, and river waters; sewage; frozen fish; feces of snakes, lizards, frogs, and rabbits; rotten eggs; and soil in petroleum zones. It has been isolated from the internal tissue of banana pseudostem, decaying banana sucker, cotton seed, bean pod, and tobacco seedling (Hugh and Gilardi, 1980). Strains similar to *Xanthomonas* and *Stenotrophomonas* have been frequently isolated from biofilms for waste gas treatment (Lipski et al., 1992; Lipski and Altendorf, 1997). Some strains have been assigned to a new species, *Stenotrophomonas nitritireducens* (Finkmann et al., 2000) (see note at the end of the list of species). From a soda lake, an alkaliphilic strain was studied by Duckworth et al. (1996).

For years, the species has been associated with a variety of infections in humans (Sutter, 1968; Gilardi, 1971), and it could be found in "contaminated tissue culture, streptomycin solution, distilled water, incubator reservoirs, respirators, nebulizers, and evacuated blood collection tubes" (Hugh and Gilardi, 1980). As mentioned above, strains of the species have been isolated from the rhizosphere and aerial parts of plants. These organisms may represent a very high proportion of the non-fermentative, Gram-negative bacteria in the soil microflora (Debette and Blondeau, 1977), and they have been isolated in large numbers from manured soils in Japan (Katoh and Suzuki, 1979) (cited by Ikemoto et al., 1980). The isolation of three strains similar to *S. maltophilia*, from gut and feces of the arthropod *Folsomia candida* was reported by Duckworth et al. (1996).

The distribution of *S. africana* in nature is not known.

ENRICHMENT AND ISOLATION PROCEDURES

Strains of *S. maltophilia* have been isolated from a bewildering variety of materials (Hugh and Gilardi, 1980) and they are second only to *P. aeruginosa* in the frequency of isolation in hospitals and clinical laboratories. Because the nutritional spectrum of the species is so restricted, it is not possible to design effective enrichment procedures based on carbon sources specifically used by strains of *S. maltophilia*. As mentioned elsewhere (Table BXII.γ.28), some carbon sources (maltose, lactose, cellobiose) are used by *S. maltophilia* strains and not by strains of other species of aerobic pseudomonads. However many other organisms found in natural samples are capable of using these substrates, which conspires against the effectiveness of enrichment procedures based on their use by *S. maltophilia*. The fact that so many strains have been isolated using traditional bacteriological procedures is indicative of their wide occurrence in appreciable numbers in many natural materials. In view of the fact that many strains require growth factors (methionine or other amino acids), direct isolation is best performed by streaking plates of any of several

TABLE BXII.γ.28. General characteristics of *S. maltophilia*

Characteristics	Typical phenotype
Methionine requirement	+
Oxidase reaction	—
Utilization of nitrate as nitrogen source	—
Growth on:	
Lactose	+
Maltose	+
L-Aspartate	—
β-Hydroxybutyrate	—
Glutarate	—
Glycerol	—
Cellobiose	+

complex media (Difco Nutrient Agar, Trypticase Soy, Brain Heart Infusion, etc).

MAINTENANCE PROCEDURES

As with most aerobic pseudomonads, *S. maltophilia* can be maintained in slants of common complex culture media, with transfers every one or two months. Although there is no information about *S. africana*, the above considerations might also apply to this species. *S. maltophilia* tolerates freeze-drying well, although freezing of cell suspensions at -20°C or lower after addition of a cryo-protecting agent like glycerol (5–10% final concentration) may be the most reliable method for long-term preservation, since some strains do not revive from freeze-dried preparations.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Screening for nutritional properties Details of the methodology for determination of the nutritional spectra of aerobic pseudomonads are available (Stanier et al., 1966; Palleroni and Doudoroff, 1972; Palleroni, 1984). A simple minimal medium that does not include strong chelating compounds (Palleroni and Doudoroff, 1972)¹ gives better results than that of earlier formulations (Terry et al., 1991). Methionine (50 $\mu\text{g}/\text{ml}$) should be added from a sterile stock solution after the medium has been autoclaved. Good heterotrophic growth can be obtained by adding to the medium a single organic compound as carbon and energy source to a final concentration of 0.1%. Some compounds may be toxic at this concentration, but even if this is reduced to 0.0125%, visible growth above the level of the control without substrate can be obtained. Precautions and useful suggestions can be found in the above-mentioned papers.

Acid production from carbohydrates The recommended medium (Hugh and Leifson, 1953)² gives clear and reproducible results. It contains a small amount of peptone, and ammonification does not interfere with the acid production. As mentioned by Palleroni and Doudoroff (1972), the results do not necessarily correlate with utilization of the substrates for growth. There may be redundancy in some of the results, since a single dehydrogenase may cause acid production from different sugars.

Hydrolysis of Tween 80 The test devised by Sierra (1957) for the detection of lipolytic activity is very simple and the results are clear. Sierra's medium³ contains Tween 80 and calcium chloride, and opacity formation around the patches of growth or the individual colonies indicates precipitation of calcium salts of the fatty acids liberated by lipases (Sierra, 1957).

Nitrate reduction Reduction of nitrate to nitrite can be tested according to Lelliott et al. (1966).

Oxidase reaction A description of this test can be found in the paper by Stanier and collaborators (Stanier et al., 1966).

Gelatinase Of the several techniques that have been proposed, the one described by Skerman (1967) is most convenient.

Egg yolk reaction The egg yolk test has been used for many years to detect lecithinase production (Esselman and Liu, 1961). However, the reaction is quite complex, although the results are in most cases clear and reproducible. The test can be performed according to published procedures (Lelliott et al., 1966; Stanier et al., 1966).

Catalase reaction The presence of catalase is simply tested by placing a drop of a hydrogen peroxide solution on top of a colony and observing the formation of bubbles of free O_2 . The observation may be facilitated by using a dissecting microscope. In cases where the reaction is very weak, the use of an oxygen electrode may be advisable (Auling et al., 1978).

Nucleic acid hybridization Several methods are in use. In some of them, one of the DNA partners is immobilized on a membrane (Johnson, 1994a; Ballard et al., 1970; Palleroni et al., 1972). More stringent conditions are obtained by hybridization in solution followed by hydrolysis of single strands using S1 nuclease (Johnson and Palleroni, 1989; Johnson, 1994a). rRNA–DNA hybridization, as originally applied (Palleroni et al., 1973), has been largely replaced by sequence studies of PCR-amplified genes (Johnson, 1994b).

Fatty acid analysis Although several analytical procedures based on similar principles are in use, the method described by Stead (1992) can be recommended for the pseudomonads.

DIFFERENTIATION OF THE GENUS *STENOTROPHOMONAS* FROM OTHER GENERA

Stenotrophomonas can be easily differentiated from other genera of aerobic pseudomonads. The salient characteristics of *Stenotrophomonas* include the requirement for growth factors (mainly methionine), the inability to use nitrate as a nitrogen source, and the restricted range of substrates that can be used as sole carbon and energy sources. A clear differentiation of *S. maltophilia* from species of various genera of aerobic pseudomonads is possible using the small group of selected characteristics shown in Table BXII.γ.28. When the occurrence of these characters is scored for various other groups of pseudomonads, a clear separation is evident (Table BXII.γ.29). As stated elsewhere (Stanier et al., 1966), these organisms constitute the most isolated group among the aerobic pseudomonads. Useful phenotypic characteristics for the differentiation of *S. maltophilia* and *S. africana* from some species of aerobic pseudomonads are given in Table BXII.γ.30. *S. nitritireducens* could not be included in the table since no pertinent phenotypic details are given in the characterization of this species.

TAXONOMIC COMMENTS

Strains of *S. maltophilia* were included in the phenotypic study of Stanier et al. (1966), and their phenotypic characteristics were precisely defined. Later studies by Palleroni et al. (1973) separated this species from all other species of *Pseudomonas* and allocated it to rRNA similarity group V, together with the species of the genus *Xanthomonas*. Based on these results, Swings et al. (1983) proposed the transfer of *P. maltophilia* to the genus *Xanthomonas*, a decision that did not receive universal approval. In a study on numerical taxonomy and protein gel electrophoresis patterns as tools for the differentiation of pseudomonads, van Zyl and Steyn (1990) showed that *S. maltophilia* and *Pseudomonas hibiscicola* constituted a cluster that was not significantly close to

1. Medium of Palleroni and Doudoroff (1972) (g/l of 0.33 M Na-K phosphate buffer, pH 6.8): NH_4Cl , 1.0; $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 0.5; ferric ammonium citrate, 0.05; and CaCl_2 , 0.005. The first two ingredients are added to the buffer and sterilized by autoclaving. The ferric ammonium citrate and CaCl_2 are added aseptically from a single stock solution that has been sterilized by filtration.

2. Hugh and Leifson medium (g/l): peptone 2.0; NaCl, 5.0; K_2HPO_4 , 0.3; agar, 3.0; brom thymol blue, 0.03; carbohydrate, 10.0; pH 7.1.

3. Sierra's medium (g/l): Bacto-peptone (Difco), 10.0; NaCl, 5.0; $\text{CaCl}_2 \cdot \text{H}_2\text{O}$, 0.1; and agar, 17.0. Tween 80 is sterilized separately and added to the medium after autoclaving to give a final concentration of 10 g/l.

TABLE BXII.γ.29. Distribution of ten selected characters among various aerobic pseudomonad species^a

Number of characters from table 1 that are shared by various species	Organisms			
	<i>S. maltophilia</i> (23 strains)	<i>Pseudomonas</i> spp. (202 strains)	<i>Comamonas</i> spp. (26 strains)	<i>Burkholderia cepacia</i> and <i>B. pseudomallei</i> (60 strains)
10	21	0	0	0
9	2	0	0	0
8	0	0	0	0
7	0	0	0	0
6	0	0	0	0
5	0	0	0	0
4	0	1	0	0
3	0	1	0	7
2	0	11	0	35
1	0	23	21	18
0	0	166	5	0

^aEach figure in the table represents the number of strains positive for a given number of the characteristics listed in Table BXII.γ.28.

the *Xanthomonas* group, and the authors found it reasonable to accept the suggestion of Palleroni (1984) for the creation of a new genus name for the strains, which were at the time labeled *Xanthomonas maltophilia*. In fact, a later paper by the same authors (van Zyl and Steyn, 1992) clearly highlighted the differences between *X. maltophilia* and the xanthomonads, and presented compelling evidence that culminated in the proposal of the new genus *Stenotrophomonas* by Palleroni and Bradbury (1993).

Xanthomonas has been classically defined as a group of plant-pathogenic organisms and, from what we know about *Stenotrophomonas*, the pathogenic propensities of the two species assigned to the genus seem to be directed to human hosts. We know very little about the relationship of *S. maltophilia* with plants, but, as indicated in an earlier section of this chapter, the association seems to be a beneficial one. But even if we discard the value of these qualifications as part of a genus definition, various other characteristics have been invoked elsewhere to support allocating (*Xanthomonas*) *maltophilia* to a different genus (Palleroni and Bradbury, 1993). Additional evidence supporting this criterion is summarized in the following paragraphs.

Restriction analysis of PCR-amplified 16S rRNA genes, performed on a collection of *Xanthomonas* and *Stenotrophomonas* strains, resulted in the definition of 19 restriction fragment length polymorphism (RFLP) groups, which subsequently could be subdivided into two main clusters, one containing all the *Xanthomonas* species, and the other, *S. maltophilia*. An examination of the relationship between the DNA similarities as determined by hybridization methods, and the nucleotide substitution rate in the small rRNA subunit, confirmed the differentiation of the two genera (Nesme et al., 1995). A rapid method of classification based on PCR has demonstrated the production of DNA fragments in *S. maltophilia* extracts different from those obtained from the *Xanthomonas* strains tested (Maes, 1993).

S. maltophilia was included in a study using the AFLP (amplified fragment length polymorphism) technique (Janssen et al., 1996). As in the case of RFLP, this promising method has resulted in dendrograms that confirm relationships derived from other molecular approaches (Janssen et al., 1996; Rademaker et al., 2000).

Analysis of fatty acid composition for a large collection of strains of the genus *Xanthomonas* by Yang et al. (1993d) revealed the presence of nine fatty acids in more than 99% of the *Xanthomonas* strains. Three of the acids ($C_{11:0 \text{ iso}}$, $C_{11:0 \text{ iso } 3OH}$, and

$C_{13:0 \text{ iso } 3OH}$), are characteristic for the genus and therefore are taxonomically important to differentiate it from related genera. *S. maltophilia* strains contained the nine typical fatty acids, but, in addition, they were characterized by the presence of the fatty acid $C_{17:0}$ cyclopropane, which was absent from *Xanthomonas* species, with the exception of *X. campestris* pathovar *hyacinthi*. There was great uniformity in the fatty acid composition of the *S. maltophilia* strains, although they had been isolated from very diverse materials, but the DNA homology values among them ranged from 37 to 100% according to Ikemoto et al. (1980).

In a study of a large collection of *S. maltophilia* strains isolated from environmental and clinical sources, Hauben et al. (1999b) found a striking phenotypic uniformity but considerable genomic heterogeneity, and ten genomic groups were observed. However, the lack of conventional phenotypic traits by which the various groups could be identified has prevented the authors from proposing a subdivision of *S. maltophilia* into different species.

Ikemoto et al. (1980) reported the results of studies on *Pseudomonas maltophilia* strains that do not require methionine. Strains of this new biovar (biovar II) differ from *S. maltophilia* (biovar I) in phenotypic properties, fatty acid composition, DNA composition, and requirement for methionine as an organic growth factor. DNA-DNA hybridization studies also point to DNA sequence differences between the two groups. Subsequently, Ha and Komagata (1984) extended the studies to the electrophoretic behavior of six enzymes, with results confirming the differences between the biovars.

Singer et al. (1994), in a comparison between phenotypic properties and esterase electrophoretic polymorphism, included another group of strains not requiring methionine, and confirmed the differences from *S. maltophilia* observed by the Japanese workers. At this point, sufficient evidence is available to suggest that biovar II of *S. maltophilia*, defined by Ikemoto et al. (1980), should be assigned to a new species of the genus. The observed differences in general properties and in DNA-DNA hybridization values clearly support this suggestion.

As previously mentioned, the phage isolated from sewage by Kelln and Warren (1971) was lytic for species of *Pseudomonas* but did not lyse *S. maltophilia* or strains of species of homology groups other than group I. The phylogenetic significance of this finding became apparent after demonstration of the genomic heterogeneity of the genus *Pseudomonas*, as classically defined.

TABLE BXII-7.30. Phenotypic properties that differentiate *S. maltophilia* and *S. africana* from species of other genera of aerobic pseudomonads

Characteristics	<i>S. maltophilia</i>	<i>S. africana</i>	<i>S. pseudomonas aeruginosa</i>	<i>P. fluorescens</i>	<i>P. putida</i>	<i>P. alcaligenes</i>	<i>P. pseudoalcaligenes</i>	<i>P. stutzeri</i>	<i>P. mendocina</i>	<i>Burkholderia cepacia</i>	<i>B. pseudomallei</i>	<i>B. mallei</i>	<i>Ralstonia pickettii</i>	<i>R. solanacearum</i>	<i>Comamonas acidovorans</i>	<i>C. testosteroni</i>
Flagellar number	>1	>1	1	>1	>1	1	1	1	1	>1	1	0	>1	1	>1	>1
PHB accumulation	-	-	-	-	-	-	v	-	-	+	+	+	+	+	+	+
Oxidase reaction	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Gelatin liquefaction	+	+	+	+	-	+	v	-	-	+	+	+	-	-	-	-
Methionine requirement	+	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Arginine dihydrolase	-	-	+	+	+	+	v	-	+	-	+	+	-	-	-	-
Denitrification	-	-	+	v	-	-	-	+	+	-	+	+	+	+	-	-
Growth on:																
Maltose	+	+	-	-	-	-	-	+	-	v	+	v	-	-	-	-
Cellobiose	+	+	-	-	-	-	-	-	-	+	+	+	-	-	-	-
Lactose	+	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Trehalose	+	+	v	v	v	-	-	-	-	+	+	+	+	-	-	-
Arginine	-	-	+	+	+	+	+	-	+	+	+	+	-	-	-	-

FURTHER READING

- Vauterin, L., J. Swings, K. Kersters, M. Gillis, T.W. Mew, M.N. Schroth, N.J. Palleroni, D.C. Hildebrand, D.E. Stead, E.L. Civerolo, A.C. Hayward, H. Maraite, R.E. Stall, A.K. Vidaver and J.F. Bradbury. 1990. Towards an improved taxonomy of *Xanthomonas*. *Int. J. Syst. Bacteriol.* 40: 312–316.

List of species of the genus *Stenotrophomonas*

1. ***Stenotrophomonas maltophilia*** (Hugh 1981) Palleroni and Bradbury 1993, 608^{VP} (*Pseudomonas maltophilia* Hugh 1981, 195; *Xanthomonas maltophilia* Swings, De Vos, Van Den Mooter and De Ley 1983, 412.)
mal.to.phil'i.a. Anglo-Saxon n. *malt* Gr. n. *philia* friend; M.L. fem. n. *maltophilia* friend of malt.
Straight or slightly curved rods, $0.5 \times 1.5 \mu\text{m}$, occurring singly or in pairs. Motile by means of polar multitrichous flagella. Do not accumulate PHB as an intracellular reserve material. Colonies may be yellow or greenish yellow. The yellow color is not due to carotenoid pigments or xanthomonadins. Reduction of nitrate to nitrite is positive, but denitrification does not occur. The oxidase reaction is negative. Strongly lipolytic. Gelatinase reaction positive. Ammonia odor can be produced by colonies on blood agar. Methionine or cystine is required for growth, although strains lacking this requirement may be isolated (Ikemoto et al., 1980).
Limited in its nutritional spectrum. Individual strains used between 24 and 28 of 146 organic compounds tested as main carbon and energy sources (Stanier et al., 1966). The utilizable compounds were: glucose, mannose, sucrose, trehalose, maltose, cellobiose, lactose, salicin, acetate, propionate, valerate, malonate, succinate, fumarate, L-malate, lactate, citrate, α -ketoglutarate, pyruvate, L-alanine, D-alanine, L-glutamate, L-histidine, and L-proline. A variable number of strains also used fructose, isobutyrate, aconitate, and *n*-propanol. Under aerobic conditions, acid is readily produced in complex media with maltose but not with glucose (Hugh and Ryschenkow, 1960). Nitrate is not used as a nitrogen source.
No growth at 4°C or 41°C. Optimum growth temperature, 35°C. Many strains can be isolated from various natural materials, and frequently from clinical specimens. According to Hugh and Gilardi (1980), *S. maltophilia* is the second most frequently isolated aerobic pseudomonad, after *P. aeruginosa*, in the clinical laboratory. Strains of the species appear to be opportunistic human pathogens. Also found in water, milk, frozen foods, and many other sources. Dettre and Blondeau (1980) were able to isolate more strains of *S. maltophilia* than of any other pseudomonad from the rhizosphere of some cultivated plants. Ikemoto et al. (1980) defined two biovars of this species and further studies perhaps will demonstrate that one of them (biotype II) deserves independent species status within the genus *Stenotrophomonas*.
The mol% G + C of the DNA is: 66.9 ± 0.8 (Bd) (Mandel, 1966).
Type strain: ATCC 13637, Hugh 810-2, DSM 50170, ICPB 2648-67, IMET 10402, NCIB 9203, NCTC 10257, NRC 729, RH 1168.
GenBank accession number (16S rRNA): M59158, X95923, AB008509.
2. ***Stenotrophomonas africana*** Drancourt, Niel and Raoult 1997, 162^{VP}
a.fri.ca'na. L. adj. *africana* from Africa.
Curved rods, $0.5 \times 1.5 \mu\text{m}$. Motile by means of two polar flagella. Colonies are gray to green. No ammonia odor is produced on blood plates. Indole and oxidase reactions are negative. Catalase positive. Basic phenotypic properties are similar to those of *S. maltophilia*, among them the basic biochemical characteristics, except for the failure to assimilate *cis*-aconitate. A total of 24 of 1377 16S rDNA base positions are different from those of *S. maltophilia*, and the DNA–DNA similarity between the two species is 35%. Isolated from the cerebrospinal fluid of a patient testing positive for HIV antibodies and exhibiting primary meningoencephalitis (Drancourt et al., 1995, 1997).
The mol% G + C of the DNA is: not known.
Type strain: MGB, CIP 104854.
GenBank accession number (16S rRNA): U62646.
3. ***Stenotrophomonas nitritireducens*** Finkmann, Altendorf, Stackebrandt and Lipski 2000, 281^{VP}
ni.tri.ti.re.du'cens. M.L. *nitritis* nitrite; L. *reducere* to reduce; L. part. pres. *reducens* reducing; M.L. part. pres. *nitritireducens* reducing nitrite.
The following description is transcribed in its entirety from the original publication by Finkmann et al. (2000). Predominant fatty acids are $\text{C}_{15:0}$ iso, $\text{C}_{17:0}$ iso $\omega 7\text{c}$, $\text{C}_{16:0}$ iso, $\text{C}_{14:0}$ iso, and $\text{C}_{15:0}$ anteiso. $\text{C}_{17:0}$ cyclo and $\text{C}_{13:0}$ iso 3OH are present. The strains reduce nitrite to nitrous oxide, nitrate is not reduced, and nitrous oxide is the end product. Esculin is not hydrolysed and lecithin is not degraded. The strains grow in the presence of 0.01% ZnCl_2 . The strains are resistant to fucidin, tetracycline, and novobiocin, but susceptible to streptomycin, nalidixic acid, and kanamycin. The strains show strong hydrolytic activity for *p*-nitrophenyl N-acetyl- β -D-glucosaminide. The strains were isolated from experimental biofilters supplied with artificial waste gases containing ammonia or ammonia and dimethyl disulfide. One strain (B1910/29.1) was isolated from an experimental biofilter supplied with the waste gas of an animal-rendering plant.
The mol% G + C of the DNA is: not known.
Type strain: L2, DSM 12575.
GenBank accession number (16S rRNA): AJ012229.
An examination of the properties reported in the above description clearly shows that a comparison of *S. nitritireducens* with *S. maltophilia* and *S. africana* is not possible, in view of the disparity of criteria used for characterization. Therefore, with the exception of nucleic acid similarity data, there is no phenotypic information that may account for assignment of *S. nitritireducens* to the genus *Stenotrophomonas*, and this species has been excluded from Table BXII.γ.30.

Genus *X. Thermomonas* Busse, Kämpfer, Moore, Nuutinen, Tsitko, Denner, Vauterin, Valens, Rosselló-Mora and Salkinoja-Salonen 2002, 480^{VP}

EWALD B.M. DENNER, PETER KÄMPFER AND HANS-JÜRGEN BUSSE

Ther.mo.mo' nas. Gr. n. *therme* heat; *monas* a unit, monad; M.L. fem. n. *Thermomonas* a thermophilic monad.

Gram-negative filamentous rods (1.0–12.5 × 0.5–0.75 µm). Motile by means of polar monotrichous flagellation or nonmotile. **Aerobic.** Endospores are not formed. **Moderately thermophilic;** temperature range for growth is 18–50°C, with an **optimum between 37 and 50°C.** **Catalase and oxidase positive.** Chemoor-ganotrophic. **Carbohydrates are usually not utilized.** Indole, Voges–Proskauer, methyl red, and Simmons citrate tests are negative. Urea is hydrolyzed. **Nitrate is not reduced to nitrite or nitrogen;** H₂S is not produced. **Ubiquinone Q-8** is the major respiratory lipoquinone system. The main component in the cellular polyamine content is **spermidine.** Polar lipid fingerprints are characterized by the presence of diphosphatidylglycerol, phosphatidylethanolamine, and phosphatidylglycerol. Cellular fatty acids are predominantly **iso-branched** with C_{15:0 iso} and C_{16:0 iso} as major components; other characteristic fatty acids are C_{11:0 iso}, C_{11:0 iso 3OH}, and C_{17:1 iso ω9c}. Based on 16S rRNA gene sequence data, *Thermomonas haemolytica* currently forms a **separate line of descent within the Xanthomonadaceae related to the genera Xanthomonas, Pseudoxanthomonas, Luteimonas, Stenotrophomonas, and Xylella.**

The mol% G + C of the DNA is: 67.1–68.7.

Type species: *Thermomonas haemolytica* Busse, Kämpfer, Moore, Nuutinen, Tsitko, Denner, Vauterin, Valens, Rosselló-Mora and Salkinoja-Salonen 2002, 480.

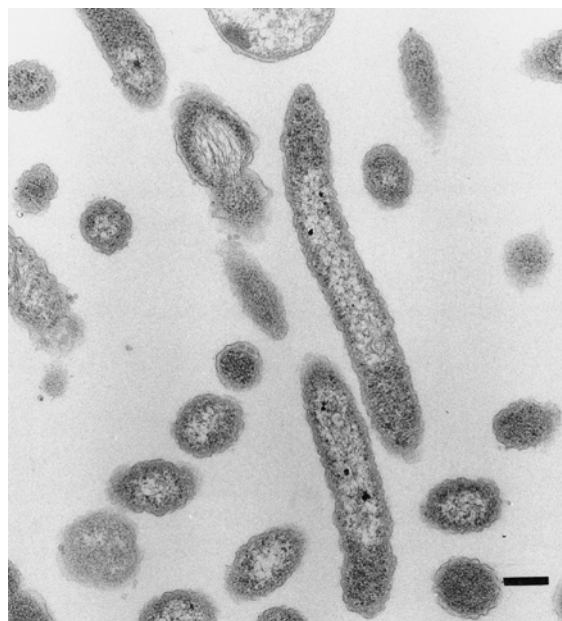
FURTHER DESCRIPTIVE INFORMATION

Currently, the genus contains only a single species, *T. haemolytica*. The four strains (A50-7-3^T, B50-7-1, B50-8-1, and D50-7-1) of the only species yet described characteristically show a marked β-hemolysis on blood agar. Cells of *T. haemolytica* are rod-shaped with a typical Gram-negative cell wall structure and have a tendency to form filaments (Fig. BXII.γ.38). *T. haemolytica* is motile by means of a polar flagellum. While tumbling can be frequently noticed, directed movement is only rarely observable. Motility of the cells is most pronounced at 37°C and higher. The flagellation of the cells can be visualized by the staining procedure of Kodaka et al. (1982), but has not yet been observed by electron microscopy.

T. haemolytica grows well on standard bacteriological media such as Tryptone soy agar (TSA), Luria–Bertani (LB) medium, or nutrient agar. Colonies on standard media are circular, smooth, and convex with an entire edge. At 50°C colonies are whitish/translucent, and at 37°C, transparent. After prolonged incubation (2–3 d) on TSA at 50°C, *T. haemolytica* produces a diffusible brownish discoloration of the medium and a characteristic ammonium-like odor. The temperature range for growth is 18–50°C; no growth occurs at 14 or 53°C.

Cellular fatty acid analyses indicate that with increasing growth temperature, over a range of 23–50°C, the proportion of unsaturated fatty acids in *Thermomonas haemolytica* decreases from 24% to 2.7% (mean of the four strains) of total fatty acids, mainly due to a decrease in the content of C_{17:1 iso ω9c}. The relative amount of C_{17:0 iso} remains constant in cells growing over the whole temperature range. This may indicate that the adaptation to growth at high temperatures by *Thermomonas haemolytica* does not occur by increasing the ratio of C_{17:0 iso} to C_{17:1 iso ω9c}. Instead, the relative proportion of the major fatty acid, C_{15:0 iso}, increases

with increasing growth temperature. Interestingly, the relative amounts of short chain fatty acids, C_{11:0 iso} and C_{11:0 iso 3OH}, also increase slightly at 50°C (Busse et al., 2002).



A



B

FIGURE BXII.γ.38. Electron micrographs of ultrathin sections of cells of *T. haemolytica* A50-7-3^T showing (A) the filamentous and (B) rod-shaped forms. Bars = 200 nm. (Reprinted with permission from H.-J. Busse et al., International Journal of Systematic and Evolutionary Microbiology 52: 473–483, 2002, ©International Union of Microbiological Societies.)

T. haemolytica also grows well on R2A agar, a low-nutrient complex medium developed previously for the enumeration and subculture of heterotrophic bacteria in potable water (Reasoner and Geldreich, 1985). *T. haemolytica* does not grow on the chemically defined yet simpler Czapek–Dox medium, which contains only sodium nitrate as inorganic source of nitrogen and sucrose as sole source of carbon.

As tested on selective/differential media, *T. haemolytica* is not able to grow in the presence of bile salt–neutral red (MacConkey agar) or in the presence of the cationic detergent cetyl trimethylammonium bromide (cetrimide agar). *T. haemolytica* is susceptible to amoxycillin/clavulanic acid, ampicillin, cephalixin, chloramphenicol, colistin sulfate, erythromycin, fusidic acid, gentamicin, kanamycin, neomycin, penicillin G, polymyxin B, streptomycin, and tetracycline, but is resistant to lincomycin, nitrofurantoin, and sulfamethoxazole/trimethoprim.

T. haemolytica characteristically uses only a limited number of organic acids and amino acids as a single source of carbon, whereas *T. haemolytica* A7-3-1^T is the most versatile strain and B50-8-1 the most inactive strain of the four *T. haemolytica* strains (Table BXII.γ.31). Strain B50-8-1 is also somewhat distinct from the other three *T. haemolytica* strains, on the basis of mol% G + C content of DNA (67.1); however, by DNA–DNA similarity, ribo-print patterns, and AFLP profiles strain B50-8-1 is univocally a strain of *T. haemolytica* (Busse et al., 2002).

ENRICHMENT AND ISOLATION PROCEDURES

The four strains of *T. haemolytica* were isolated at different times from papermaking chemicals (kaolin and pigment slurries) that were plated onto Plate Count Agar (Difco) with subsequent incubation at 50°C (Väisänen et al., 1998).

TABLE BXII.γ.31. Physiological and biochemical characteristics of the *Thermomonas haemolytica* strains^a

Characteristic	A50-7-3 ^T	B50-7-1	D50-7-1	B50-8-1
<i>Assimilation of:</i>				
Acetate	+	–	+	–
Propionate	+	+	–	–
<i>cis</i> -Aconitate	+	+	–	–
L-Malate	+	–	–	–
Mesaconate	+	+	–	–
Oxoglutarate	+	–	–	–
Pyruvate	+	+	+	–
L-Alanine	+	+	–	–
L-Aspartate	+	+	–	–
L-Phenylalanine	+	–	–	–
L-Proline	–	+	–	–
L-Serine	+	–	–	–
<i>Hydrolysis of:</i>				
Gelatin	+	+	–	+

^aAll *Thermomonas haemolytica* strains assimilate DL-3-hydroxybutyrate, and L-leucine, whereas none of the strains assimilate N-acetyl-D-glucosamine, L-arabinose, *p*-arbutin, D-cellobiose, D-fructose, D-galactose, gluconate, D-glucose, D-mannose, D-maltose, α-D-melibiose, L-rhamnose, D-ribose, sucrose, salicin, D-trehalose, D-xylose, adonitol, *l*-inositol, maltitol, D-mannitol, D-sorbitol, putrescine, *trans*-aconitate, adipate, 4-aminobutyrate, azelate, citrate, fumarate, glutarate, itaconate, DL-lactate, suberate, β-alanine, L-histidine, L-ornithine, L-tryptophan, 3-hydroxybenzoate, 4-hydroxybenzoate, or phenylacetate. *p*NP-α-D-glucopyranoside, *p*NP-phenylphosphonate, 2-deoxythymidine-5'-*p*NP-phosphate, L-alanine-*p*NA, L-glutamate-γ-3-carboxy-*p*NA, and L-proline-*p*NA are hydrolyzed, but esculin, *p*NP-β-D-galactopyranoside, *p*NP-β-D-glucuronide, *p*NP-β-D-glucopyranoside, bis-*p*NP-phosphate, and *p*NP-phosphoryl-choline are not. Enzymatic tests for urease, alkaline phosphatase, esterase (C4), esterase lipase (C8), lipase (C14), leucine arylamidase, valine arylamidase, chymotrypsin, phosphatase acid, and naphthol-AS-BI-phosphohydrolase are uniformly positive.

MAINTENANCE PROCEDURES

Slant cultures grown on standard bacteriological media should not be held for longer than 1 week at 4°C. Cells may also stored at –70°C in broth media, e.g., nutrient or LB broth containing 15–20% glycerol as cryoprotectant or under liquid nitrogen.

DIFFERENTIATION OF THE GENUS *THERMOMONAS* FROM OTHER GENERA

Thermomonas can be clearly differentiated from other genera of the *Xanthomonadaceae* on the basis of 16S rRNA gene sequence, cellular fatty acid composition, and several physiological and bio-

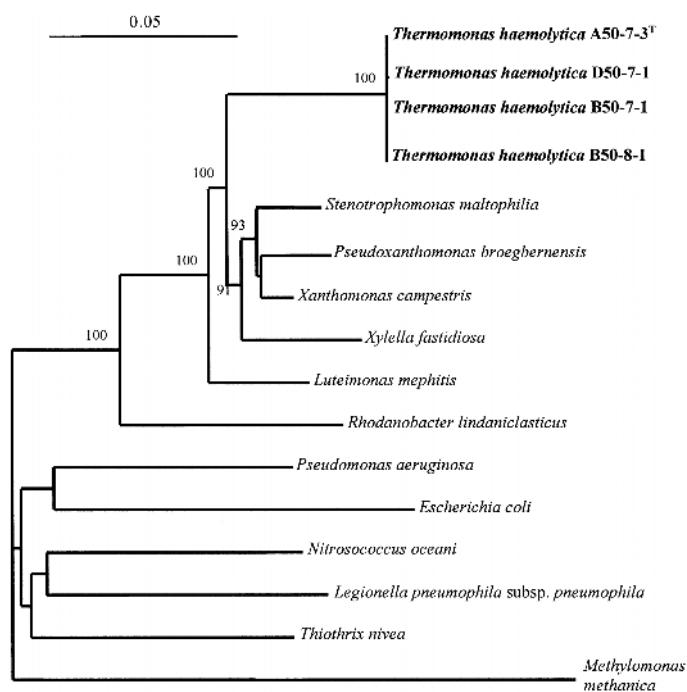


FIGURE BXII.γ.39. Phylogenetic position of *Thermomonas haemolytica* to related genera of the *Xanthomonadaceae*, and species representing distinct lineages within the *Gammaproteobacteria*. The dendrogram was generated from evolutionary distances, calculated from pair-wise dissimilarities, using the FITCH program contained in the PHYLIP package (Felsenstein, 1993). Bootstrap estimates of confidence, derived from 1000 resamplings, are shown for branchings with values greater than 80%, within the *Stenotrophomonas*–*Xanthomonas*–*Xylella* phylogenetic lineage. The topography of the branching order of the lineage within the dendrogram was supported by identical branching orders generated using an alternative treeing algorithm of maximum-likelihood (FastDNAm1, version 1.0) (data not shown) and by estimated high bootstrap proportions of confidence for the respective branchings. GenBank/ EMBL/ DDBJ accession numbers: *Escherichia coli*, M24836; *Legionella pneumophila* subsp. *pneumophila*, M59157; *Luteimonas mephitis*, AJ012228; *Methylobacterium methanica*, M95660; *Nitrosococcus oceanii*, M96395; *Pseudomonas aeruginosa*, Z76651; *Pseudoxanthomonas broegbernensis*, AJ012231; *Rhodanobacter lindaniclasticus*, AF039167; *Stenotrophomonas maltophilia*, X95923; *Thermomonas haemolytica*, A50-3-7^T (AJ300185); *Thermomonas haemolytica*, B50-7-1, AJ300186 (partial sequence, see Busse et al., 2002); *Thermomonas haemolytica* B50-8-1, AJ300188 (partial sequence, see Busse et al., 2002); *Thermomonas haemolytica* D50-7-1, AJ300189 (partial sequence, see Busse et al., 2002); *Thiothrix nivea*, L40993; *Xanthomonas campestris*, X95917; *Xylella fastidiosa*, AF192343. Bar = 5 substitutions at any nucleotide position per 100 nucleotide positions (estimated evolutionary distance from the point of divergence of the 16S rRNA gene sequences).

chemical characteristics (Table BXII.γ.32). In particular, *Thermomonas* species exhibit a higher temperature optimum of 37–50°C.

TAXONOMIC COMMENTS

Comparative evolutionary distance analysis of the 16S rDNA sequence shows that the genus *Thermomonas* represents a separate line of descent within the *Gammaproteobacteria*. The nearest phylogenetic relatives, albeit relatively distant (16S rDNA sequence similarities <95%), are species of the genera *Xanthomonas*, *Pseu-*

doxanthomonas, *Stenotrophomonas*, *Luteimonas*, and *Xylella* (Fig. BXII.γ.39).

Very recently, an isolate SGM-6 was obtained from a hot spring at Sao Gemil in central Portugal, showing a 16S rDNA gene sequence similarity of 97.4% to *T. haemolytica* A50-7-3^T and able to grow at temperatures as high as 55°C. This isolate may represent another species of the genus *Thermomonas*, or at least a novel strain of *T. haemolytica* (M. Da Costa and F. Rainey, personal communication). Meanwhile, a second *Thermomonas* species has been described Alves et al., 2003a.

TABLE BXII.γ.32. Characteristics differentiating *Thermomonas haemolytica* from other related taxa within the *Xanthomonadaceae*^a

Characteristic	<i>Thermomonas haemolytica</i> ^b	<i>Xanthomonas campestris</i> ^c	<i>Pseudoxanthomonas broegbernensis</i> ^d	<i>Stenotrophomonas maltophilia</i> ^e	<i>Luteimonas mephitis</i> ^f	<i>Xylella fastidiosa</i> ^g
Temperature optimum (°C)	37–50	28	28	35	28	26–28
Nitrate reduction	–	–	–	+	–	–
Esculin hydrolysis	–	+	–	+	–	–
<i>Susceptibility to:</i>						
Ampicillin	+	–	–	–	–	+
Penicillin G	+	–	–	–	–	–
Erythromycin	+	–	–	–	–	–
Kanamycin	+	+	–	+	V+	+
Neomycin	+	–	+	–	V–	–
Streptomycin	+	+	–	–	–	–
<i>Predominant fatty acid:</i>						
C _{15:0} iso	+	+	+	+	+	–
C _{15:0} anteiso	–	+	+	+	–	+
C _{16:0}	–	–	+	–	–	–
C _{16:0} iso	+	–	–	–	–	–
C _{16:1}	–	+	+	–	–	+
C _{17:0}	–	–	–	–	+	+
C _{17:0} iso	–	–	–	–	–	–
C _{17:1}	–	–	–	–	+	–
C _{17:1} iso	–	–	+	–	–	–
<i>Hydroxy fatty acid:</i>						
C _{10:0} 2OH	–	–	–	–	–	+
C _{11:0} iso 2OH	–	–	+	–	–	–
C _{11:0} iso 3OH	+	+	+	+	+	–
C _{12:0} 3OH	–	+	–	+	–	–
C _{13:0} 2OH	–	–	–	+	–	–
C _{13:0} iso 3OH	–	+	–	+	–	–
<i>Major polar lipids:</i> ^h						
Diphosphatidylglycerol	+	+	–	+	–	–
Phosphatidylethanolamine	+	+	–	+	–	–
Phosphatidylglycerol	+	+	–	+	–	–
Phosphatidylmonomethylethanolamine	–	+	–	–	–	–
Unidentified phospholipid	–	+	–	–	–	–
Quinone system	Q-8	Q-8	Q-8	Q-8	Q-8	–
<i>Major polyamines:</i> ⁱ						
Cadaverine	–	–	–	+	–	–
Spermidine	+	+	–	+	–	–

^aSymbols: +, positive for all strains; –, negative for all strains; nd, no data; V+, most strains are resistant; V–, most strains are susceptible.

^bData from Busse et al. (2002).

^cData from Oyaizu and Komagata (1983), Busse and Auling (1988), Auling et al. (1991), Yang et al. (1993b, d).

^dData from Finkmann et al. (2000).

^eData from Oyaizu and Komagata (1983), Palleroni (1984), Busse and Auling (1988), Stead (1992), Yang et al. (1993b, d), Palleroni and Bradbury (1993), Vauterin et al. (1995, 1996b), Finkmann et al. (2000).

^fData from Finkmann et al. (2000).

^gData from Wells et al. (1987).

^hData on polar lipids for *Xanthomonas campestris* and *Stenotrophomonas maltophilia* from Busse et al. (2002).

ⁱNo data for *Pseudoxanthomonas broegbernensis*, *Luteimonas mephitis*, and *Xylella fastidiosa*.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *THERMOMONAS*

Only one species, *Thermomonas haemolytica*, is currently known.

List of species of the genus Thermomonas

1. ***Thermomonas haemolytica*** Busse, Kämpfer, Moore, Nuutinen, Tsitko, Denner, Vauterin, Valens, Rosselló-Mora and Salkinoja-Salonen 2002, 480^{VP}
hae.mo.ly' ti.ca. Gr. n. *haema* blood; Gr. adj. *lyticus* dissolving; M.L. fem. adj. *haemolytica* blood dissolving.

Morphological, physiological, biochemical, and chemotaxonomic characteristics are the same as those described for the genus. Other characteristics are listed in Tables BXII.γ.31 and BXII.γ.32. Isolated from kaolin slurry used in the papermaking industry.

The mol% G + C of the DNA is: 67.1–68.7 (HPLC); type strain, 68.5.

Type strain: A50-7-3, DSM 13605, LMG 19653.

GenBank accession number (16S rRNA): AJ300185.

Additional Remarks: Reference strains include B50-71 (DSM 13598, LMG 19655; GenBank accession number AJ300186), B50-8-1 (DSM 13599, LMG 19654; GenBank accession number AJ296318), D50-7-1 (DSM 13610, LMG 19656; GenBank accession number AJ296319).

Genus XI. Xylella Wells, Raju, Hung, Weisburg, Mandelco-Paul and Brenner 1987, 141^{VP}

GERARD S. SADDLER AND JOHN F. BRADBURY

Xy.lel' la. Gr. n. *xylon* wood; M.L. dim. ending *-ella*; M.L. fem. n. *Xylella* a small wood.

Single, straight rods, 0.25–0.35 × 0.9–3.5 μm, that can form long filamentous strands under some cultural conditions. Gram negative. **Nonmotile**, lacking flagella. Endospores not produced. **Aerobic**, having a strictly respiratory type of metabolism with oxygen as the terminal electron acceptor. **Colonies are cream to white**. Two colonial variants are frequently encountered: convex to pulvinate, smooth, opalescent with entire margins; and umbonate, rough with finely undulated margins. **Oxidase negative and catalase positive**. Optimum temperature for growth, 26–28°C; optimum pH for growth, 6.5–6.9. Nutritionally fastidious. Found mainly in the xylem of plant tissue.

The mol% G + C of the DNA is: 51.0–52.4 (*T_m*); 52–53.1 (Bd).

Type species: ***Xylella fastidiosa*** Wells, Raju, Hung, Weisburg, Mandelco-Paul and Brenner 1987, 141.

FURTHER DESCRIPTIVE INFORMATION

Cell morphology Long filamentous strands can be observed in many strains, but filaments are encountered more frequently when sampling from smooth colonies; rough colonial variants tend to exhibit only single rods (Wells et al., 1987). Flagella are not present and motility has not been observed. Cells exhibit rounded ends, though tapering can be observed at one end only. Irregular ridges or folds in the cell surface can be seen. Fibril-like structures are evident and, although peritrichous, predominate at the poles. There is some evidence of morphological heterogeneity, because strains of *X. fastidiosa* causing phony peach disease are thought to be slightly shorter (0.35 × 2.3 μm) and exhibit more pronounced ridges, or rippling of the cell wall

when compared against strains causing Pierce's disease (Nyland et al., 1973). Staining with methylene blue is effective for the visualization of citrus variegated chlorosis strains *in planta* (De Lima et al., 1997).

Cell wall composition The major fatty acid components present are: pentadecanoic acid (C_{15:0}), hexadecanoic acid (C_{16:0}), hexadecenoic acid (C_{16:1}) and heptadecanoic acid (C_{17:0}) (Wells and Raju, 1984; Wells et al., 1987). The hydroxy fatty acids 2-hydroxydecanoic (C_{10:0} 2OH) and 3-hydroxydodecanoic (C_{12:0} 3OH) acids are reported to be present (Wells et al., 1987).

Fine structure The cell wall shows periodic, spiral, or annulated structural arrangement with the thickness varying from 25 to 35 nm in the furrows to 45–55 nm around the ridges (Nyland et al., 1973; Chagas, et al., 1992). The wall consists of an outer, an inner (each comprised of 3-layered unit membrane structure), and a middle peptidoglycan layer. Progressive changes in the cell wall occur with age; initially the inner and outer layer are thin, becoming denser with time, the outer layer increasing in density first (Mollenhauer and Hopkins, 1974). As the cells age, the cell wall structure can be more difficult to visualize and this is particularly true when examining senescent or dead cells. Cell content consists of DNA threads, ribosomes, and osmophilic granules of unknown function (Chagas et al., 1992). The distribution of cell constituents alters with age, with ribosomes aggregating along the periphery of aging cells. Fibril-like structures can also be observed.

Colonial characteristics Two colony types can be observed: smooth and rough, both are circular and opalescent. When grown on BCYE¹ medium, colonies are small, 0.6 mm in diameter after 10 days at 27°C, expanding to 1.5 mm after 30 days. On this medium, rough colonies exhibit green or red margins when viewed under reflected light. Cell replication is by binary fission (Wells et al., 1987; Chagas et al., 1992).

Nutrition, metabolism, and metabolic pathways Strains are aerobic, chemoorganotrophic, and use oxygen as the terminal electron acceptor. Many strains are nutritionally fastidious and require specialized media such as BCYE and PW² for growth (Wells et al., 1987). Little information on enzymic activity exists; however, protease activity has been determined by growing both virulent and weakly virulent strains of *X. fastidiosa* on PD3³ medium (Davis et al., 1981c) amended with gelatin (Fry et al., 1994). Detailed study revealed the presence of at least two proteases, designated P1 and P2, of 54 and 50 kDa, respectively. Production reached a maximum with onset of the stationary phase, P2 becoming more, and P1 becoming less, predominant over time.

Genetics, mutants and plasmids The genetics of this genus have attracted limited interest thus far. The production of a cosmid clone bank (Goodwin, 1989), and evidence for the lack of homology between the *hrp* gene cluster of *Xanthomonas vesicatoria* and DNA from *Xylella fastidiosa*, (Leite et al., 1994b) have contributed to what limited information exists. The reader is directed towards the Bioinformatics Laboratory at the University of Campinas* for details of a project to sequence the entire genome of *X. fastidiosa* (Setubal, personal communication).

Plasmids have been observed in some strains of *X. fastidiosa* (Chen et al., 1992a). A restriction map has been generated for plasmid pXFPW1, which is 1.3 kb in size and was isolated from an *X. fastidiosa* strain causing periwinkle wilt disease. A copy number of 60 was estimated, and restriction analysis demonstrated a high degree of homology with two other plasmids from strains of Pierce's disease and mulberry leaf scorch disease. In contrast, a cryptic plasmid from *X. fastidiosa* ATCC 35868 was found in only one other strain of *X. fastidiosa*—ATCC strain 35878—and further searches revealed some nucleotide sequence homology with plasmid pNKH43 from *Stenotrophomonas maltophilia* (Pooler et al., 1997a).

Molecular data Restriction fragment length polymorphism (RFLP) analysis has been used to study diversity within *X. fastidiosa* by examining 24 strains from eight different hosts (Chen et al., 1992b). Two genomic libraries were constructed in pUC18 from a PD strain and a plum leaf scald strain. By studying 24

probe-enzyme combinations, 67 characters could be determined and PD strains, in particular, exhibited a high degree of uniformity. Subsequent studies produced similar results, this time using RAPD fingerprinting with 14 random primers on 17 different strains of *X. fastidiosa* (Chen et al., 1995); again, a high degree of similarity among the PD strains was evident, although clear differences between this group and *Quercus* leaf scorch, plum leaf scald strains, and a periwinkle wilt strain indicates a high degree of heterogeneity within the species overall. Analogous results were obtained by Albibi et al. (1998), using two random primers; OPA-03 and OPA-11 (Operon Technologies Inc.). With this method, PD strains could be distinguished from other strains of *X. fastidiosa* (citrus variegated chlorosis, mulberry leaf scorch, periwinkle wilt, plum leaf scald, and phony peach), as well as strains of *Xanthomonas campestris* pathovar *vesicatoria* and *Escherichia coli*. The use of RAPD markers has been further refined to develop a detection method, as demonstrated by Pooler and Hartung (1995b). Primers were developed for the specific detection of strains of citrus variegated chlorosis (CVC). A CVC-specific region of the *X. fastidiosa* genome that contains a 28-nucleotide insertion was identified, and single base changes that distinguish CVC from other *X. fastidiosa* strains. This method was used to develop a sensitive and specific assay for detecting *X. fastidiosa* in potential insect vectors, by the prior selection of the bacterium using immunomagnetic separation (Pooler et al., 1997b).

Intraspecific heterogeneity within *X. fastidiosa* has also been observed using other markers, such as PCR amplification with tRNA consensus primers (Beretta et al., 1997). With this method, it is possible to distinguish between CVC strains and mulberry leaf scorch strains. Further, a 7.4-kb *Eco*R1 fragment of genomic DNA of the type strain of *X. fastidiosa*, ATCC 35879, has been used as a probe and found to be conserved in 18 other strains (Minsavage et al., 1994). PCR primers developed from a 1.0-kb internal *Eco*RV portion of the fragment allowed the differentiation of the species into two groups or "pathovars". A PCR assay has been developed for the specific detection of *X. fastidiosa* strains from a wide variety of hosts, through the amplification of a 400-bp product for the variable regions V1 and V6 of the 16S rDNA sequence (Firrao and Bazzi, 1994).

Antigenic structure Polyclonal antibodies have been used in conjunction with ELISA and immunofluorescence for the detection of *X. fastidiosa* strains (Sherald and Lei, 1991; Zaccardelli et al., 1993; Hartman et al., 1995). However, not all strains in the species can be detected using this method. In a further refinement, polyclonals were successfully obtained by screening for unique protein bands of PD strains of *X. fastidiosa* (Bazzi et al., 1994b). The antibodies directed against the two best-suited protein bands (20.7 kDa and 19.8 kDa) did not show any cross-reaction in a Western blot, a solid phase ELISA, or using indirect fluorescent antibody staining, when tested against other bacteria commonly found on grapevine. Using these "monospecific" antibodies in conjunction with indirect ELISA, the detection limit was found to be 8.3×10^5 cells/ml. Similar levels of detection (3×10^5 cells/ml) using a series of 10 monoclonal antibodies specific to pear leaf scorch (PLS) strains of *X. fastidiosa*, also by indirect ELISA (Leu et al., 1998), have been reported. The monoclonals proved highly specific: 9 of the 10 did not cross-react with 14 other bacterial strains belonging to 9 genera. In Western blot analysis, all the monoclonal antibodies recognized a 46.9-kDa polypeptide in all 12 *X. fastidiosa* strains, and a specific 21.5 kDa polypeptide in PLS strains.

1. BCYE medium (Wells et al., 1981) (g/l distilled water): yeast extract, 10.0; activated charcoal, 2.0; L-cysteine HCl·H₂O, 0.4 (filter sterilized [0.2 µm pore size] and added after autoclaving); ferric pyrophosphate (soluble), 0.25 (filter sterilized [0.2 µm pore size] and added after autoclaving); ACES buffer, 10.0; agar, 17.0.

2. PW medium (Davis et al., 1981b) (g/l distilled water): Phytone peptone (BBL), 4.0; trypticase peptone (BBL), 1.0; hemin chloride, 0.01; MgSO₄·7H₂O, 0.4; K₂HPO₄, 1.2; KH₂PO₄, 1.0; phenol red, 0.02; L-glutamine, 4.0 (filter sterilized [0.2 µm pore size] and added after autoclaving); bovine serum albumin fraction V, 6.0 (filter sterilized [0.2 µm pore size] and added after autoclaving); agar, 12.0.

3. PD3 medium (Davis et al., 1981c) (g/l distilled water) Tryptone (Difco), 4.0; Soytone (Difco), 2.0; trisodium citrate, 1.0; disodium succinate, 1.0; hemin chloride, 0.01 (10 ml of 0.1% w/v solution in 0.05N NaOH); potato starch (soluble) 2.0; MgSO₄·7H₂O, 1.0; K₂HPO₄, 1.5; KH₂PO₄, 1.0; Bacto agar (Difco), 15.0.

*Editorial Note: The website for the Bioinformatics Laboratory at the University of Campinas is <http://www.lbi.dcc.unicamp.br/>.

Antibiotic and drug sensitivity Data from *in vitro* studies have shown that different disease strains of *X. fastidiosa* exhibit a differential response to carbenicillin, penicillin G, and gentamycin when these antibiotics are added to CS-20⁴ medium (Hopkins, 1988b). Data from *in vivo* studies of leaf scorch of pear trees (*Pyrus pyrifolia*) in the low-altitude areas of central Taiwan (Leu and Su, 1993) have shown that symptom remission can be achieved by injection of oxytetracycline.

Pathogenicity and ecology Symptoms produced in host plants vary, the most characteristic being a necrosis of interveinal leaf tissue, producing a characteristic scorching effect in a wide range of woody plants. Of primary importance is Pierce's disease of grapevine, which can cause 100% destruction in some vineyards, but which is normally not a serious pathogen (Goodwin and Meredith, 1988). The bacteria block xylem vessels and cause water stress, leading to leaf scorch symptoms, but no wilting is observed as leaf stomata close. Although transpiration is reduced, diseased leaves remain water stressed and various functions are impaired, resulting in carbohydrate imbalance, loss of chlorophyll and disruption of cell membranes. Physiological changes associated with water stress and senescence have been compared between leaves of healthy grapevines (*Vitis vinifera* cv. Chardonnay) and grapevines infected with the PD strains (Goodwin et al., 1988a, b). Vegetative growth of diseased vines was less than that of healthy vines under natural vineyard conditions. Although stomatal resistance of diseased leaves with marginal necrosis was similar to that of healthy leaves during early morning, the stomatal resistance of diseased leaves increased greatly later in the day, while that of healthy leaves remained relatively constant. The diurnal patterns of transpiration and photosynthesis demonstrated that these processes were inhibited by PD strains. The reduced photosynthesis of diseased leaves was at least partly due to inhibition of nonstomatal-related aspects of CO₂ fixation. Compared with healthy leaves, diseased leaves had higher concentrations of abscisic acid, glucose, fructose, Ca²⁺ and Mg²⁺, but a lower concentration of K⁺. Sucrose accumulated in green tissue of a diseased leaf, but the concentrations of sucrose and starch were lower in diseased chlorotic tissue than in diseased green or healthy leaf tissue. Chlorophyll reduction in diseased leaves was also associated with increases in electrolyte leakage, lipid peroxidation, and superoxide anion. Based on these physiological changes, it was proposed that the symptoms of PD were the result of relatively mild but prolonged water stress accelerating leaf senescence. Subsequent studies have demonstrated that PD strains were responsible for the production of occlusions in xylem vessels of susceptible hosts (Fry and Milholland, 1990).

Pierce's disease of grapevine can be transmitted by xylem-feeding leafhoppers (Hill and Purcell, 1997). Acquisition of PD strains has been demonstrated in the blue-green sharpshooter (*Graphocephala atropunctata*) after four days' incubation with infected grapes. Subsequent transmission to healthy plants can be achieved after 10 days. Previous studies have demonstrated that harvested fruit clusters from diseased grapes did not serve as inoculum sources for *G. atropunctata* (Purcell and Saunders,

1995). Only when *G. atropunctata* was given access to diseased grapevines did it subsequently transmit PD strains to healthy grapevine. Transmission efficiency by *G. atropunctata* remained high at seven days' post-acquisition and beyond (Hill and Purcell, 1995a). Bacterial population size in the heads of the inoculative vectors also increased up to day seven, but did not consistently increase further. Studies on grapevine (*Vitis vinifera*) alongside other *G. atropunctata* host plants (*Rubus discolor*, *Artemisia douglasiana*, *Echinochloa crusgalli*, and *Cynodon dactylon*) showed that distal movement from the inoculation site could only be detected in grapevine and *Rubus discolor* (Hill and Purcell, 1995b). These findings highlight the epidemiological importance of the diseased host as a reservoir for further dissemination and, in addition, agree with earlier work, which showed that strains isolated from natural hosts such as *Rubus* spp. could induce symptoms in grapevine (Hopkins, 1988a).

Similar results were obtained by studying CVC strains of *X. fastidiosa* in citrus nurseries in Brazil (Garcia et al., 1997) where the increases in vector populations of *Dilobopterus costalimai*, *Aerogonia terminalis*, and *Oncometopia facialis* and disease incidence were found to be linked. CVC strains, as well as PD strains, appear to be systemic, because xylem vessels of diseased plant petioles harbor large numbers of cells (Hartung et al., 1994).

The significance of disease reservoirs, such as weeds, has also been studied. Weeds collected from plum orchards infested with leaf scald were examined, and the presence of the pathogen was confirmed by isolation and by DAS-ELISA in *Brachiaria plantaginea*, *Digitaria* sp., *Facelis retusa*, *Hypochoeris brasiliensis*, *Leonurus sibiricus*, *Lolium multiflorum*, *Paspalum urvillei*, *Pennisetum clandestinum*, *Richardia* sp., *Taraxacum officinale*, and *Vernonia* sp. (Leite et al., 1997). Phony peach disease strains can be detected in symptomatic or asymptomatic hosts (Aldrich et al., 1992). Populations of *X. fastidiosa* in the root xylem fluid of symptomatic peach trees were high and the distribution of these bacteria was uniform both along the length of roots and among roots within the root ball; in contrast, distribution of the bacteria within asymptomatic trees was discontinuous.

ENRICHMENT AND ISOLATION PROCEDURES

It has long been considered that *X. fastidiosa* is fastidious and unable to grow on standard bacteriological media, despite reports that Nutrient Agar can support growth (Chang et al., 1990; Fry et al., 1990). More commonly, a range of media have been used successfully to culture *X. fastidiosa*, namely CS20, PD1,⁵ PD2,⁶ PD3, and PW.

MAINTENANCE PROCEDURES

Lyophilization and storage on silica gel at -20°C are effective long-term storage methods (Sleesman and Leben, 1978; Hopkins, 1988b). Long-term culturing of *X. fastidiosa* frequently re-

4. CS-20 medium (Chang and Schaad in Hopkins, 1988b) (g/l distilled water): Tryptone, 2.0; Soytone, 2.0; (NH₄)₂HPO₄, 0.85; hemin chloride, 15 ml (0.1% w/v solution in 0.05N NaOH); KH₂PO₄, 1.0; MgSO₄·7H₂O, 0.4; L-glutamine, 6.0; dextrose, 1.0; potato starch (soluble), 2.0; L-histidine, 1.0; phenol red, 5.0 ml (0.2% solution prepared by dissolving 0.2 g in 30 drops 5N NaOH then making the volume to 100 ml in H₂O); agar, 12.0.

5. PD1 medium (Davis et al., 1980) (g/l distilled water): Tryptone (Difco), 7.0; Soytone (Difco), 3.0; disodium succinate, 1.0; hemin chloride, 0.01 (10 ml of 0.1% w/v solution in 0.05N NaOH); MgSO₄·7H₂O, 1.0; FeCl₃·6H₂O, 0.002; MnSO₄·4H₂O, 0.002; (NH₄)₂SO₄, 1.0; Bacto agar (Difco), 15.0; bovine serum albumin fraction V, 2.0 (10 ml of 20% v/v, added after autoclaving to medium held at 45–50°C).

6. PD2 medium (Davis et al., 1980) (g/l distilled water): Tryptone (Difco), 4.0; Soytone (Difco), 2.0; trisodium citrate, 1.0; disodium succinate, 1.0; hemin chloride, 0.01 (10 ml of 0.1% w/v solution in 0.05N NaOH); MgSO₄·7H₂O, 1.0; K₂HPO₄, 1.5; KH₂PO₄, 1.0; Bacto agar (Difco), 15.0; bovine serum albumin fraction V, 2.0 (10 ml of 20% v/v added after autoclaving to medium held at 45–50°C).

sults in the loss of virulence, e.g., after 18 months of weekly culturing on PD2 medium (Hopkins, 1984).

DIFFERENTIATION OF THE GENUS *XYLELLA* FROM OTHER GENERA

Characteristics useful for differentiating *Xylella* from other members of the family *Xanthomonadaceae* are listed in Table BXII.γ.17 in Genus *Xanthomonas*. In addition, a technique for identifying *X. fastidiosa* involving direct isolation of endophytic bacteria from propagation material, combined with one-dimensional SDS-PAGE of the total soluble proteins of the bacterial cell envelope, is described by Bazzi et al. (1994a). The gel electrophoretic pattern of this pathogen is characterized by a constant distribution of 14 major diagnostic bands.

TAXONOMIC COMMENTS

Analysis of 16S rRNA gene sequence data placed this genus in the *Gammaproteobacteria*, related to *Xanthomonas campestris* and *Stenotrophomonas maltophilia* (Wells et al., 1987). Similar findings have been presented by Kersters et al. (1996). In a subsequent study, an average sequence similarity of 94.7% was reported between *X. fastidiosa* and seven type strains of the genus *Xanthomonas*; the sequence similarity with *S. maltophilia* was 94.1% (Moore et al., 1997). Results obtained from restriction maps of PCR-amplified 16S rRNA genes of *Xylella*, *Xanthomonas*, and *Stenotrophomonas* strains highlighted relationships between the three

genera (Nesme et al., 1995). Nineteen restriction fragment length polymorphisms (RFLPs) were distinguished and strains were recovered in two main clusters, one of which contained *Xanthomonas* spp., the other *Stenotrophomonas* spp. The genera *Xanthomonas* and *Stenotrophomonas* formed a coherent group, with the genus *Xylella* recovered as a single member cluster and clearly more distantly related.

X. fastidiosa is a heterogeneous species producing a variety of symptoms on many hosts of economic significance: Pierce's disease of grapevine, citrus variegated chlorosis, phony peach, periwinkle wilt; alfalfa dwarf; ragweed stunt; and almond, coffee, elm, maple, mulberry, oak, plum, and sycamore leaf scorch. Differences between pathogenic populations can be observed using traditional methods (Table BXII.γ.33), RAPDs, (Chen et al., 1995; Pooler and Hartung, 1995a, b; Albibi et al., 1998), RFLP (Chen et al., 1992b), and monoclonal antibodies (Leu et al., 1998). Evidence from these data suggests that a pathovar structure may be appropriate to circumscribe infraspecific variation within *X. fastidiosa*, but clearly further work is required.

FURTHER READING

Hopkins, D.L. 1989. *Xylella fastidiosa*: xylem-limited bacterial pathogens of plants. Ann. Rev. Phytopathol. 27: 271–290.
Purcell, A.H. 1997. *Xylella fastidiosa*, a regional problem or global threat? J. Plant Pathol. 79: 99–105.
Purcell, A.H. and D.L. Hopkins. 1996. Fastidious xylem-limited bacterial plant pathogens. Ann. Rev. Phytopathol. 34: 131–151.

TABLE BXII.γ.33. Differentiation of some pathogenic strains of *Xylella fastidiosa*^{a, b}

Characteristics	Disease-associated strains					
	Elm leaf scorch	Periwinkle wilt	Pierce's disease	Phony peach	Plum leaf scald	Ragweed stunt
<i>Growth on:</i>						
BCYE, PW, CS-20	+	+	+	+	+	+
PD3	+	–	+	–	–	–
<i>Antibiotic resistance^c:</i>						
Carbenicillin, 12.5 µg/ml	+	+	–	+	+	nd
Penicillin, 6.25 µg/ml	+	+	+	–	–	nd
Gentamicin, 80 µg/ml	–	+	–	–	–	nd

^aFor symbols see standard definitions; nd, not determined.
^bAdapted from Chang and Schaad, unpublished in Hopkins, 1988b.
^cCS-20 used as basal medium.

List of species of the genus *Xylella*

1. *Xylella fastidiosa* Wells, Raju, Hung, Weisburg, Mandelco-Paul and Brenner 1987, 141^{VP}
fas.tid.i.o' sa. M.L. fem. adj. *fastidiosa* highly critical; referring to the nutritional fastidiousness of the organism, particularly on primary isolation.

The characteristics are those described for the genus. Gelatin is hydrolyzed and gelatinase is produced. Many strains produce β-lactamase. β-galactosidase, coagulase, lip-

ase, amylase, and phosphatase negative; indole and H₂S are not produced.
Differential characteristics of some pathogenic strains of *Xylella fastidiosa* are listed in Table BXII.γ.33.
The mol% G + C of the DNA is: 51.0–52.4 (*T_m*) and 52–53.1 (Bd).
Type strain: PCE-RR, ATCC 35879, DSM 10026, ICMP 11140.
GenBank accession number (16S rRNA): AF192343.

Order IV. **Cardiobacteriales** ord. nov.

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

Car.di.o.bac.te.ri.a'les. M.L. neut. n. *Cardiobacterium* type genus of the order; -ales ending to denote order; M.L. fem. n. *Cardiobacteriales* the *Cardiobacterium* order.

The order *Cardiobacteriales* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rDNA sequences; the order contains the family *Cardiobacteriaceae*.

Description is the same as for the family *Cardiobacteriaceae*.

Type genus: **Cardiobacterium** Slotnick and Dougherty 1964, 271^{AL}.

Family I. **Cardiobacteriaceae** Dewhirst, Paster, La Fontaine and Rood 1990, 431^{VP}

FLOYD E. DEWHIRST AND BRUCE J. PASTER

Car.di.o.bac.te.ri.a'ce.ae. M.L. neut. n. *Cardiobacterium* type genus of the family; -aceae ending to denote a family; M.L. fem. pl. n. *Cardiobacteriaceae* the family of *Cardiobacterium*.

Straight rods, 0.5–1.7 × 1–6 µm, with rounded ends. Cells occasionally occur in pair, chains, or rosettes. Gram negative but with a tendency to resist Gram decolorization. Nonmotile, as cells do not possess flagella, but may possess type IV pili and exhibit twitching motility. Aerobic and oxidase positive except for members of the anaerobic genus *Dichelobacter*. Catalase negative. Urease negative. Fermentative type of chemoorganotrophic metabolism except for members of the genus *Dichelobacter*, which are arginine chemoorganotrophs. Acid but not gas is produced from glucose, sucrose, maltose, mannose, and fructose in the fermentative species. Nitrates are not reduced. H₂S is produced (as determined by the lead acetate method). No growth occurs on MacConkey agar. Species in the family possess the following rare or unique signatures in their 16S rRNAs: uracil at positions 18, 1051, 1400; cytosine at positions 1124 and 1253; and adenine at position 917 (*E. coli* numbering). The family is composed of three genera—the type genus *Cardiobacterium* and the genera *Suttonella* and *Dichelobacter*.

The mol% G + C of the DNA is: 45–60.

Type genus: **Cardiobacterium** Slotnick and Dougherty 1964, 271^{AL}.

Differentiation from other taxa Based on 16S rRNA sequence analysis, it was recognized that *Kingella indologenes* and *Bacteroides nodosus* were taxonomically misplaced and that they

were most closely related to *Cardiobacterium hominis* (Dewhirst et al., 1990). A neighbor-joining phylogenetic tree based on 16S rRNA sequence comparisons is shown in Fig. BXII.γ.40. The three organisms form a monophyletic group in the *Gammaproteobacteria*. *Bacteroides nodosus* clearly differs from *C. hominis* and *K. indologenes* in that it is an asaccharolytic anaerobe. It was therefore placed in its own genus as *Dichelobacter nodosus*. *Cardiobacterium hominis* and *K. indologenes* are quite similar phenotypically, but the mol% G + C content of their DNA differs by 11%, and *K. indologenes* was placed in its own genus as *Suttonella indologenes*. Strains for two additional *Cardiobacterium* species have been isolated, but not formally named (see Fig. BXII.γ.40). These three genera were placed in the family *Cardiobacteriaceae* based upon their forming a closely related (92% similarity) monophyletic group. The *Cardiobacteriaceae* cluster is supported in 100% of 200 bootstrap resamplings. The family branches deeply in the *Gammaproteobacteria*, being less than 87% similar to any other taxa. The most closely related organisms are *Thiobacillus neapolitanus*, *Halothiobacillus hydrothermalis* (shown on tree) and *T. halophilus*. However, these species clearly belong in a separate family because they have a chemolithotrophic metabolism, possess none of the 16S rRNA base signatures of the *Cardiobacteriaceae*, and have less than 87% 16S rRNA sequence similarity. Table BXII.γ.34 shows phenotypic traits that differentiate *Cardiobacteriaceae* genera from other genera.

Genus I. **Cardiobacterium** Slotnick and Dougherty 1964, 271^{AL}

BRUCE J. PASTER AND FLOYD E. DEWHIRST

Car.di.o.bac.te'ri.um. Gr. n. *cardia* heart; Gr. n. *bakterion* small rod; M.L. neut. n. *Cardiobacterium* bacterium of the heart.

Straight rods 0.5–0.75 × 1.0–3.0 µm, with rounded ends. Occasional long filaments, 7.0–35.0 µm may occur. **Pleomorphic**. Cells are arranged singly, in pairs, in short chains, and in rosette clusters. Gram negative, but retention of crystal violet may occur in the swollen ends or central portions of cells. Nonmotile. **Facultatively anaerobic**. CO₂ is required by some strains on isolation. Aerobic growth is scant unless humidity is elevated. Growth in candle jars or under anaerobic conditions is not dependent on an elevated humidity. Optimum temperature, 30–37°C. Colonies on blood agar are smooth, convex, and opaque. **Chemoorgano-**

trophic, having a strictly fermentative type of metabolism. Acid but not gas is produced from fructose, glucose, mannose, sorbitol, mannitol, and sucrose. Lactic acid is the major product of glucose fermentation; smaller amounts of pyruvate, formate, and propionate are formed. Oxidase positive. **Catalase negative**. Small amounts of **indole** are formed. **Nitrates are not reduced**. No growth occurs on MacConkey agar. Urease negative. Ornithine decarboxylase (ODC) negative. Occur in nasal flora of humans; isolated from the blood of humans with bacterial endocarditis.

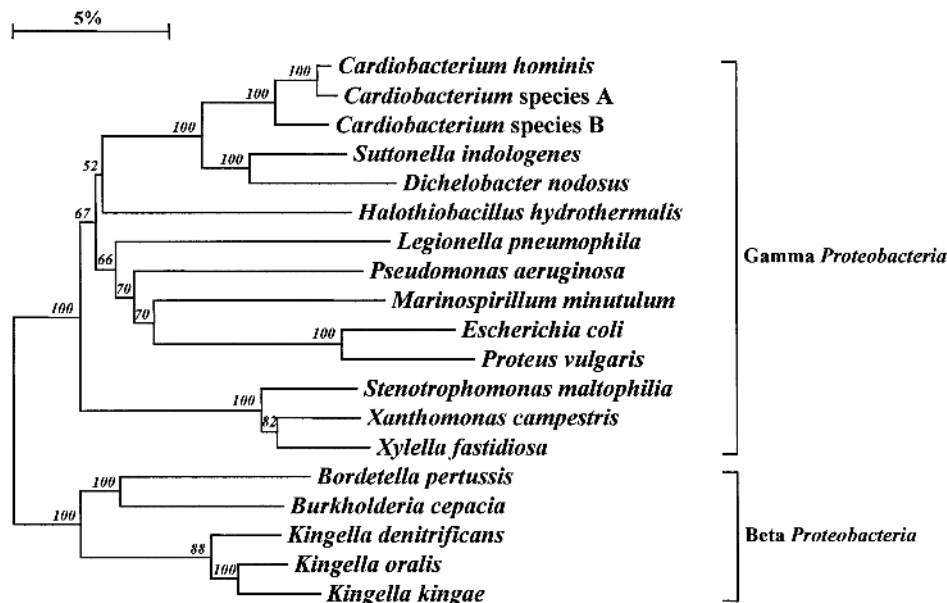


FIGURE BXII.γ.40. Phylogeny of *Cardiobacteriaceae*. The phylogenetic relationships of the three genera, namely *Cardiobacterium*, *Suttonella*, and *Dichelobacter*, of the family *Cardiobacteriaceae*, compared to other members of the *Betaproteobacteria* and *Gamma**proteobacteria*. Bar = 5% difference in nucleotide sequences. The neighbor-joining method was used for tree construction. Two hundred bootstrap trees were generated, and bootstrap confidence levels (shown as percentages above the nodes) were determined. GenBank accession numbers for sequences of members of the *Cardiobacteriaceae*: *Cardiobacterium hominis*, M35014; *Cardiobacterium* sp. A, AF144697; *Cardiobacterium* sp. B, AF144696; *Suttonella indologenes*, M35015; *Dichelobacter nodosus*, M35016. Accession numbers for the sequences of the other species (all type strains) are available in GenBank.

TABLE BXII.γ.34. Phenotypic traits that differentiate *Cardiobacteriaceae* genera from other genera^a

Characteristic	<i>Cardiobacterium</i>	<i>Dichelobacter</i>	<i>Suttonella</i>	<i>Bacteroides</i>	<i>Eikenella</i>	<i>Kingella</i>	<i>Moraxella</i>	<i>Pasteurella</i>	<i>Porphyromonas</i>
Aerobic	+	—	+	—	—	+	+	+	—
Oxidase	+	—	+	—	+	+	+	+	—
Catalase	—	—	—	D	—	—	D	+	D
Indole production	+	—	+	—	—	—	—	D	+
Nitrate reduction	—	—	—	nd	+	D	D	+	nd
Acidifies glucose	+	—	+	+	—	+	—	+	—
Acidifies D-sorbitol	D	—	—	D	—	—	—	D	—
γ-Glutamyltransferase activity	+	nd	+	nd	—	—	D	D	nd
Ornithine decarboxylase	—	+	—	—	+	—	D	D	—
Branched chain fatty acids	—	—	—	+	—	—	—	—	+

^aFor symbols see standard definitions; nd, not determined.

The mol% G + C of the DNA is: 59–60.
Type species: ***Cardiobacterium hominis*** Slotnick and Dougherty 1964, 271.

FURTHER DESCRIPTIVE INFORMATION

Cells of *Cardiobacterium* grown on tryptose blood agar (containing 5% human blood) are pleomorphic and have rounded ends. The usual cell dimensions are ~0.5–0.6 × 1.0–2.2 μm. Occasionally filaments are formed, which vary in length from 7.0–35.0 μm. Teardrop-shaped cells may occur. One or both ends of *Cardiobacterium* cells are frequently enlarged, and the crystal violet of the Gram stain tends to be retained in these areas. Other cells may show crystal violet retention in the central portions (Slotnick and Dougherty, 1964).
The degree of pleomorphism is apparently influenced by the medium on which *Cardiobacterium* is grown. Pleomorphism has been reported in cultures grown on agar media, which do not contain yeast extract. On yeast extract-containing media, the cells

have been found to be generally uniform Gram-negative rods that are 0.5 × 2.0 μm (Savage et al., 1977).
Sudanophilic bodies and metachromatic granules have been reported with the use of the Sudan Black B and Albert’s stain, respectively (Slotnick and Dougherty, 1964). However, Midgley et al. (1970) were unable to confirm these observations.
The cell wall is of the Gram-negative type with distinct layers—a dense outer layer, a unit membrane, and a dense inner layer. The substructure of the crystalline surface layer (S-layer) consists of closely packed units, which are nearly spherical and have an average diameter of about 3.5 nm (Reyn et al., 1971). Cells have 20–40 nm thick polar caps consisting of a dense, tuft-like material. In the periphery of the cytoplasm, numerous unit membrane profiles have been observed and are thought to be intrusions of the plasma membrane (Reyn et al., 1971).
After incubation for 24 h at 37°C on tryptose blood agar (containing 5% human blood), colonies are punctiform. After 48 h the maximum size is 1–2 mm. The colonies are circular,

convex, smooth, moist, glistening, opaque, and butyrous, and have an entire edge. No clear zones of hemolysis are formed around the colonies (Slotnick and Dougherty, 1964).

Strains of *C. hominis* may be grown in a defined medium¹ described by Slotnick and Dougherty (1965). Essential components of the medium are pantothenate, niacinamide, thiamine, threonine, proline, leucine, histidine, glycine, and arginine. Biotin, pyridoxine, tyrosine, valine, and glutamate are not required but must be included for optimum growth. Glucose is not an essential carbon source or the sole energy source, but it is fermented and does increase the growth yield. The major end product of glucose fermentation is lactic acid (Slotnick and Dougherty, 1964). Maltose, sucrose, fructose, and mannose are also fermented and can be substituted effectively for glucose in the medium. Lactose, arabinose, and galactose are not fermented or utilized for growth. Low concentrations (~1.5–2.0 µg/ml) of riboflavin or flavin mononucleotide inhibit the growth of *C. hominis*. This inhibitory effect is neutralized by the presence of excess leucine in the medium.

Growth at 30°C and 37°C is equally good. Growth at 25°C is sporadic and light. No growth occurs above 42°C or at 22°C (Slotnick and Dougherty, 1964; Midgley et al., 1970).

The optimal pH range for growth is 7.0–7.2. *C. hominis* grows poorly in air unless the humidity is increased by a method such as placing filter paper strips, saturated with water, in a closed container. In a candle jar or under anaerobic conditions, it is not necessary to increase the humidity. The optimal atmosphere contains 3–5% CO₂ (Slotnick and Dougherty, 1964). Midgley et al. (1970) found that three of four strains grew equally well in a candle jar and in a sealed jar without increased CO₂. The fourth strain grew less well in the sealed jar than in the candle jar.

Slotnick and Dougherty (1964) produced antisera to four strains. All of the strains reacted to a high titer in each of the unabsorbed sera; however, by absorption studies the four strains were shown not to be antigenically identical. No antigenic relationship was demonstrated to members of the genera *Brucella*, *Pasteurella*, *Bordetella*, *Moraxella*, *Haemophilus*, *Streptobacillus*, *Corynebacterium*, *Bacteroides*, *Neisseria*, *Escherichia*, or *Lactobacillus*.

Savage et al. (1977) tested five strains and found the minimum inhibitory concentrations for all strains to be less than 2 µg/ml for ampicillin, carbenicillin, cephalothin, chloramphenicol, penicillin, tetracycline, streptomycin, kanamycin, gentamicin, and colistin.

In humans, *C. hominis* is a relatively rare cause of bacterial endocarditis. Dental caries and periodontal disease have been suggested as predisposing factors (Berbari et al., 1997; el Khizzi et al., 1997). A single case of abdominal abscess and bacteremia due to *Cardiobacterium hominis* and *Clostridium bifermentans* has been reported (Rechtman and Nadler, 1991). Tucker et al. (1962) found no evidence of infection in mice, guinea pigs, rabbits, hamsters, or pigeons after injection of 10⁷–10⁸ viable organisms intravenously, intraperitoneally, or subcutaneously.

C. hominis is apparently part of the normal nasal and pharyngeal flora of humans. It was isolated from the nose or throat of 68 of 100 individuals examined by Slotnick et al. (1964). The

organism was also isolated from cervical and vaginal cultures obtained from 2 of 159 subjects examined (Slotnick, 1968).

ENRICHMENT AND ISOLATION PROCEDURES

Cardiobacterium hominis has been isolated from blood in several different media under both aerobic and anaerobic conditions. Media that have been used successfully are brain heart infusion broth containing sodium polyanetholsulfonate (SPS) and *p*-aminobenzoic acid, casein soy broth with SPS, thioglycolate broth with SPS, and glucose broth (Midgley et al., 1970; Savage et al., 1977).

Slotnick et al. (1964) isolated *C. hominis* from the nose and throat of humans on trypticase soy agar plates containing 5% human blood. The plates were incubated at 37°C for 48–72 h in candle jars. Because of difficulty in recognizing the colonies of *C. hominis*, smears of selected areas of the plates were stained with fluorescent antibody. Isolation of strains was accomplished by subculturing from the areas that showed positive fluorescent staining reactions.

MAINTENANCE PROCEDURES

Strains of *C. hominis* may be preserved by lyophilization. Freezing also may preserve them. Cells from 18–24 h old agar cultures suspended in defibrinated rabbit blood have survived for more than 15 years at –50°C.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Indole formation is an important characteristic for the identification of *C. hominis*. Indole production, however, may be weak and may not be detected by test procedures that do not first concentrate the indole by xylene extraction.

Tryptone broth and heart infusion broth are suitable media for the indole test. A portion of broth culture should be tested after incubation of 18–24 h. If the reaction is negative, the remaining portion of the broth culture should be tested after incubation for an additional 24 h. To perform the test, add approximately 0.5–1 ml of xylene to 3–4 ml of broth culture. Shake vigorously to extract the indole. After the xylene layer has formed at the surface, add either Ehrlich's or Kovacs' reagent, allowing the reagent to flow down the side of the slightly tilted tube.

DIFFERENTIATION OF THE GENUS *CARDIOBACTERIUM* FROM OTHER GENERA

Characteristics by which *C. hominis* can be distinguished from other Gram-negative rod-shaped organisms that have similar physiological characteristics have been described by Bruun et al. (1984b) and are shown in Table BXII.γ.34 of the chapter *Cardiobacteriaceae*. The cellular fatty acid composition of *Cardiobacterium hominis* differs from *Eikenella corrodens*, *Kingella denitrificans*, and *Kingella kingae* by the absence of 3-hydroxylauric (C_{12:0 3OH}) acid, from *Suttonella indologenes* by the presence of 3-hydroxypalmitic (C_{16:0 3OH}) acid, and from *E. corrodens* by the presence of 3-hydroxymyristic (C_{14:0 3OH}) acid (Wallace et al., 1988). *Kingella denitrificans* and *Kingella kingae* contained myristic (C_{14:0}) and palmitic (C_{16:0}) acids as major acids, whereas *cis*-vaccenic (C_{18:1 ω7c}) and palmitic acids were the major acids in *Suttonella indologenes*, *C. hominis*, and *E. corrodens*.

TAXONOMIC COMMENTS

Cardiobacterium hominis is the only named species in the genus; however, two unnamed *C. hominis*-like species have been described (see Other Organisms). Based on 16S rRNA sequence

1. Defined medium has the following composition (per liter of distilled water): MgSO₄·7H₂O, 0.4 g; MnSO₄·H₂O, 3 mg; NaCl, 50 mg; Na₂HPO₄, 2.84 g; KH₂PO₄, 2.72 g; ZnSO₄·7H₂O, 4.43 mg; FeSO₄·7H₂O, 4 mg; CuSO₄·5H₂O, 0.5 mg; glucose, 5.0 g; arginine, 160 mg; glutamic acid, 200 mg; glycine, 176 mg; histidine, 126 mg; leucine, 426 mg; proline, 100 mg; threonine, 276 mg; valine, 185 mg; tyrosine, 35 mg; calcium pantothenate, 10 mg; niacinamide, 10 mg; pyridoxine hydrochloride, 20 mg; thiamine hydrochloride 10 mg; and biotin, 1 mg. Final pH is 7.0.

analysis, the genera *Cardiobacterium*, *Suttonella*, and *Dichelobacter* compose the family *Cardiobacteriaceae* (Dewhirst et al., 1990). The family branches deeply in the *Gammaproteobacteria* at the juncture of the *Gammaproteobacteria* and *Betaproteobacteria*. Neighboring families include “*Xanthomonadaceae*”, “*Piscirickettsiaceae*”, and “*Francisellaceae*”. A phylogenetic tree including *Cardiobacterium hominis* and related bacteria is shown in Fig. BXII.γ.40 in the *Cardiobacteriaceae* chapter.

ACKNOWLEDGMENTS

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FURTHER READING

Dewhirst, F.E., B.J. Paster, S. La Fontaine and J.I. Rood. 1990. Transfer of *Kingella indologenes* (Snell and Lapage 1976) to the genus *Suttonella* gen. nov. as *Suttonella indologenes* comb. nov.; transfer of *Bacteroides nodosus* (Beveridge 1941) to the genus *Dichelobacter* gen. nov. as *D. nodosus* comb. nov.; and assignment of the genera *Cardiobacterium*, *Dichelobacter*, and *Suttonella* to *Cardiobacteriaceae* fam. nov. in the gamma division of *Proteobacteria* on the basis of 16S rRNA sequence comparisons. *Int. J. Syst. Bacteriol.* 40: 426–433.

Snell, J.J.S. and S.P. Lapage. 1976. Transfer of some saccharolytic *Moraxella* species to *Kingella* Henriksen and Bøvre 1976, with descriptions of *Kingella indologenes* sp. nov. and *Kingella denitrificans* sp. nov. *Int. J. Syst. Bacteriol.* 26: 451–458.

List of species of the genus *Cardiobacterium*

1. ***Cardiobacterium hominis*** Slotnick and Dougherty 1964, 271^{AL}
ho' mi. nis. L. gen. n. *hominis* of man.
The characteristics are as described for the genus.

Occurs in nasal and oral flora of humans; isolated from the blood of humans with endocarditis.
The mol% G + C of the DNA is: 59–60 (*T_m*).
Type strain: 6573, ATCC 15826, DSM 8339, NCTC 10426.
GenBank accession number (16S rRNA): M35014.

Other Organisms

Strains potentially representing two additional *Cardiobacterium* species have been described (Rothenpieler et al., 1986). These species are shown as *Cardiobacterium* sp. A and *Cardiobacterium* sp. B in Fig. BXII.γ.40 in the *Cardiobacteriaceae* chapter. *Cardiobacterium* sp. A is represented by the single strain SSI AB2167 (FsK 3118; M. Kilian, human dental plaque; Denmark, 1985). This strain has been called *Suttonella indologenes*-like and has 97% 16S rRNA sequence similarity with *C. hominis*. *Cardiobacterium* sp. B is represented by at least two strains: CCUG 13150 (SSI AB 2102; G. Dahlen; human periodontitis; Sweden, 1982) and CCUG 31207 (SSI AB 2168; FsK3120; M. Kilian; human dental plaque; Denmark, 1985). These strains have also been called *Cardiobac-*

terium hominis-like and were designated by Enevold Falsen as EF Group 18 strains (Culture Collection, University of Göteborg). There are additional EF Group 18 strains in the CCUG collection. *Cardiobacterium* sp. B has 95% 16S rRNA sequence similarity with *C. hominis*. The phenotypic traits differentiating these strains from *C. hominis* are shown in Table BXII.γ.35. Further characterization of strains in the genus *Cardiobacterium* is needed prior to naming strains these additional species. Inclusion of *Cardiobacterium* spp. A and B would necessitate modification of genus description to include indole-negative and alkaline phosphatase-positive species.

TABLE BXII.γ.35. Differentiation of unnamed *Cardiobacterium* species from *C. hominis*

Characteristic	<i>Cardiobacterium hominis</i>	<i>Cardiobacterium</i> sp. A	<i>Cardiobacterium</i> sp. B	<i>Suttonella indologenes</i>
Ferments D-mannitol or D-sorbitol	+	–	+	–
Indole	+	+	–	+
Alkaline phosphatase	–	–	+	+

Genus II. ***Dichelobacter*** Dewhirst, Paster, La Fontaine and Rood 1990, 430^{VP}

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Di. che' lo. bac. ter. Gr. adj. *Dichelos* cloven hoofed; N.L. n. *bacter* a rod; N.L. masc. n. *Dichelobacter* cloven-hoofed rod, because this organism is the rod-shaped bacterium that causes footrot in sheep, goats, and cattle.

Large straight or slightly curved rods, 3–6 × 1.0–1.7 μm, with rounded ends. Freshly isolated cells often have terminal swellings, this feature is less pronounced after subculture. **Gram negative** but with a tendency to resist decolorization. Organisms stained with Loeffler methylene blue have prominent polychromatic granules toward the poles and at intermediate sites within the cells. **Grows slowly only under strictly anaerobic conditions but is not rapidly killed by exposure to oxygen.** The cells have large numbers of thin polar fimbriae or pili that vary in number according to changes in colony morphology. The pili belong to

the type IV or N-methylphenylalanine family; consequently, **the cells exhibit the twitching motility and spreading of colonies that is characteristic of cells with type IV pili. The organism also produces extracellular serine proteases that are relatively heat stable.** Ammonia is produced from arginine, asparagine, serine, and threonine. Positive for ornithine decarboxylase, phosphatase (weak), H₂S production, selenite reduction, and proteolytic activity on gelatin, casein, and albumin. **Negative for acid or gas production from carbohydrates;** ammonia production from phenylalanine, cysteine, citrulline, and ornithine; starch hydrol-

ysis; esculin hydrolysis; indole production; nitrate reduction; growth in 0.1% bile; hemolysis; arginine decarboxylase; catalase; oxidase; urease; DNase; coagulase; lipase; lecithinase; and hyaluronidase. The genus contains a single species, *Dichelobacter nodosus*. Further details on the properties of the genus are available elsewhere (Beveridge, 1941, Cato et al., 1979, Holdeman et al., 1984a, Skerman, 1989; Dewhirst et al., 1990).

The mol% G + C of the DNA is: 45.

Type species: Dichelobacter nodosus (Beveridge 1941) Dewhirst, Paster, La Fontaine and Rood 1990, 426 (*Bacteroides nodosus* (Beveridge 1941) Mráz 1963, 85; *Fusiformis nodosus* Beveridge 1941, 23.)

FURTHER DESCRIPTIVE INFORMATION

Dichelobacter is one of three genera that compose the family *Cardiobacteriaceae*. Based on 16S rRNA sequence data the members of this family are placed as deeply branching members of the *Gammaproteobacteria*. The 16S rRNA sequences of these genera, *Cardiobacterium*, *Suttonella*, and *Dichelobacter*, have approximately 93% identity (Dewhirst et al., 1990).

Little is known about the composition of the cell wall. The lipopolysaccharide (LPS) of *D. nodosus* is similar to enterobacterial LPS in that it contains 2-keto-3-deoxyoctonic acid (KDO), glucose, galactose, and heptose. The LPS is not highly toxic, although it does exhibit some characteristic endotoxin properties (Stewart, 1977). It is not immunoprotective (Stewart, 1978). Virtually nothing is known about the genes involved in LPS biosynthesis, although one gene, *lpsA*, whose product appears to interact with the acidic inner LPS core, has been identified (Billington et al., 1995).

The predominant fine structural feature is the presence of polar fimbriae or pili. These long filamentous structures are ~6 nm in diameter and up to 10 µm in length (Stewart, 1973; Walker et al., 1973). The fimbriae consist of identical subunits of a 17-kDa protein that has the typical sequence and structure of pilins belonging to the type IV pilin family (Elleman, 1988; Mattick, 1989). Several genes that appear to be involved in fimbrial biogenesis have been cloned and analyzed, including the gene encoding the prepilin peptidase (Johnston et al., 1995, 1998).

The organism is relatively slow growing and after 2–6 d of anaerobic growth on agar media produces flat spreading colonies 1–3 mm in diameter that are circular, concave, semiopaque, and slightly mottled, with a shiny, bumpy surface, and a pitted edge. Large colonies have concentric circles with the inner circle pitting the agar (Cato et al., 1979). However, there is considerable variation in colonial morphology, depending on both the isolate and the growth medium. Colonies may have a highly papillated or beaded central zone overlying a pit-like depression etched into the agar surface, producing a sunken appearance, a pale, granular mid-zone surrounded by a peripheral zone with a ground glass texture, and a fimbriate edge. Changes in colony type within a strain can occur with subculturing (Skerman, 1989). Variations in the basic features of colony morphology by different strains are described (Stewart et al., 1986).

D. nodosus does not ferment carbohydrates but catabolizes arginine, a property that has been used in the formulation of the growth medium TAS (Skerman, 1989). Maximum growth requires trypticase and 0.02–0.05 M arginine. Serine also enhances growth, and in some strains so does lysine (Skerman, 1975). The organism is routinely grown in rich medium and little is known about its specific growth requirements or its metabolism. Growth is most rapid at 37°C and within a pH range of 6.8–

7.8 (optimal range 7.2–7.8). In broth cultures, growth is evenly dispersed with a light, turbid, or silky texture and is usually visible within 24–48 h. The organism hydrolyses gelatin, produces ammonia from amino acids, and has ornithine decarboxylase activity (Skerman, 1989). *D. nodosus* isolates are susceptible *in vitro* to a wide range of antimicrobial agents.

The organism is naturally transformable with 7 of 17 strains that have been tested able to be transformed with plasmid DNA preparations. Natural transformation can be used to make chromosomal mutants by allelic exchange (Kennan et al., 1998, 2001). The genome of *D. nodosus* strain VCS1703A is currently being determined (see <http://www.tigr.org/>). The genome of the reference *D. nodosus* strain A198 (198A, VCS1001, or ATCC 27521) has been mapped; it consists of a 1.54-Mb circular chromosome that has three rRNA operons (La Fontaine and Rood, 1996). Key features are that the putative virulence genes are not localized to one region of the genome and that the genome contains a large 27-kb virulence-related locus, the *vrl* region (Billington et al., 1999), which has a markedly different distribution of *EagI* and *StuI* restriction sites. Only one extrachromosomal element, the 10-kb plasmid pJIR896, has been identified. This plasmid carries a copy of another virulence-associated locus, the *vap* region, as well as the insertion sequence *IS1253*, which is also found on the *D. nodosus* chromosome (Billington et al., 1996b).

The primary antigenic determinants are the type IV fimbriae. *D. nodosus* isolates can be divided into nine major serogroups (A to I) based on their ability to react with antisera prepared against purified fimbriae (Claxton, 1989). The serogroups can be further divided into two major classes based on their fimbrial subunit structure and the genetic organization of the fimbrial subunit gene region (Elleman, 1988; Hobbs et al., 1991; Mattick et al., 1991). Class I isolates (serogroups A, B, C, E, F, G, and I) have a single gene *fimB* located between the fimbrial subunit gene *fimA* and the *clpB* gene. By contrast, in the same genomic position, class II isolates (serogroups D and H) have three genes, *fimC*, *fimD*, and *fimZ*. Mutation of the fimbrial subunit gene, *fimA*, has shown that production of fimbrial subunits is essential for fimbrial biogenesis, twitching motility, natural competence, protease secretion, and virulence in sheep (Kennan et al., 2001). These studies also showed that the *fimB* gene was not required for fimbrial biogenesis or virulence.

D. nodosus is the essential causative agent of footrot in ruminants, especially in sheep, goats, cattle, and deer. The ovine disease manifests as a continuum from virulent through intermediate to benign footrot. The severity of the disease depends on both the climatic conditions, being optimal under warm, moist conditions, and the virulence of the *D. nodosus* isolates. Clinical isolates are therefore often referred to as being virulent, intermediate, or benign. The only virulence factor that has been shown to be essential for virulence is the fimbrial subunit (Kennan et al., 2001). However, the production of extracellular serine proteases and the presence of the *vrl* genomic island have also been postulated to be involved in virulence (Billington et al., 1996a). The role of these and other virulence factors in the disease process remains to be determined.

Confirmed isolates of *D. nodosus* have been obtained only from ruminants such as sheep, goats, cattle, and deer (Egerton, 1989; Stewart, 1989). The organism does not survive for longer than 14 d in feces, soil, or pasture (Stewart, 1989). Although there have been instances of *D. nodosus* supposedly being isolated from other sources, such as human specimens, these strains were not

confirmed as *D. nodosus* (Cato et al., 1979; Holdeman et al., 1984a).

ENRICHMENT AND ISOLATION PROCEDURES

Specimens are collected with a cotton-tipped swab, wooden applicator stick, or scalpel blade, by scraping active footrot lesions containing exudate. The specimens are then deposited into a variety of transport media (Skerman, 1989) and promptly sent to the laboratory within 24–48 h of collection where they are inoculated onto hoof agar (Thomas, 1958), trypticase arginine serine (TAS) agar (Skerman, 1975), or modified TAS agar incorporating hemoglobin-HEPES solution (Depiazzi et al., 1990) containing 4% agar. The plates are incubated at 37°C for 2–4 d in anaerobic jars containing CO₂/H₂ (10:90). A selective blood Eugonagar medium, using lincomycin, also has been devised (Gradin and Schmitz, 1977). Primary cultures can be inoculated by heavy surface streaking with lines 10 mm apart onto well-dried plates (Skerman, 1989) or by stab inoculation (Depiazzi et al., 1990). Surface colonies grow as translucent flat granular colonies, often with concentric zones with a fimbriate edge (Thorley, 1976). In the streaking method, colonies are picked off each primary plate with a sterile straight wire and subcultured onto a 4% hoof agar, TAS agar, or blood Eugonagar plate to obtain pure colonies and then incubated anaerobically for 3–4 d at 37°C. In the stab inoculation method, an agar section is cut from the isolation plate with the aid of a stereo microscope and using a nichrome bent, spatula-shaped wire. The agar then is placed colony side down onto the surface of a modified TAS maintenance agar plate and incubated anaerobically for 2 d at 37°C. The addition of hoof particles to modified TAS agar enhances the growth of spreading colonies and facilitates the isolation of *D. nodosus*. Small Petri dishes (55 mm) are used instead of conventional plates (90 mm), which enables less culture medium to be used and three times as many agar plates to be incubated in each anaerobic jar. If a pure colony is not obtained after incubation of the primary isolation plate, then with a sterile spatula the agar is flipped upside down into the lid so that the underside colony can be sampled and subcultured (Depiazzi et al., 1990). Alternatively, a sterile microscope cover glass, dipped into a bacterial suspension of a contaminated bacterial colony and rinsed to remove nonadherent bacteria, can be used to purify *D. nodosus* colonies. The cover glass is moved in a zigzag manner over the agar surface to inoculate the adherent *D. nodosus* cells. These modifications enable the time interval between submission and final reporting to be reduced to 8 d.

MAINTENANCE PROCEDURES

For routine maintenance of *D. nodosus* cultures under anaerobic conditions, either hoof agar, TAS agar, modified TAS agar, or blood Eugonagar (Gradin and Schmitz, 1977) can be used. For anaerobic broth cultures, TAS broth, HEPES TAS broth, or modified Eugonbroth (BBL) is recommended. To facilitate prompt growth in liquid medium from agar cultures, the broth can be inoculated with an aqueous vortexed suspension of pure colonies or a small portion of agar. Although *D. nodosus* is an anaerobe, fully grown plate cultures can tolerate exposure to air for up to 12 h on the laboratory bench (Skerman, 1989). Cultures will remain viable in anaerobic jars for several weeks at room temperature (Skerman, 1975). For long-term storage of cultures, *D. nodosus* may be preserved by lyophilization of 3–4-d-old cells suspended in horse serum containing 7.5% (w/v) glucose, skim milk, 0.25 M sucrose, or TAS broth, or alternatively by adding

20% glycerol to TAS broth suspensions and freezing at –70°C. Cryopreserved preparations are known to survive for more than 20 years.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Ovine footrot lesions differ in severity according to the virulence of the strain of *D. nodosus* involved; the two major clinical forms are termed virulent and benign footrot (Stewart, 1989). There are also strains of *D. nodosus* that are intermediate in virulence. However, for purposes of uniformity in footrot diagnosis, footrot is described as virulent or benign, with strains of intermediate virulence being regarded as effectively virulent. Because the extent of disease expression is also influenced by the host (e.g., the breed of sheep), the season, and the environment, laboratory diagnosis is increasingly being used to assist the clinical diagnosis of virulent or benign footrot. Although there is a correlation between virulence of wild-type isolates and colony type, this distinction is not sufficiently reliable as a laboratory test for differentiating virulent and benign isolates. The gelatin gel test is the recommended laboratory diagnostic test for differentiating virulent and benign footrot (Palmer, 1993); it measures the thermostability of proteases in 2–4-d-old broth cultures. In general, isolates of *D. nodosus* capable of producing virulent footrot produce proteolytic enzymes that are stable to heating at 67°C, whereas proteases from benign isolates are inactivated at this temperature. Proteolytic activity is detected by clearing of a gelatin substrate incorporated into agarose plates. The gelatin gel test has now superseded the elastase test because the latter may take 7–28 d for a definitive result. Virulent isolates cause digestion of elastin particles beneath the streak line in elastin TAS agar plates, whereas benign isolates do not produce clearing (Stewart, 1979). In the alternative zymogram protease test, the electrophoretic pattern of the protease isoenzymes is determined in a nondenaturing polyacrylamide slab gel electrophoresis system followed by detection with a gelatin/agarose contact overlay assay. Benign and virulent strains have different and characteristic profiles (Every, 1982; Kortt et al., 1983; Palmer, 1993). There is a range of different zymogram patterns for isolates with either stable (S) proteases or unstable (U) proteases, with the predominant pattern being the S1 and U1 pattern, respectively (Palmer, 1993). The zymogram test may be useful for classifying isolates where there are anomalous results with the gelatin gel test and virulence assessment.

More recently, two new technologies have become available for footrot diagnosis. These are the protease monoclonal antibody (Mab) ELISA (Stewart et al., unpublished) and the gene probe/PCR-based assay (Katz et al., 1991; Liu, 1994; Rood et al., 1996). The virulent and benign Mabs are able to distinguish antigenic differences between the proteases produced by virulent and benign strains, and there is a high level of agreement between the results for the gelatin gel test and the protease Mab ELISA. This difference in epitope specificity may result from a single amino acid change in one of the virulent and benign acidic proteases (Riffkin et al., 1995). The gene probe or PCR tests detect virulence-associated genomic DNA unrelated to the protease genes. The DNA probes, which specifically detect the *vrl* and *vap* gene regions, divide *D. nodosus* isolates into three major categories (1, 2, and 3). The PCR test is based on specific oligonucleotide primers for the *vrl* and *vap* specific plasmids, pJIR314B and pJIR318, respectively (Rood et al., 1996). The PCR test further divides isolates in gene probe categories 1 and 2 into groups L and H. Isolates in all three categories can belong to

probe category + or – depending on the presence or absence of an additional gene region. It is recommended that rapid Mab tests for the proteases be used for differentiation of virulent and benign footrot and that the PCR assay should be used for molecular epidemiological studies.

DIFFERENTIATION OF THE GENUS *DICHELOBACTER* FROM OTHER GENERA

Dichelobacter nodosus, previously *Bacteroides nodosus*, can be differentiated from members of the genera *Bacteroides*, *Prevotella*, and *Porphyromonas* by fatty acid analysis. *D. nodosus* possesses C_{16:1 ω7c}, C_{18:1 ω7c}, and C_{16:0} fatty acids, while the genera in the phylum “*Bacteroidetes*” contain large amounts of the methyl branched-chain fatty acids C_{15:0 iso} and C_{15:0 anteiso}. Genera in the “*Bacteroidetes*” also possess sphingolipids. *D. nodosus* contains type IV pili, which have been found only in *Proteobacteria*. *D. nodosus* can be differentiated from other genera in the family *Cardiobacteriaceae* in that it is anaerobic, oxidase negative, indole negative, and ornithine decarboxylase positive.

TAXONOMIC COMMENTS

Dichelobacter nodosus is the only species in the genus *Dichelobacter*. The genera *Dichelobacter*, *Suttonella*, and *Cardiobacterium* compose the family *Cardiobacteriaceae*. The family branches deeply in the *Gamma*proteobacteria at the juncture of the *Gamma*proteobacteria

and *Betaproteobacteria*. Neighboring families include “*Xanthomonadaceae*”, “*Piscirickettsiaceae*”, and “*Francisellaceae*”. A phylogenetic tree including *Dichelobacter nodosus* is shown in the *Cardiobacteriaceae* chapter.

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List of species of the genus *Dichelobacter*

1. ***Dichelobacter nodosus*** (Beveridge 1941) Dewhirst, Paster, La Fontaine and Rood 1990, 426^{VP} (*Bacteroides nodosus* (Beveridge 1941) Mráz 1963, 85; *Fusiformis nodosus* Beveridge 1941, 23.)

no.do'sus. L. adj. *nodosus* full of knots, refers to the shape of the cells.

The characteristics are as described for the genus. The organism is found on the hooves and interdigital skin of ruminants, especially sheep and cattle. The growth of most strains on an appropriate medium, such as TAS medium, is more luxuriant than that of ATCC 25549. The genome of strain A198 is circular, 1.54 Mb in size, and has three rRNA operons (La Fontaine and Rood, 1996). The GenBank accession number of the 16S rRNA molecule from the reference strain A198 is M35016 (Dewhirst et al., 1990). There is no 16S rDNA sequence available from the type strain.

The organism is the essential causative agent of footrot in sheep and other ruminants (Stewart, 1989). The fimbriae are the major protective antigens; immunization of sheep with purified recombinant fimbriae will protect against infection but only by members of the homologous serogroup (Egerton et al., 1987; Stewart and Elleman, 1987). The major virulence antigens appear to be the type IV fimbriae and the extracellular serine proteases produced by the organism. Virulent isolates produce four extracellular acidic serine proteases (V1–V3, V5), pI 5.2–5.6, and a basic protease (BprV), pI ~9.5. The benign protease isoenzymes, B1–B4, have distinctly different electrophoretic mobilities

and pIs, elastolytic activity, and heat stability, but are very similar to V1–V3 from virulent strains in amino acid composition, and peptide profiles, and apparent *M_r*, proteases V5 and B5 are identical. Benign strains secrete a distinct basic protease BprB), pI 8.6, that is very closely related to the virulent basic protease. In summary, there are three distinct types of *D. nodosus* proteases, V5/B5, V1–V3/B1–B4, and the basic protease. The gene sequences for the two acidic protease genes, *aprV5* and *aprV2/B2*, and the basic protease genes, *bprV/bprB*, have been determined. These serine proteases are members of the subtilisin family of proteases and presumably arose by gene duplication from an ancestral precursor gene. They show significant sequence similarity to the extracellular serine protease of *Xanthomonas campestris*, a plant pathogen. In addition, there is a 27-kb genomic region, the *vrl* locus, that appears to be preferentially associated with more virulent isolates. An additional integrated region, the *vap* locus, is found in all isolates that carry the *vrl* region, and in other isolates (Billington et al., 1996a). The genome of *D. nodosus* strain VCS1703A is currently being sequenced by the Institute for Genomic Research (TIGR). The genome has been assembled into 264 contigs and can be searched using BLAST at TIGR (<http://www.tigr.org/>) or NCBI (<http://www.ncbi.nlm.nih.gov/blast/>).

The mol% G + C of the DNA is: 45.

Type strain: ATCC 25549.

Additional Remarks: The reference strain is ATCC 27521 (A198, 198A, or VPI5731-1).

Genus III. *Suttonella* Dewhirst, Paster, La Fontaine and Rood 1990, 429^{VP}

FLOYD E. DEWHIRST AND BRUCE J. PASTER

Sut. ton. el' la. L. dim. ending *-ella*; N.L. fem. n. *Suttonella* named after R.G.A. Sutton.

Large straight rods, $1.0 \times 2\text{--}3 \mu\text{m}$, with rounded ends. Cells occasionally occur in pair, chains, or rosettes. Gram negative but with a tendency to resist Gram decolorization. Nonmotile as determined by normal tests, but may be fimbriated (i.e., have pili) and exhibit twitching motility. Aerobic. Aerobic growth is enhanced by high humidity and CO_2 , which makes the organism appear to be facultatively anaerobic. **Oxidase positive** (when tested with tetramethyl-*p*-phenylene diamine). **Catalase negative**. **Indole positive** (activity may be weak). Urease negative. DNase negative. Ornithine and lysine decarboxylase negative. **Alkaline phosphatase positive**. Trypsin (*N*-benzoyl-DL-arginine-2-naphthylamide) negative. Phosphohydrolase (naphthyl-AS-BI-phosphate) negative. Chemoorganotrophic, having a fermentative type of metabolism. Acid but not gas is produced from fructose, glucose, maltose, D-mannose, and sucrose. The maltose reaction is weak and delayed (positive at 28 d), but is negative as determined by rapid (4–24 h) protocols. **Nitrates are not reduced**. Nitrites are reduced by some strains. Resazurin is reduced. H_2S is produced (as determined by the lead acetate method, but not by the triple sugar iron method). No growth occurs on MacConkey agar.

The mol% G + C of the DNA is: 49.

Type species: *Suttonella indologenes* (Snell and Lapage 1976) Dewhirst, Paster, La Fontaine, and Rood 1990, 430 (*Kingella indologenes* Snell and Lapage 1976, 456.)

FURTHER DESCRIPTIVE INFORMATION

Initial isolates of *Suttonella indologenes* were considered novel *Moraxella* strains (van Bijsterveld, 1970; Sutton et al., 1972; Bøvre et al., 1974). Snell and Lapage (1976) placed these saccharolytic and indole producing strains in the genus *Kingella* as the novel species *Kingella indologenes*. DNA–DNA hybridizations (Rothenpieler et al., 1986) and 16S rRNA sequence analysis (Dewhirst et al., 1990) demonstrated that *Kingella indologenes* was not related to other *Kingella* species in the family *Neisseriaceae*, but rather branched between the *Betaproteobacteria* and *Gamma proteobacteria* (rRNA homology groups I and II) with *Cardiobacterium hominis*. Subsequent 16S rRNA sequence analysis demonstrated that *Bacteroides nodosus* also clustered with these species. These three species were placed in the family *Cardiobacteriaceae*, and *Kingella indologenes* and *Bacteroides nodosus* were renamed *Suttonella indologenes* and *Dichelobacter nodosus*, respectively (Dewhirst et al., 1990).

Suttonella grows slowly and after 24 h colonies are 0.1–0.5 mm in diameter. After 72 h, colonies are 1–1.5 mm. Pitting of media or spreading edges has been seen in fresh colonies, but may be lost on storage (Snell and Lapage, 1976).

The major cellular fatty acids are as follows: $\text{C}_{18:1 \omega 7\text{C}}$ (27%), $\text{C}_{16:0}$ (26%), $\text{C}_{16:1 \omega 7\text{C}}$ (19%), $\text{C}_{14:0}$ (11%), $\text{C}_{12:0}$ (5%), and $\text{C}_{12:0 \text{ 3OH}}$ (2%) (Wallace et al., 1988).

Ultrastructure has not been examined, but related genera *Cardiobacterium* and *Dichelobacter* both possess crystalline surface layers and intrusive membranes (Reyn et al., 1971; Stewart and Egerton, 1979). *Suttonella indologenes* possesses *N*-methylphenylalanine, or type IV pili.

Growth and biochemical characteristics from several references were summarized by Bruun et al. (1984b) and by Weyant et al. (1996). Growth at 22, 30, and 37°C. Acid is produced from

dextrin, fructose, glucose, maltose, D-mannose, and sucrose. Growth in the presence of 4% NaCl. Production of acid phosphatase, γ -glutamyltransferase, indole, and hydrogen sulfide (by lead acetate paper). Hydrolysis of casein, Tween 20, and Tween 40. No growth at 5, 40, 42, or 45°C, or in the presence of 6% NaCl, 10% or 40% bile. No acid is produced from D-adonitol, L-arabinose, cellobiose, dulcitol, ethanol, D-galactose, glycerol, *i*-inositol, lactose, D-mannitol, melibiose, raffinose, L-rhamnose, salicin, starch, D-sorbitol, trehalose, or D-xylose. No liquefaction of gelatin or serum. No production of β -galactosidase, urease, ornithine decarboxylase, lysine decarboxylase, or trypsin. No citrate utilization or growth on hydroxybutyrate. No hydrolysis of esculin, starch, Tween 80, or arginine. No reduction of nitrate.

Currently, the only genetic information in GenBank for *Suttonella indologenes* is the 16S rRNA sequence under accession number M35015. The closely related organism *Dichelobacter nodosus* has been much more extensively studied, and over 60 sequences are in GenBank (see the chapter on the genus *Dichelobacter*). This sequence information should be extremely useful in seeking homologous genes in *Suttonella indologenes*.

Little is known regarding the pathogenicity of *Suttonella indologenes*. It has been isolated from patients with eye infections and endocarditis (van Bijsterveld, 1970; Sutton et al., 1972; Jenny et al., 1987).

ENRICHMENT AND ISOLATION PROCEDURES

No particular methods have been developed specifically for isolation of *S. indologenes*. Strains have been recovered on standard blood plates grown under microaerophilic conditions.

MAINTENANCE PROCEDURES

Strains of *S. indologenes* may be preserved by lyophilization.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Indole formation may be weak and should be determined using extraction with xylene or chloroform and detection with Ehrlich's reagent (*p*-dimethylaminobenzaldehyde). The spot indole reagent *p*-aminocinnamaldehyde may fail to detect weak indole production.

DIFFERENTIATION OF THE GENUS *SUTTONELLA* FROM OTHER GENERA

See Table BXII.γ.34 in the chapter on the family *Cardiobacteriaceae*. *Suttonella* may be distinguished from *Cardiobacterium* by a positive reaction for alkaline phosphatase, and negative reactions for D-sorbitol and D-mannitol fermentation. *Suttonella* may be distinguished from *Eikenella* and *Kingella* by indole production and γ -glutamyltransferase activity. *Suttonella* may be distinguished from many other genera by the unusual combination of being oxidase and indole positive and catalase and nitrate reduction negative.

TAXONOMIC COMMENTS

Suttonella indologenes is the only species in the genus *Suttonella*. The genera *Suttonella*, *Dichelobacter*, and *Cardiobacterium* compose the family *Cardiobacteriaceae*. The family branches deeply in the *Gamma proteobacteria* at the juncture of the *Gamma proteobacteria*

and *Beta*proteobacteria. Neighboring families include “*Xanthomonadaceae*”, “*Piscirickettsiaceae*”, and “*Francisellaceae*”. A phylogenetic tree including *Suttonella indologenes* can be seen in Fig. BXII.γ.40.

ACKNOWLEDGMENTS

Floyd E. Dewhirst and Bruce J. Paster were supported by NIH grants DE10374 and DE11443.

FURTHER READING

Dewhirst, F.E., B.J. Paster, S. La Fontaine and J.I. Rood. 1990. Transfer of *Kingella indologenes* (Snell and Lapage 1976) to the genus *Suttonella*

gen. nov. as *Suttonella indologenes* comb. nov.; transfer of *Bacteroides nodosus* (Beveridge 1941) to the genus *Dichelobacter* gen. nov. as *D. nodosus* comb. nov.; and assignment of the genera *Cardiobacterium*, *Dichelobacter*, and *Suttonella* to *Cardiobacteriaceae* fam. nov. in the gamma division of *Proteobacteria* on the basis of 16S rRNA sequence comparisons. *Int. J. Syst. Bacteriol.* 40: 426–433.

Jenny, D.B., P.W. Letendre and G. Iverson. 1987. Endocarditis caused by *Kingella indologenes*. *Rev. Infect. Dis.* 9: 787–789.

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List of species of the genus *Suttonella*

1. ***Suttonella indologenes*** (Snell and Lapage 1976) Dewhirst, Paster, La Fontaine, and Rood 1990, 430^{VP} (*Kingella indologenes* Snell and Lapage 1976, 456.)
in.dol.o'ge.nes. M.L. n. *indolum* indole; M.L. suff. *genes* producing; M.L. adj. *indologenes* indole-producing.

The characteristics are as described for the genus. Previously referred to as the “Bijsterveld/Sutton strains”.
The mol% G + C of the DNA is: 49 (*T_m*).
Type strain: ATCC 25869, DSM 8309, NCTC 10717.
GenBank accession number (16S rRNA): M35015.

Order V. **Thiotrichales** ord. nov.

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

Thi.o.tri.cha'les. M.L. fem. n. *Thiothrix* type genus of the order; -ales suffix to denote order;
M.L. fem. n. *Thiotrichales* the *Thiothrix* order.

The order *Thiotrichales* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rRNA sequences; the order contains the families *Thiotrichaceae*, *Piscirickettsiaceae*, and *Francisellaceae*.

Order is morphologically, metabolically and ecologically diverse. Includes obligate intracellular parasites of fish (*Pisciric-*

kettsia) and of insects, arachnids, mammals, and man (*Francisella*). Other genera include chemoorganotrophs, methylotrophs, and chemolithotrophic sulfur-oxidizing organisms.

Type genus: **Thiothrix** Winogradsky 1888, 39^{AL} emend. Howarth, Unz, Seviour, Blackall, Pickup, Jones, Yaguchi and Head 1999, 1824.

Family I. **Thiotrichaceae** fam. nov.

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

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Most genera deposit sulfur. Several form filaments. Motility,

if present, is by gliding, except for *Thiospira*, which possesses flagella.

Type genus: **Thiothrix** Winogradsky 1888, 39^{AL} emend. Howarth, Unz, Seviour, Blackall, Pickup, Jones, Yaguchi and Head 1999, 1824.

Genus I. *Thiothrix* Winogradsky 1888, 39^{AL} emend. Howarth, Unz, Seviour, Blackall, Pickup, Jones, Yaguchi and Head 1999, 1824

RICHARD F. UNZ AND IAN M. HEAD

Thi'o.thrix. Gr. n. *thium* sulfur; Gr. n. *thrix* hair; M.L. fem. n. *Thiothrix* sulfur hair.

Rods, 0.7–1.5 × 1.2–2.5 μm, depending on species, seriate in rigid, multicellular **filaments (trichomes)** of uniform (as long as 500 μm) or slightly tapering (base to tip and typically ranging from 100–200 μm in length with shorter trichomes present in agitated as opposed to static cultures) diameter. Gram negative or Gram variable. Capsules not produced. Trichomes clearly en-

sheathed in some but not all species. False branching never observed. **Rod-shaped gonidia with rounded ends formed at the apical region of the trichome. Gonidia motile by gliding.** The base of the trichome may have a holdfast. Rosettes of trichomes typically but not always produced. Only rosette-forming strains

and *Beta*proteobacteria. Neighboring families include “*Xanthomonadaceae*”, “*Piscirickettsiaceae*”, and “*Francisellaceae*”. A phylogenetic tree including *Suttonella indologenes* can be seen in Fig. BXII.γ.40.

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FURTHER READING

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in.dol.o'ge.nes. M.L. n. *indolum* indole; M.L. suff. *genes* producing; M.L. adj. *indologenes* indole-producing.

The characteristics are as described for the genus. Previously referred to as the “Bijsterveld/Sutton strains”.
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GenBank accession number (16S rRNA): M35015.

Order V. **Thiotrichales** ord. nov.

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

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Order is morphologically, metabolically and ecologically diverse. Includes obligate intracellular parasites of fish (*Pisciric-*

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Family I. **Thiotrichaceae** fam. nov.

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

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Most genera deposit sulfur. Several form filaments. Motility,

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Genus I. *Thiothrix* Winogradsky 1888, 39^{AL} emend. Howarth, Unz, Seviour, Blackall, Pickup, Jones, Yaguchi and Head 1999, 1824

RICHARD F. UNZ AND IAN M. HEAD

Thi'o.thrix. Gr. n. *thium* sulfur; Gr. n. *thrix* hair; M.L. fem. n. *Thiothrix* sulfur hair.

Rods, 0.7–1.5 × 1.2–2.5 μm, depending on species, seriate in rigid, multicellular **filaments (trichomes)** of uniform (as long as 500 μm) or slightly tapering (base to tip and typically ranging from 100–200 μm in length with shorter trichomes present in agitated as opposed to static cultures) diameter. Gram negative or Gram variable. Capsules not produced. Trichomes clearly en-

sheathed in some but not all species. False branching never observed. **Rod-shaped gonidia with rounded ends formed at the apical region of the trichome. Gonidia motile by gliding.** The base of the trichome may have a holdfast. Rosettes of trichomes typically but not always produced. Only rosette-forming strains

produce gonidia. Resting stages not known. Flagella absent, but a tuft of monopolar fimbriae may be present.

Aerobic or microaerophilic. Facultatively autotrophic, chemorganotrophic, and mixotrophic species; the latter requiring any of several small organic compounds, e.g., acetate, lactate, propionate, pyruvate, succinate, fumarate, and oxalacetate, as well as a reduced inorganic sulfur source. **Sulfur globules deposited** within invaginations of the cytoplasmic membrane of cells grown in the presence of a reduced inorganic sulfur compound. **By light microscopy, sulfur globules appear to be internal.**

Found in sulfide-containing flowing water and in activated-sludge wastewater treatment systems.

The mol% G + C of the DNA is: 44–52.

Type species: ***Thiothrix nivea*** (Rabenhorst 1865) Winogradsky 1888, 39 (*Beggiatoa nivea* Rabenhorst 1865, 94).

FURTHER DESCRIPTIVE INFORMATION

Advances have occurred in the taxonomy of the genus *Thiothrix* since the last edition of the *Manual* and it is no longer possible to rely on morphology alone to positively identify *Thiothrix* spp., especially, the polymorphic *T. eikelboomii* and *T. defluvii*. Notwithstanding dimensional differences, cells of *Thiothrix* spp. exhibit geometries that are typically cylindrical (Fig. BXII.γ.41a) as for *T. nivea*, *T. unzii*, and *T. fructosivorans*, and variously cuboidal, barrel-shaped, discoid, and bead-like (Fig. BXII.γ.41d and e) with respect to *T. eikelboomii* and *T. defluvii*. Trichomes are rigid in most *Thiothrix* species; however, *T. eikelboomii* and *T. defluvii* trichomes can form distinct loops or knots (Fig. BXII.γ.42), suggesting flexibility in the filaments. Several species do not contain ensheathed trichomes; however, when present, the sheath (Fig. BXII.γ.43) appears to be made up of several layers (Fig. BXII.γ.44a) and extends the full length of the trichome (Fig. BXII.γ.41c). Base to tip differentiation in cells of trichomes (Fig. BXII.γ.45) is prominent in some species. The cell wall is typically Gram negative (Fig. BXII.γ.44b), although, *T. eikelboomii* may demonstrate Gram stain variability and the formation of transverse septa in the process of cell division (Fig. BXII.γ.41e). Sulfur globules are deposited within invaginations of the cytoplasmic membrane (Fig. BXII.γ.46) and are therefore external to the cytoplasm, as in *Beggiatoa*. Poly-β-hydroxybutyrate (PHB; Fig. BXII.γ.41b and f) and polyphosphate intracellular inclusions (Fig. BXII.γ.47) may be produced. Larkin (1989) reported unconfirmed nitrogen fixation by several JP strains, including the neotype strain of *T. nivea*. Rosettes of trichomes are observed in all freshly isolated species but the property may not be stable upon repeated subculture in the laboratory. Rosettes may form from outgrowth of gonidia collected at the tip of a preexisting trichome or aggregated in the suspending fluid (Fig. BXII.γ.48). Initial attachment of the gonidia involves fimbriae (Fig. BXII.γ.49), with holdfast material being produced after attachment (Fig. BXII.γ.47). A life cycle is proposed by Larkin and Shinabarger (1983) and Larkin (1989).

Isolates of *Thiothrix* spp. develop visible colonies on LT agar after 2–3 d incubation at 28°C. Colonies are less than 2 mm in diameter, white to off-white, hard to the touch, flat to slightly raised with rough to slightly wavy or thread-like margins (Fig. BXII.γ.50). Liquid cultures are turbid to flocculent depending on species and contain freely dispersed gonidia and small rosettes. In addition, length of trichomes appears affected by culture conditions with shorter (<200 μm) trichomes occurring in shaken cultures and long (500 μm) trichomes present in static cultures.

Winogradsky (1888) first described *Thiothrix nivea* and sug-

gested the possible presence of autotrophy within the genus *Thiothrix*. Present evidence for autotrophic potential in *Thiothrix* spp. includes ribulose biphosphate carboxylase/oxygenase (RubisCo) activity (Unz and Williams, 1989; Odintsova et al., 1993) and carbon dioxide fixation (Strohl and Schmidt, 1984; McGlannan and Makemson, 1990). Chemolithoautotrophic growth is reported for "*Thiothrix ramosa*" (Odintsova et al., 1993) and *Thiothrix* strain CT3 (Tandoi et al., 1994) and *in situ* measurements of the physiology of *Thiothrix* cells in activated sludge further implicates autotrophy for these organisms (Nielsen et al., 2000). *T. nivea* has not been cultured chemolithoautotrophically or shown to fix carbon dioxide although DNA probes have detected genes encoding for RubisCo in the organism (Shively et al., 1986).

Thiothrix spp. are found in flowing sulfide-containing waters of circumneutral pH (Bahr and Schwartz, 1956; Brigmon et al., 1994) and low oxygen tension (Bahr and Schwartz, 1956; Bland and Staley, 1978) and in activated sludge systems treating septic or sulfide-bearing wastewaters (Farquhar and Boyle, 1971; Williams and Unz, 1985b; Unz and Williams, 1988). It may be the dominant filamentous sulfide-oxidizing organism in suitable waters (Lackey et al., 1965; McGlannan and Makemson, 1990). *Thiothrix* spp. have been observed in symbiotic (Brigmon and De Ridder, 1998) and epiphytic (Larkin et al., 1990) association with aquatic animals. The ability of *Thiothrix* spp. to attach to objects and to itself undoubtedly allows it to remain situated in a suitable location in flowing water.

ENRICHMENT AND ISOLATION PROCEDURES

Several methods have been reported for the enrichment of *Thiothrix* spp. In the procedure of Larkin (1989), fine-tipped forceps may be used to select tufts of suspected *Thiothrix* trichomes from *Thiothrix*-containing material contained in a Petri dish while under observation with a dissecting microscope and transmitted light. Bundles of trichomes are transferred to a second Petri dish containing about 5–10 ml of a salt solution (SS)¹ (Strohl and Larkin, 1978b), agitated with the forceps, and then transferred to another Petri dish containing the SS solution. The procedure is repeated through four or five transfers. A few drops from each dilution are transferred with a Pasteur pipette to either MP agar (SS plus 0.0001% sodium acetate, 0.03% Na₂S and 1.5% agar; Strohl and Larkin, 1978b) or MY agar (SS plus 0.01% each of sodium acetate, nutrient broth powder, and yeast extract, plus 0.03% Na₂S, and 1.5% agar; Larkin, 1980) in separate Petri dishes. Each dish is held on an angle so that the drops will flow across the agar surface. The excess is then withdrawn from the other side with the pipette. The dishes are incubated at about 25–30°C and are examined daily under transmitted light with the aid of a dissecting microscope. Colonies with a hairy or filamentous edge are transferred with sterile toothpicks to fresh media and are restreaked until pure.

Williams and Unz (1985a) transferred 1 ml amounts of activated sludge mixed liquor to standard test tubes containing 5 ml of solid media (0.15% glucose and/or 0.06% Na₂S·9H₂O plus 2.0% Bacto-agar) and an overlay of 15 ml of a mineral salts-vitamin solution (MSV).² Tubes are incubated at 22–25°C. Sur-

1. Salt solution contains: NH₄Cl, 0.02%; K₂HPO₄, 0.001%; MgSO₄, 0.001%; CaSO₄, 20 ml/l of a saturated solution; and trace element solution, 5 ml/l. pH is about 7.5. (Trace element solution contains: ZnSO₄·7H₂O, 0.001%; MnSO₄·4H₂O, 0.002%; CuSO₄·5H₂O, 0.00000005%; H₃BO₃, 0.001%; Co(NO₃)₂, 0.0001%; Na-molybdate, 0.0001%; EDTA, 0.02%; and FeSO₄·7H₂O, 0.07%.)

2. Mineral salts solution contains (per liter): (NH₄)₂SO₄, 0.5 g; MgSO₄·7H₂O, 0.1 g; CaCl₂·2H₂O, 0.05 g; K₂HPO₄, 0.11 g; KH₂PO₄, 0.085 g; FeCl₃·6H₂O, 0.002 g; EDTA, 0.003 g; and vitamin mix (Eikelboom, 1975), 1.0 ml. pH is about 7.5.

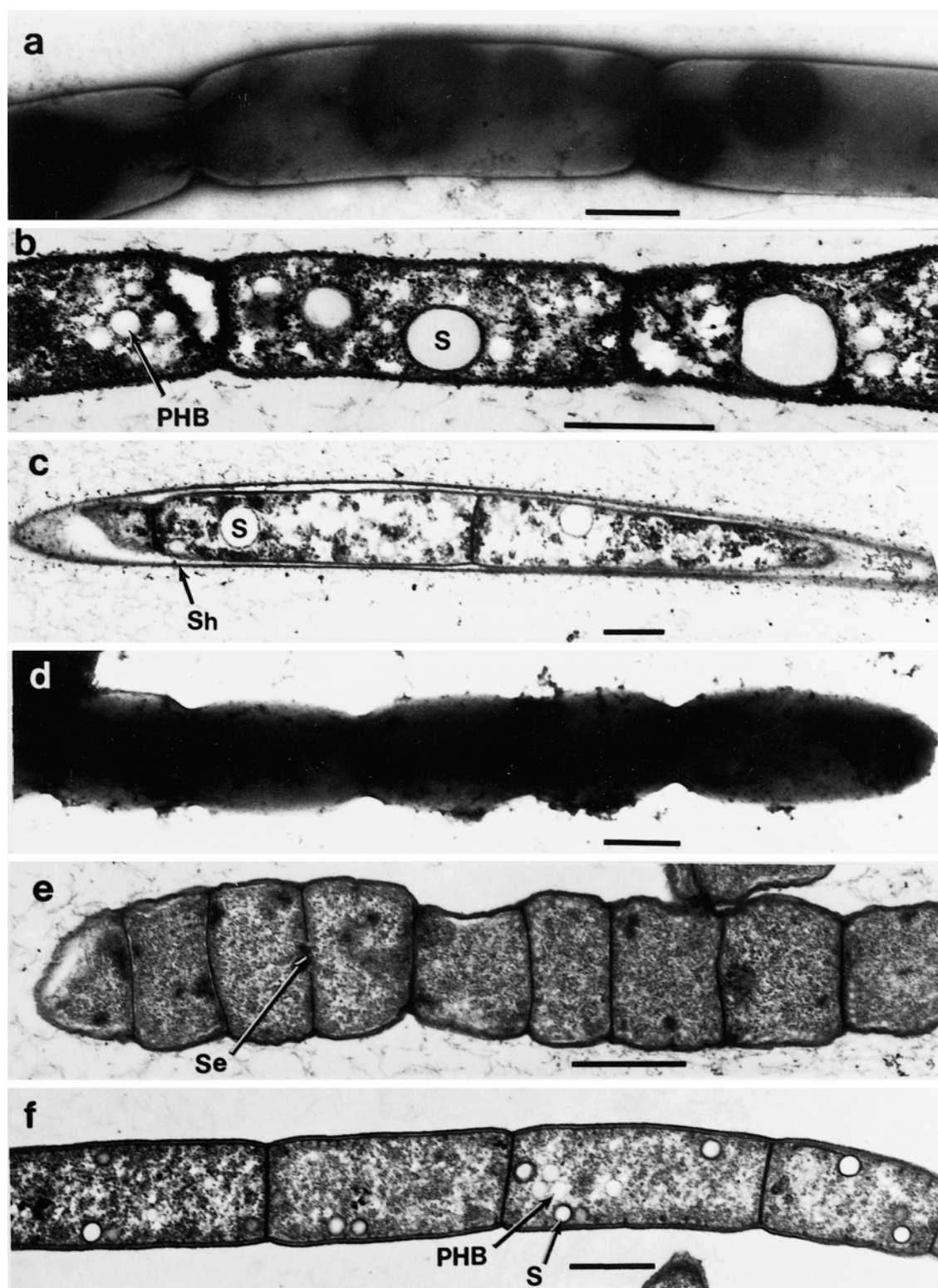


FIGURE BXII.γ.41. General ultrastructural characteristics of trichomes of *Thiothrix* species. *Thiothrix unzii* (a, whole cell; b, thin section); *Thiothrix fructosivorans* (c, thin section); *Thiothrix eikelboomii* (d, whole cell; e, f, thin section). Abbreviations: *Sh*, sheath; *Se*, septum; *S*, sulfur inclusion; *PHB*, poly-β-hydroxybutyrate. Bars = 1.0 μm. (Reproduced with permission from T.M. Williams et al., *Applied and Environmental Microbiology* 53: 1560–1570, 1987, ©American Society for Microbiology.)

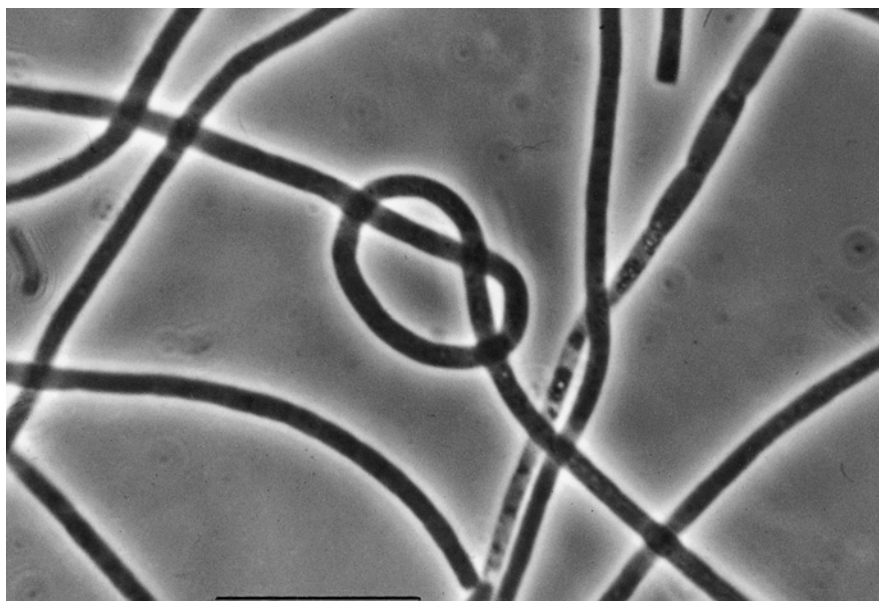


FIGURE BXII.γ.42. Knot produced in trichome by *Thiiothrix eikelboomii*. Bar = 10 μm. (Reproduced with permission from T.M. Williams and R.F. Unz, *Applied and Environmental Microbiology* 49: 887–898, 1985, ©American Society for Microbiology.)

face specimens are examined periodically by microscopy for evidence of the development of *Thiiothrix* trichomes. Isolation of *Thiiothrix* spp. is initiated by placing several loopfuls of enrichment culture films or dilute (10^{-1}) activated sludge in Petri dishes containing 7 ml of sterile MSV and, with the aid of a stereomicroscope (15–45 $\times$ magnification), selecting 40–50 single filaments or rosettes to be washed by serial transfer six to seven times in fresh MSV. Washed trichomes are streaked on glucose–sulfide³ or LT⁴ medium.

MAINTENANCE PROCEDURES

The type species can be maintained in semisolid (0.15% agar) deeps of either MP or MY medium, with transfers at intervals of about 3–4 weeks. Other *Thiiothrix* spp. survive monthly transfers in LTH⁵ medium (sodium lactate supplied at 1.0 g/l) when stored at 10°C and cryopreservation at –83 to –90°C.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Larkin (1989) employed MP broth or appropriate modifications to test sole carbon (0.05%), nitrogen (0.02 or 0.05%), and sulfur (0.03%) sources in support of sustained growth of *T. nivea* strains through three successive transfers. Other *Thiiothrix* species were nutritionally evaluated in MSV employing as a positive result the criterion of optical density equal to or greater than 4 $\times$ that of controls (Williams and Unz, 1989).

3. Glucose–sulfide medium contains (per liter): glucose, 0.15 g; $(\text{NH}_4)_2\text{SO}_4$, 0.5 g; $\text{Ca}(\text{NO}_3)_2$, 0.01 g; $\text{K}_2\text{HPO}_4 \cdot 7\text{H}_2\text{O}$, 0.05 g; KCl, 0.05 g; CaCO_3 , 0.1 g; $\text{Na}_2\text{S} \cdot 9\text{H}_2\text{O}$, 0.187 g; vitamin mix, 1.0 ml; and agar, 15.0 g, when needed. pH is about 7.5.

4. LT medium contains per liter of MSV: sodium lactate, 0.5 g; $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$, 0.5 g; and agar, 12.0 g. pH is about 7.5.

5. LTH medium contains per liter of MSV: sodium lactate, 0.5 g; $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$, 0.5 g; and HEPES buffer (*N*-2-hydroxyethylpiperazine-*N'*-2-ethanesulfonic acid), 0.01 M. pH is about 7.5.

DIFFERENTIATION OF THE GENUS *THIOTRICH* FROM OTHER GENERA

Recent inclusion of new species within *Thiiothrix* has exacerbated the problem of differentiation of the genus from other trichome-forming bacteria on morphological and physiological grounds. *Thiiothrix* differs from *Beggiatoa* in that the latter is less nutritionally diverse, does not form gonidia, and trichomes are not ensheathed or in rosettes although they exhibit a gliding motility. *Thiiothrix* is separated from *Thioploca* by the production of rosettes and the lack of gliding trichomes in the former and by the presence of multiple filaments within a single sheath and a multi-layered cell wall in *Thioploca*. The unavailability of axenic cultures of *Thioploca* precludes further physiological comparison with *Thiiothrix*. Analyses of 16S rDNA sequences recovered from purified *Thioploca* biomass collected from natural environments has shown that *Thioploca* spp. are distantly related to *Thiiothrix* spp., although both genera belong to the *Gammaproteobacteria* (Teske et al., 1995; Ahmad et al., 1999). *Thioploca* spp. are more closely related to *Beggiatoa* spp. than to *Thiiothrix* spp. (Teske et al., 1999; Ahmad et al., 1999). *Thiiothrix* may be differentiated from *Leucothrix*, which it resembles morphologically, by the presence of a sheath in some species of the former and the unequivocal ability to deposit sulfur globules. It should be noted, however, that some *Leucothrix* strains may deposit intracellular sulfur (Brock, 1992; Dul' tseva et al., 1996; Grabovich et al., 1999); hence, this property alone may be insufficient to distinguish *Leucothrix* from *Thiiothrix*. Although both genera exhibit gonidia formation, *Thiiothrix* produces gonidia from only the end of a trichome, whereas the entire trichome of *Leucothrix* may fragment into gonidia.

TAXONOMIC COMMENTS

The type species of *Thiiothrix* is *Thiiothrix nivea*. The original specimen for which the type species was named is nonexistent and *T. nivea* JP2 (DSM 5202) is the approved neotype strain. Sub-

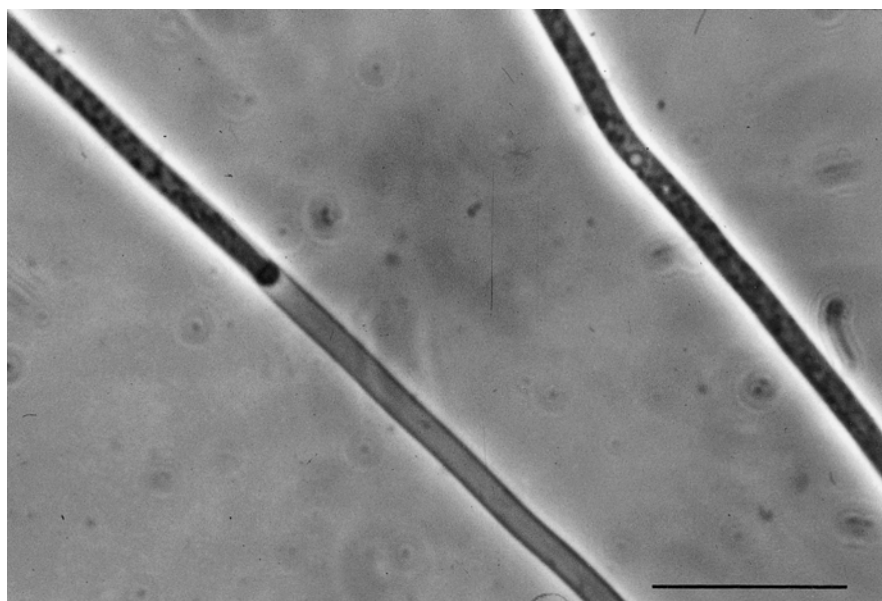


FIGURE BXII.γ.43. Sheath produced by trichome of *Thiothrix fructosivorans*. Bar = 10 μ m. (Reproduced with permission from T.M. Williams and R.F. Unz, *Applied and Environmental Microbiology* 49: 887–898, 1985, ©American Society for Microbiology.)

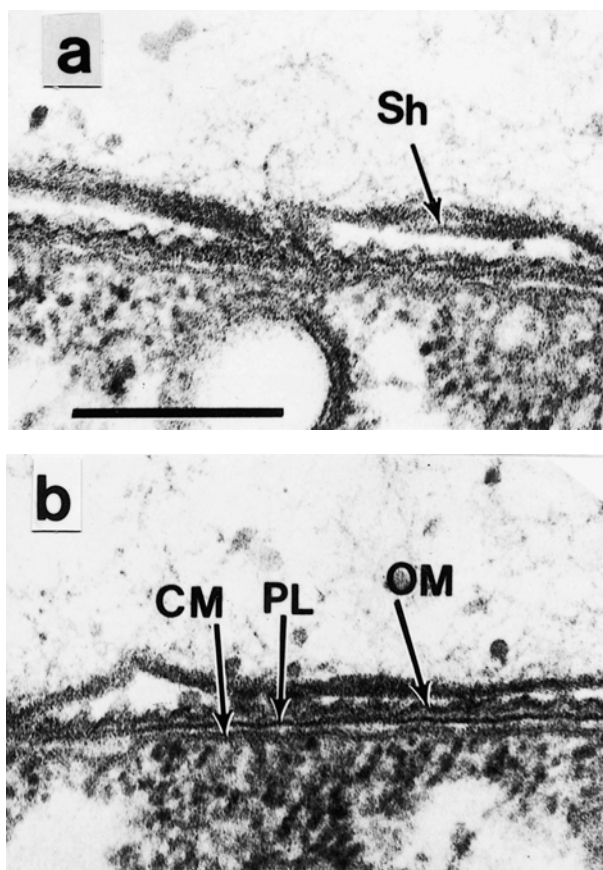


FIGURE BXII.γ.44. *a*, Ultrastructure of sheath surrounding trichome of *Thiothrix fructosivorans*. Abbreviation: Sh, sheath. Bar = 250 nm. *b*, Ultrastructure of cell in trichome of *Thiothrix fructosivorans*. Abbreviations: CM, cytoplasmic membrane; PL, peptidoglycan layer; OM, outer membrane. (Reproduced with permission from T.M. Williams et al., *Applied and Environmental Microbiology* 53: 1560–1570, 1987, ©American Society for Microbiology.)

sequent descriptions of *Thiothrix* spp. have largely been based on morphological criteria that have not progressed beyond descriptions at the genus level.

Morphology has been a primary criterion in the taxonomy of many filamentous bacteria and it has been suggested that this may be sufficient for the characterization of *Thiothrix* spp. (Polz et al., 1996). However, features considered diagnostic for *Thiothrix* spp. (holdfast and rosette formation and the production of gliding gonidia) are not exclusive as may be noted for *Leucothrix mucor* (Williams and Unz, 1985b; Brock, 1992) and the Eikelboom type 021N bacteria associated with bulking activated sludge. Woese (1987) has stressed that morphological characteristics are of little value as indicators of phylogenetic relationships among a majority of bacteria and axenic cultures of *Thiothrix* species have been observed to undergo change in morphology of trichomes (Shuttleworth and Unz, 1991, 1993; Brignon et al., 1995). Despite extensive use of *in situ* morphology and staining reactions to identify filamentous bacteria associated with activated sludge bulking and foaming (Eikelboom, 1975; Jenkins et al., 1993), these characteristics can vary depending on environmental conditions and are singularly of limited diagnostic value. Furthermore, bacterial morphotypes termed “*Leucothrix cohaerens*” (Pringsheim, 1957; Cyrus and Sladka, 1970) and *Thiothrix* sp. forms I and II (Farquhar and Boyle, 1971) have been equated with Eikelboom type 021N bacteria (Eikelboom, 1975). The taxonomic position of the polymorphic Eikelboom type 021N bacteria and their relationship to the genus *Thiothrix* has been questioned (Williams and Unz, 1985a, 1989; Ziegler et al., 1990). In recognition of the limitations of morphological characterization in the taxonomy of many filamentous bacteria, molecular biological approaches based on comparative 16S rRNA analyses have been used in the attempt to clarify relationships among such bacterial morphotypes.

Analysis of 16S rRNA sequences derived from six *Thiothrix* spp. (including one *Thiothrix* sp. form I) and 8 Eikelboom type 021N bacteria revealed that they fell within a single monophyletic

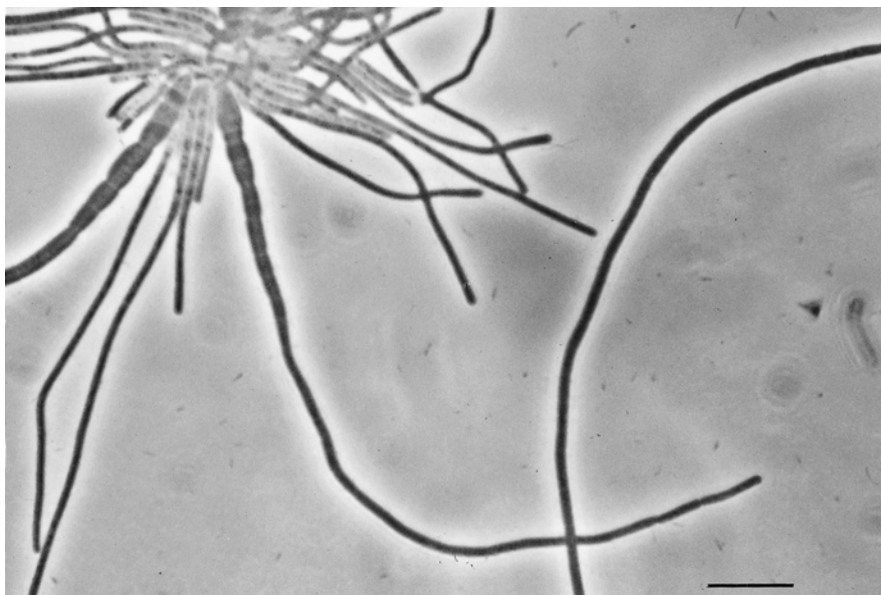


FIGURE BXII.γ.45. Base to tip differentiation in cell dimensions in a trichome of *Thiiothrix fructosivorans*. Bar = 10 μm. (Reproduced with permission from T.M. Williams and R.F. Unz, *Applied and Environmental Microbiology* 49: 887–898, 1985, ©American Society for Microbiology.)

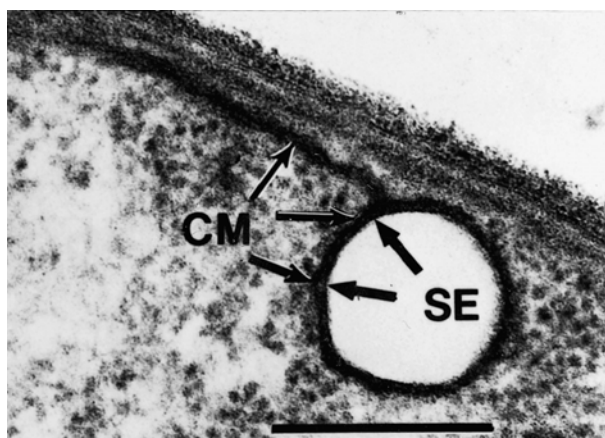


FIGURE BXII.γ.46. Sulfur inclusion in invaginated cytoplasmic membrane of a cell of *Thiiothrix eikelboomii*. Abbreviations: CM, cytoplasmic membrane; SE, sulfur inclusion envelope. Bar = 250 nm. (Reproduced with permission from T.M. Williams et al., *Applied and Environmental Microbiology* 53: 1560–1570, 1987, © American Society for Microbiology.)

group suggesting a clear relationship between Eikelboom type 021N and *Thiiothrix* spp. (Howarth et al., 1999). Consequently, four new *Thiiothrix* species have been validly described with sequence identities below 97.5% (Howarth et al., 1999) and include strains categorized as *incertae sedis* in the previous edition of the *Manual*. Strains previously categorized as Eikelboom type 021N are now accommodated within the species, *T. eikelboomii*, of which strain AP3 (ATCC 49788) is the type strain and a single *Thiiothrix* sp. form I is named *T. defluvii* (type strain Ben57). *T. unzii* is represented by type strain A1 (ATCC 49747) that, like *T. nivea*, requires a source of reduced sulfur and organic carbon for growth. *T. fructosivorans* is a chemoorganotrophic species rep-

resented by the type strain Q (ATCC 49748) and strain I (ATCC 49749), which is essentially identical to the type strain. Despite the close phylogenetic relationship between *T. fructosivorans*, “*T. ramosa*”, and *Thiiothrix* sp. CT3 (>97% 16S rRNA sequence identity), *T. fructosivorans* is not known to be capable of autotrophic growth, notwithstanding evidence of RubisCo activity (Unz and Williams, 1989), whereas “*T. ramosa*” and *Thiiothrix* sp. CT3 both grow autotrophically (Odintsova et al., 1993; Tandoi et al., 1994). Analysis of 16S rDNA sequences from a further 15 isolates, identified morphologically as Eikelboom type 021N, revealed the organisms to be most closely related to *Thiiothrix* species (Fig. BXII.γ.51), although considerable additional phylogenetic diversity was discovered (Kanagawa et al., 2000). In addition to organisms related to *T. nivea* (group 0), several isolates related to *T. eikelboomii* (group II sensu Kanagawa et al., 2000), a completely distinct group (group I sensu Kanagawa et al., 2000), and a group related to, but significantly divergent from, *T. defluvii* (group III sensu Kanagawa et al., 2000) were resolved. Although these isolates showed a considerable degree of divergence in their 16S rRNA sequences, they had very homogeneous mol% G + C content (43.9–46.1) and all shared a characteristic deletion in helix 18 of their ribosomal RNA (positions 455–477 by *E. coli* numbering), which further supports their relationship to named *Thiiothrix* spp. (Howarth et al., 1999; Kanagawa et al., 2000). Accordingly, groups I and III (Kanagawa et al., 2000) should be designated as new species.

During the preparation of this chapter the group of Takahiro Kanagawa described groups I and III as two new *Thiiothrix* species: *Thiiothrix disciformis* and *Thiiothrix flexilis* (Aruga et al., 2002). *Thiiothrix disciformis* corresponds to group I in Fig. BXII.γ.51. B3-1 is the type strain. *Thiiothrix flexilis* corresponds to group III in Fig. BXII.γ.51 and EJ2M-B is the type strain. These organisms are morphologically recognizable as Eikelboom Type 021N and filaments exhibit considerable variability in cell morphology with discoid and ovoid cells often being present, as are elongate and

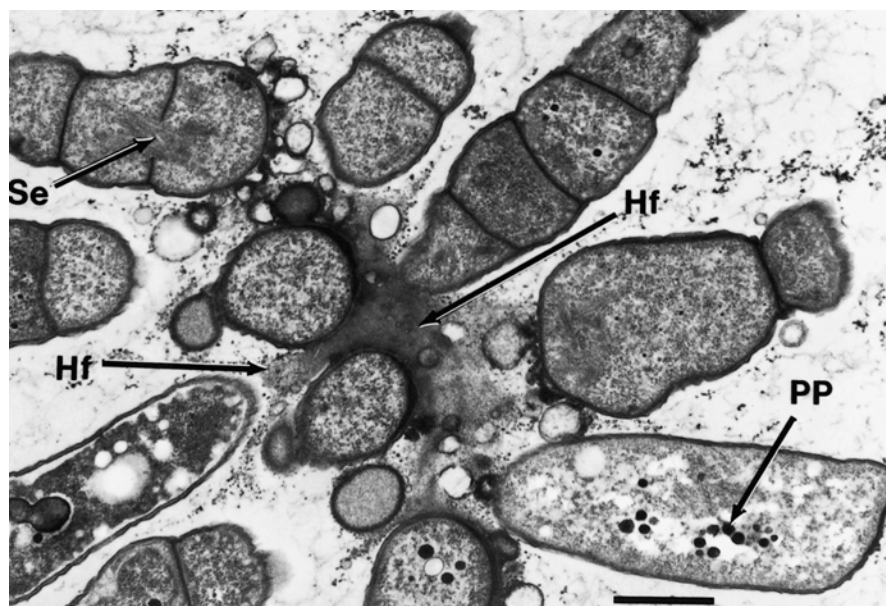


FIGURE BXII.γ.47. Thin section through rosette of *Thiothrix eikelboomii*. Abbreviations: *Hf*, holdfast material; *PP*, polyphosphate; *Se*, cell septum. Bar = 1.0 μm. (Reproduced with permission from T.M. Williams et al., *Applied and Environmental Microbiology* 53: 1560–1570, 1987, ©American Society for Microbiology.)

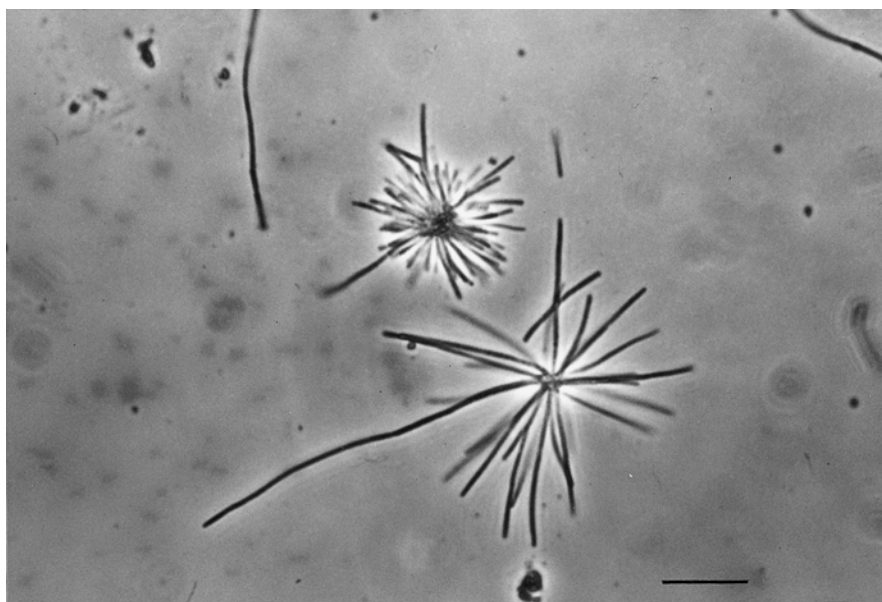


FIGURE BXII.γ.48. Rosettes of trichomes representative of *Thiothrix* spp. *Thiothrix eikelboomii*. Bar = 10 μm. (Reproduced with permission from T.M. Williams and R.F. Unz, *Applied and Environmental Microbiology* 49: 887–898, 1985, ©American Society for Microbiology.)

swollen cells. Although elemental sulfur is deposited, reduced sulfur is not required for growth. Phenotypically the species are quite similar but show differences in carbon source utilization, the extent of sulfur deposition, nitrate reducing ability, and cell dimensions. They are, nevertheless, phylogenetically very distinct (Fig. BXII.γ.51). Readers are directed to the original publication for details of the phenotypic and genotypic analyses of these newly described species (Aruga et al., 2002).

The different phylogenetic groupings within the genus *Thiothrix* cannot be distinguished morphologically (Kanagawa et al., 2000). This further calls into question the validity of identification of trichome-forming microorganisms solely on morphological criteria. This is consistent with observations made using fluorescence *in situ* hybridization (FISH) where not all organisms morphologically identified as *Thiothrix* spp. hybridized with probes targeting *T. nivea* or Eikelboom type 021N bacteria (Per-

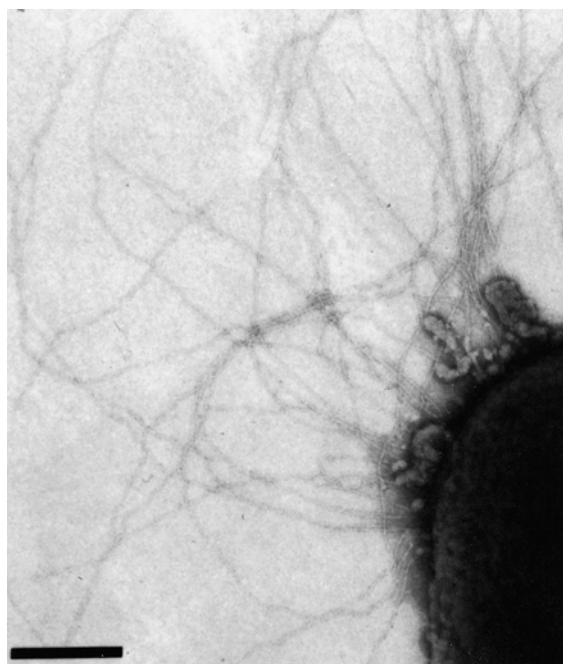


FIGURE BXII.γ.49. Fimbriae of a gonidium of *Thiothrix unzii*. Negative stained preparation. Bar = 0.5 μm. (Reproduced with permission from T.M. Williams et al., *Applied and Environmental Microbiology* 53: 1560–1570, 1987, ©American Society for Microbiology.)

nelle et al., 1998). Also, trichomes morphologically identified as Eikelboom type 021N may bind with TNI probe specific for *T. nivea* but not an Eikelboom type 021N-specific probe (Nielsen et al., 1998), although results may, in part, be due to the poor specificity of the probes that were designed based on limited available sequence data. Nevertheless, some organisms with *Thiothrix*/Eikelboom type 021N-like morphology have proven to be phylogenetically very distantly related to *Thiothrix* spp., e.g., strain Ben47, which is most closely related to the *Alphaproteobacteria* (Howarth et al., 1999). A series of probes, designed for use in FISH or other rRNA hybridization assays to identify different groups of *Thiothrix* in environmental samples, are listed with their specificity indicated in Table BXII.γ.36. Readers are directed to the original publication for detailed information on the application of these diagnostic oligonucleotide probes (Kanagawa et al., 2000). Characteristics distinguishing *Thiothrix* from morphologically similar bacteria are listed in Table BXII.γ.37.

ACKNOWLEDGMENTS

The authors acknowledge the original contributions of John Larkin contained in the contents of this writing and are most grateful to Takahiro Kanagawa for providing data on *Thiothrix disciformis* and *Thiothrix flexilis* prior to publication.

FURTHER READING

Larkin, J.M. and W.R. Strohl. 1983. *Beggiatoa*, *Thiothrix*, and *Thioploca*. *Annu. Rev. Microbiol.* 37: 341–367.

List of species of the genus *Thiothrix*

1. ***Thiothrix nivea*** (Rabenhorst 1865) Winogradsky 1888, 39^{AL}
(*Beggiatoa nivea* Rabenhorst 1865, 94.)
ni' ve.a. L. adj. *nivea* snow-white.

Rods, about 1.0–1.5 μm in diameter, seriate in multicellular filaments (trichomes) of uniform diameter. Gram negative. Trichomes clearly ensheathed. Optimum temperature

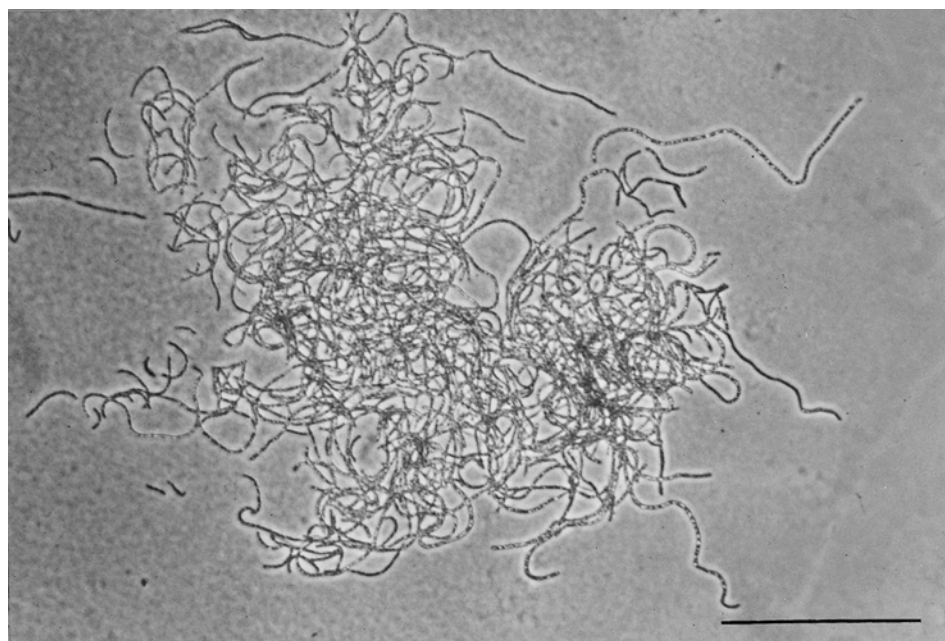


FIGURE BXII.γ.50. Colony morphology exemplified by *T. unzii*. LT medium. Bar = 60 μm. (Reproduced with permission from T.M. Williams and R.F. Unz, *Applied and Environmental Microbiology* 49: 887–898, 1985, ©American Society for Microbiology.)

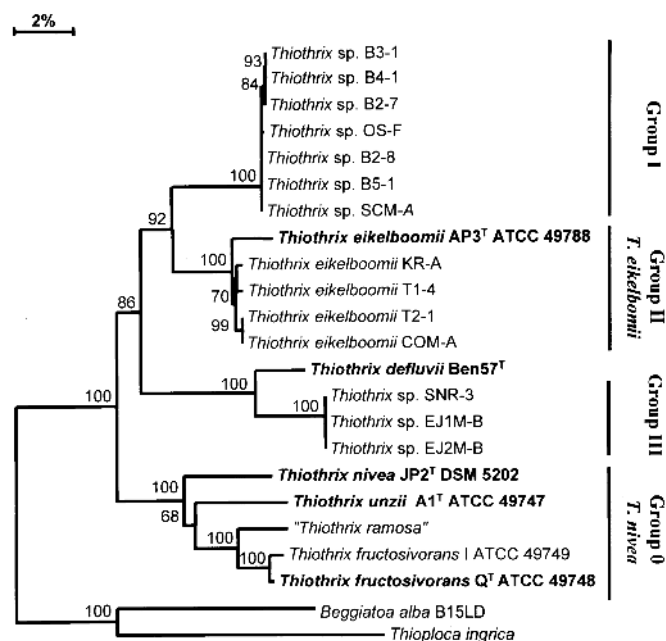


FIGURE BXII.γ.51. Phylogenetic distance tree derived from 16S rRNA sequences of *Thiiothrix* spp. The tree was generated from an alignment comprising 1148 unambiguously aligned nucleotides. Figures at nodes represent percentage bootstrap values. Bootstrap values below 50% have been omitted. Sequences from *Beggiatoa alba* B15LD and *Thioploca ingrica* were used as an outgroup to root the tree. The type strains of all validly named species are shown in bold typeface. Bar = 2 nucleotide changes per 100 nucleotides in the 16S rRNA sequence.

is 25–30°C; maximum about 32–34°C; minimum about 6–8°C. Mixotrophic. Additional characteristics of the neotype strain, which are representative of the species, appear in

the description of the genus and the data in Table BXII.γ.38.

The mol% G + C of the DNA is: 52 (T_m).

Type strain: JP2, DSM 5205.

GenBank accession number (16S rRNA): L40993.

2. *Thiiothrix defluvii* Howarth, Unz, Seviour, Blackall, Pickup, Jones, Yaguchi and Head 1999, 1824^{VP}

de.flu'vi.i. L. neut. n. *defluvium* sewage; L. gen. n. *defluvii* of sewage.

Rods, variable in shape (cylindrical, barrel-shaped, frequently elongate and swollen) in multicellular trichomes with base to tip differentiation; apical cells are 1.0–2.0 µm in diameter and 5.0–10.0 µm in length, whereas cells at the base of trichomes are 2.0–4.0 µm in diameter and 0.5–10 µm in length. Trichomes have no sheath but may form knots; rosettes, holdfasts, and gliding gonidia present in some strains. Gram negative or Gram variable. Growth occurs in the temperature range of 10–30°C but not at 4°C or 37°C. Isolates are extremely slow growing and biochemical properties of this organism have not been determined.

Isolated from activated sludge treatment plant located in Australia

The mol% G + C of the DNA is: unknown.

Type strain: Ben57.

GenBank accession number (16S rRNA): AF127020.

3. *Thiiothrix disciformis* Aruga, Kamagata, Kohno, Hanada, Nakamura and Kanagawa 2002, 1315^{VP}

dis.ci.for'mis. L. masc. n. *discus* a disc; L. n. *forma* shape; N.L. adj. *disciformis* disc-shaped, the main cell form.

Readers are referred to the original publication for a detailed description of this species (Aruga et al., 2002).

The mol% G + C of the DNA is: 43.9–44.7 (HPLC).

TABLE BXII.γ.36. Diagnostic oligonucleotide probes for characterization of *Thiiothrix* spp.^a

Probe	Target Group	Sequence (5' → 3')	Target site (16S rRNA, <i>E. coli</i> numbering)
G1B	Group I	TGTGTTTCGAGTTCCTTGC	1029–1046
G2M	Group II	GCACCACCGACCCCTTAG	842–859
G3M	Group III ^b	CTCAGGGATTCTGCCAT	996–1013
G123T	Group 0-III	CCTTCCGATCTCTATGCA	697–714

^aData from Kanagawa et al., 2000.

^bIn addition to group III organisms *T. eikelboomii* AP3 also contains the target site for this probe.

TABLE BXII.γ.37. Characteristics used to differentiate *Thiiothrix* spp. from related, morphologically similar genera^{a,b}

Characteristic	<i>Thiiothrix</i>	<i>Beggiatoa</i>	<i>Leucothrix</i>	<i>Thioploca</i>
Colorless	+	+	+	+
Strains deposit elemental sulfur	+	+	– ^c	+
Sheath	+	–	–	+
Holdfast	+	–	+	–
Gliding gonidia	+	–	+	–
Filament formation	+	+	+	+
Rosette formation	+	–	+	–
Swollen cells produced in filaments	+	–	+	–
Knots produced	+	–	+	–
mol% G + C	43–53	35–39	46–51	nd
Deletion in helix 18 of 16S rRNA	+	–	–	–

^aNote that these characteristics are illustrative only and many isolates may not exhibit all of these characteristics.

^bFor symbols see standard definitions; nd, not determined.

^cMost reports suggest that *Leucothrix* spp. do not deposit intracellular sulfur; an organism described as "*Leucothrix thiophila*" has been shown to grow lithoheterotrophically and deposit elemental sulfur (Dul' tseva et al., 1996).

TABLE BXII.γ.38. Characteristics of the type strains of species of the genus *Thiothrix*^a

Characteristic	1. <i>T. nivea</i> ^b DSM 5205 (JP2)	2. <i>T. eikelboomii</i> ^c ATCC 49788 (AP3)	3. <i>T. fructosivorans</i> ^c ATCC 49748 (Q)	4. <i>T. unzii</i> ^c ATCC 49747 (A1)
<i>Utilization as sole carbon source:</i> ^{d,e}				
Acetate, malate, oxalacetate, pyruvate	+	+	+	+ (w, malate)
Formate, fumarate, lactate, propionate, succinate	—	+	+	+
α-Ketoglutarate, butyrate, hydroxybutyrate, valerate, benzoate		+	—	—
D-Fructose	—	+	+	—
D-Glucose	—	+	—	—
D-Galactose		+	—	—
D-Mannose	—	w	—	—
Sucrose	—	+	+	—
D-Maltose	—	+	—	—
D-Trehalose		+	—	—
D-Melezitose		+	—	—
D-Raffinose		w	—	—
Amygdalin, salicin		+	—	—
D-Mannitol,		+	—	—
i-Inositol,		w	—	—
D-Sorbitol		w	—	—
<i>Utilization as carbon and nitrogen source:</i> ^{d,f}				
L-Glutamate	—	+	—	—
L-Tyrosine, L-histidine, DL-alanine, Casamino acids		+	—	—
<i>Utilization as sole nitrogen source:</i> ^{d,g}				
DL-Tryptophan, DL-aspartate, DL-glutamine, L-asparagine, L-glutamate, Casamino acids		+	+	+
Hydroxy-L-proline		—	—	+
DL-Methionine, DL-valine, DL-isoleucine		+	—	+
L-Tyrosine, L-histidine		+	—	w
L-Cysteine, L-proline		+	+	—
L-Arginine, L-leucine, DL-phenylalanine		+	—	—
Glucosamine		—	+	+
Urea		+	+	+
Ammonia	+	+	+	+
Nitrate	+	—	+	+
NO ₃ [−] → NO ₂ [−] only	+	+	+	+
Utilization of orthophosphate and Na ₅ P ₃ O ₁₀		+	+	+
Polyphosphate stored		+	+	+
Sudanophilic granules formed		+	+	+
Oxidase	+	+	+	+
Catalase	—	—	+	—
Caseinase		—	—	—
Gelatinase		+	+	+
Sulfide or thiosulfate required sulfur source	+	—	—	+

^aSymbols: See standard definitions; w, weak.^bData obtained from Larkin (1989).^cData obtained from Williams (1985).^dWith sulfide or thiosulfate present in the case of *T. nivea* and *T. unzii*.^eCarbon sources not utilized by type strains in addition to those indicated in table: *T. nivea* (citrate, *cis*-aconitate, phosphoenolpyruvate, carbonate, glyoxylate, methylamine, D-ribose, D-arabinose, D-cellobiose, L-rhamnose, D-raffinose, D-sorbose, D-melebiose, D-xylose, ethanol, methanol, propanol, glycerol, i-erythritol); *T. unzii*, *T. fructosivorans*, and *T. eikelboomii* (citrate, L-rhamnose, D-xylose, L-fucose, D-ribose, mannoheptulose, L-sorbose, D-arabinose, glyceraldehyde phosphate, D-melibiose, D-gentiobiose, D-cellobiose, starch, inulin, esculin, glycerol, erythritol, galacturonic acid, glucuronic acid, gluconic acid, tributyrin, Tween 80, ethanol, *n*-propanol, *n*-butanol, *n*-amyl alcohol, methanol, isoamyl alcohol, phenol, *m*-toluate, glycolate, oleate, palmitate, myristate).^fCarbon and nitrogen sources not utilized by type strains in addition to those indicated in table: *T. unzii*, *T. fructosivorans*, and *T. eikelboomii* (DL-lysine, DL-cystine, DL-threonine, hydroxy-L-proline, glycine, L-arginine, DL-tryptophan, L-cysteine, L-proline, DL-phenylalanine, DL-serine, DL-methionine, DL-valine, DL-aspartate, DL-glutamine, L-asparagine, DL-isoleucine, L-leucine).^gNitrogen sources not utilized by type strains in addition to those indicated in table: *T. nivea* (nitrite, dinitrogen); *T. unzii*, *T. fructosivorans*, and *T. eikelboomii* (nitrite, L-lysine, L-cystine, DL-threonine, glycine, DL-serine, DL-alanine).

Type strain: B3-1, DSM14473, JCM11364.

GenBank accession number (16S rRNA): AB042532.

4. ***Thiothrix eikelboomii*** Howarth, Unz, Seviour, Blackall, Pickup, Jones, Yaguchi and Head 1999, 1825^{VP}
ei.kel.boom' i.i. M.L. gen. n. *eikelboomii* of Eikelboom; named for D.H. Eikelboom, who pioneered morphological identification of filamentous bacteria in wastewater.

Rods may vary in shape (cuboidal, barrel-shaped, cylindrical, discoid, bead-like) and size depending on location in multicellular trichomes; apical cells are 0.6–0.8 × 1.0–1.5 μm and frequently bead-like, whereas cells at the base of trichomes are 1.0–3.0 μm in diameter and 0.4–0.7 μm in length and discoid. Trichomes have no sheath but may form knots; rosettes and a holdfast present in some strains.

The type strain does not form rosettes. Gliding gonidia occur only in rosette-forming strains and have a tuft of monopolar fimbriae but lack flagella. Gram negative or Gram variable. Growth occurs in the temperature range of 10–33°C but not at 37°C and in the pH range of 6.5–8.5. Chemoorganotrophic. Additional characteristics of the type strain, which are representative of the species, are given in Table BXII.γ.38.

Isolated from activated sludge treating domestic wastewater.

The mol% G + C of the DNA is: 44.1–46.1 (HPLC).

Type strain: AP3, ATCC 49788.

GenBank accession number (16S rRNA): AB042819.

5. **Thiothrix flexilis** Aruga, Kamagata, Kohno, Hanada, Nakamura and Kanagawa 2002, 1315^{VP}
fle.xi'lis. L. adj. *flexilis* pliable.

Readers are referred to the original publication for a detailed description of this species (Aruga et al., 2002).

The mol% G + C of the DNA is: 44.0–44.4 (HPLC).

Type strain: EJ2M-B, DSM14609, JCM11135.

GenBank accession number (16S rRNA): AB042545.

6. **Thiothrix fructosivorans** Howarth, Unz, Seviour, Blackall, Pickup, Jones, Yaguchi and Head 1999, 1824^{VP}
fruc.to.si.vo'rans. M.L. neut. n. *fructosum* fructose; L. part. pres. *vorans* eating; M.L. adj. *fructosivorans* fructose-eating.

Rods, 1.2–2.5 × 2.7–4.5 µm, in multicellular trichomes. Gram negative, trichomes ensheathed, rosettes formed,

holdfast present, gliding gonidia with monopolar fimbriae but no flagella. Growth occurs in the temperature range of 4–28°C but not at 33°C and in the pH range of 6.5–8.5. Chemoorganotrophic. Additional characteristics of the type strain, which are representative of the species, appear in Table BXII.γ.38.

Isolated from activated sludge treating domestic wastewater.

The mol% G + C of the DNA is: unknown.

Type strain: Q, ATCC 49748.

GenBank accession number (16S rRNA): L79962.

7. **Thiothrix unzii** Howarth, Unz, Seviour, Blackall, Pickup, Jones, Yaguchi and Head 1999, 1824^{VP}
un'zi.i. M.L. gen. n. *unzii* of Unz; named for R.F. Unz.

Rods, 0.7–1.5 × 1.5–3.0 µm, in multicellular trichomes. Gram negative, trichomes without sheaths, rosettes formed, holdfast present, gliding gonidia with monopolar fimbriae but no flagella. Growth occurs in the temperature range of 4–33°C but not at 37°C and in the pH range of 6.5–8.5. Mixotrophic. Additional characteristics of the type strain, which are representative of the species, appear in Table BXII.γ.38.

Isolated from activated sludge treating domestic wastewater.

The mol% G + C of the DNA is: unknown.

Type strain: A1, ATCC 49747.

GenBank accession number (16S rRNA): L79961.

Other Organisms

Two species of *Thiothrix*, “*T. ramosa*” (Odintsova and Dubinina, 1990) and “*T. arctophila*” (Dul’ tseva and Dubinina, 1994) are proposed but invalidly published. The general morphology of the illegitimate species befits the traditional description of the genus *Thiothrix* with respect to presence of nonmotile, multicellular trichomes and the ability to form gonidia and rosettes. However, in addition to gonidia, both species form gliding hormogonia through fragmentation of trichomes, and the hormogonia of “*T. ramosa*” may develop into pseudobranching filaments. Further, “*T. ramosa*” exhibits chemolithoautotrophic growth with thiosulfate or carbon disulfide as sole electron donors, mixotrophic growth on lactate and any of several organic sulfur compounds, including carbon disulfide, and chemoorganotrophic growth on lactate alone (Odintsova et al., 1993). Additional information on these organisms and other *Thiothrix* isolates is given in the following paragraphs.

1. “*T. ramosa*”

ra.mo'sa. L. n. *ramosum* branch.

Three strains, namely TA, T1, and T2, are known. Cells 0.7–2.5 × 2.5–5.0 µm in ensheathed trichomes up to 100 µm in length. Gram negative. Intracellular sulfur inclusions formed in media containing thiosulfate or sulfide; however, reduced sulfur is not required for growth. Strictly aerobic. Optimum growth temperature is 28°C with sparse growth at 20°C and no growth at 4°C. Optimum growth pH is 7.2–7.3. All strains utilize acetate, malate, succinate, fumarate, and oxalacetate, whereas isocitrate, glycolate, lactate, α-ketoglutarate, and mannitol utilization varies among strains. Sugars, alcohols, and amino acids are not utilized with the exception of mannitol by strain T2. No growth on peptone,

gelatin, casein, or yeast extract. Vitamins are required for growth. Nitrogen sources for all strains include ammonium salts, peptone, and glutamine; asparagine and aspartate are utilized by strain T1. Nitrates are not reduced to nitrite. Catalase and oxidase positive. Obtained from a hydrogen sulfide spring in Latvia.

The mol% G + C of the DNA is: 48–52.4.

2. “*Thiothrix arctophila*”

arcto'phi'la. L. adj. *arcto* cool; Gr. adj. *philus* loving; M.L. *arctophila* cool loving.

Only strain IN is known. Cells 0.7–2.6 × 3.0–5.0 µm in trichomes of variable length covered with a protein–polysaccharide sheath. Gram negative. Intracellular sulfur inclusions formed in media containing thiosulfate, sulfide, tetrathionate, or elemental sulfur. Strictly aerobic. Optimum growth temperature is 15°C with no growth at 3°C and 34°C. Optimum growth pH is 7.2. Carbon sources supporting growth are acetate, malate, pyruvate, succinate, fumarate, lactate, oxalacetate, propionate, maltose, melezitose, and butanol. Compounds not supporting growth are malonate, citrate, glyoxylate, formate, benzoate, caproate, glycolate, α-ketoglutarate, carbonate, glucose, lactose, fructose, sucrose, ribose, mannose, arabinose, cellobiose, rhamnose, raffinose, xylose, galactose, trehalose, gentiobiose, melibiose, sorbitol, methanol, ethanol, propanol, asparagine, methionine, serine, glutamate, alanine, aspartate, methylamine, glycerol, cysteine, mannitol, peptone, yeast extract, starch, Tween 80, urea, and phenol. Indole not produced from tryptophan. Nitrogen sources utilized are ammonia, nitrate, nitrite, glutamine, casein hydrolysate,

and peptone. Nitrogen sources not utilized are valine, glycine, phenylalanine, lysine, leucine, methionine, proline, cysteine, asparagine, aspartate, glutamate, urea, and dinitrogen. Nitrate is reduced to nitrite. Vitamins are required for growth. Nitrogen sources for all strains include ammonium salts, peptone, and glutamine; asparagine and aspartate are utilized by strain T1. Nitrates are not reduced to nitrite. Catalase negative and oxidase positive. Obtained from a combined wastewater treatment plant.

The mol% G + C of the DNA is: 55.0 (T_m).

3. *Thiothrix* CT3.

Thiothrix CT3 is a poorly characterized isolate recovered from an activated sludge in Italy and reported to be facultatively chemolithoautotrophic on thiosulfate, mixotrophic on acetate and thiosulfate, and chemoorganotrophic on acetate (Tandoi et al., 1994).

Several isolates of *Thiothrix* and Eikelboom type 021N, in addition to the type strains, were isolated and studied in

the course of investigations leading to the naming of the species *T. unzii*, *T. fructosivorans*, and *T. eikelboomii*. Some of these may no longer be extant; however, certain ones are referred to in publications by other workers and are described in the last edition of the *Manual*. Strain A3 is essentially identical to the type strain of *T. unzii* and strain I (ATCC 49749) is essentially identical to the type strain of *T. fructosivorans*. Isolates TH1 and TH3 share many features of *T. fructosivorans* but are distinguished by the inability of the latter strain to utilize as many organic carbon compounds or any carbohydrates, ability to utilize several more amino acids and nitrate as nitrogen sources, display cell dimensions similar to *T. unzii*, require sodium ion for growth, and tolerate relatively high (3–4%) sodium chloride concentrations. A large number of Eikelboom type 021 N isolates were included in studies on the type strain of *T. eikelboomii* (Williams, 1985; Williams and Unz, 1985a, 1989), and notwithstanding minor morphological and nutritional differences, appear to befit the description of *T. eikelboomii*.

Genus II. *Achromatium* Schewiakoff 1893, 1^{AL}

HANS-DIETRICH BABENZIE, FRANK OLIVER GLÖCKNER AND IAN M. HEAD

A.chro.ma'ti.um. Gr. pref. *a* not; Gr. n. *chromatium* color, paint; M.L. neut. n. *Achromatium* that which is not colored.

Spherical to ovoid or cylindrical unicells with hemispherical ends. In natural populations, cells of widely different sizes appear. Cells vary in size from spheres of about 10 μm in diameter to giant cylindrical forms up to $30 \times 125 \mu\text{m}$ (extremes are dividing stages); **cells are on average $15 \times 50 \mu\text{m}$.** Many size classes may be present within a sample. However, genetically different subpopulations that overlap in their dimensions can occur in a single habitat. **Division of cells by binary constriction. Gram negative. Calcium carbonate (calcite) is deposited intracellularly.** Additionally, the **cells contain globules of S^0 .**

Movement, if any, of a peculiar slow, rolling, jerky type or gliding on solid surfaces. Not motile by means of flagella. The **cell surface** is covered by a glycocalyx. **Catalase negative.** Pure cultures not available.

The **natural habitat is the oxic–anoxic boundary** in the top 0–5 cm of sediments, suggesting that the cells may use **oxygen as the electron acceptor**. The habitats for *Achromatium* include different aquatic ecosystems around the world, e.g., lakes, ponds, wetlands, bog lakes, rivers, saline environments, brackish waters, and marine environments. **Resting stages are not known.**

Phylogenetic analyses including sequences from different geographical locations confirm that *Achromatium* (e.g., *A. oxaliferum* populations) **forms a distinct, deeply branching monophyletic clade in the Gammaproteobacteria** (Fig. BXII.γ.52).

The topology of the clade indicates that beyond the two main clusters, A and B, at least three genetically distinct lineages exist (Fig. BXII.γ.52), and it is likely that several *Achromatium* spp. remain to be described. Several 16S rRNA-targeted oligonucleotide probes are available for *in situ* identification and differentiation of co-existing *Achromatium* spp. by techniques like fluorescence *in situ* hybridization (FISH).

The type species was originally found in the mud of the Neuhofer Altrhein, Germany.

The mol% G + C of the DNA is: not known.

Type species: *Achromatium oxaliferum* Schewiakoff 1893, 1.

FURTHER DESCRIPTIVE INFORMATION

From experiments with cell enrichments, it is now apparent that members of the genus encompass more than one physiological type. Some of them are capable of oxidizing reduced sulfur species to sulfate, suggesting that they obtain energy from this process, in which S^0 is deposited as an intracellular intermediate. The substrate spectrum appears to vary in different populations, and substrate utilization may involve a chemolithoautotrophic potential (assimilation of ^{14}C -labeled bicarbonate) or the uptake of organic C-compounds (e.g., acetate, protein hydrolysate) by a chemoorganoheterotrophic, chemolithoheterotrophic, or mixotrophic type of metabolism. Like the genus *Beggiatoa*, to which it is related, *Achromatium* appears to consist of a nutritionally heterogeneous group of bacteria. Without pure cultures, it is not possible to determine whether this physiological diversity exists within a single population, with induction or repression of different metabolic pathways under different environmental conditions, or if it instead corresponds with genetic heterogeneity. Twenty nearly full-length 16S rRNA sequences and three partial sequences, which were amplified from DNA extracted from highly purified cell enrichments from fresh water sediments of various habitats, have been deposited in public databases. The accession numbers are L42543 and L48222–L48227 (Head et al., 1996), AJ010593–AJ010596 (Glöckner et al., 1999) and AF129548–AF129559 (Gray et al., 1999).

The prevailing temperature at their natural habitats is around 10°C.

The use of comparative 16S rRNA analysis has clarified the position of *Achromatium* and has shown that natural communities of *Achromatium oxaliferum* comprise genetically distinct subpopulations. With this information, the physiological diversity of *Achromatium* spp. becomes understandable.

The *Achromatium oxaliferum*-derived sequences are consistently recovered with the *Chromatium* assemblage but are without strong

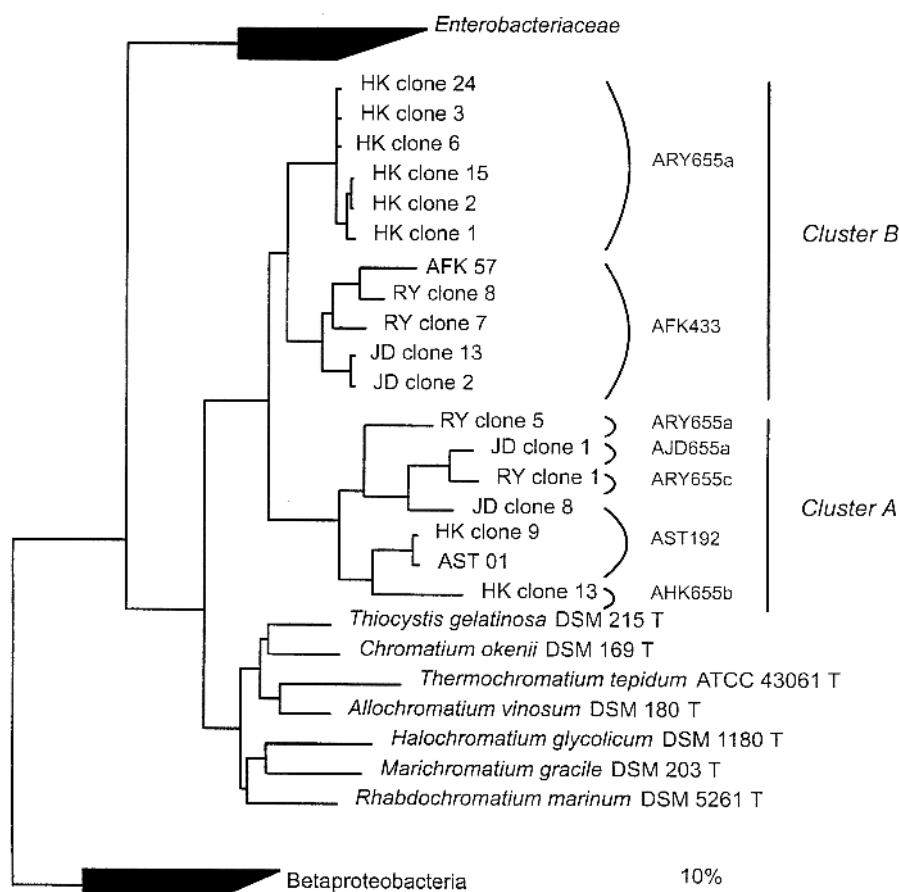


FIGURE BXII.γ.52. Phylogenetic tree based on comparative analysis of the 16S rRNA genes from all currently available full-length sequences from *Achromatium* and from selected sequences among the *Betaproteobacteria* and *Gammaproteobacteria*. The tree is a consensus tree based on maximum likelihood analysis, with highly variable positions excluded from the calculations. Sequences were obtained from *Achromatium* cells purified from different locations, including Hell Kettles (Co. Durham, UK), Rydal Water (Cumbria, UK), Jenny Dam (Cumbria, UK), Lake Fuchskuhle (Brandenburg, Germany), and Lake Stechlin (Brandenburg, Germany). These locations are designated with the suffixes HK, RY, JD, AFK, and AST, respectively. It has been proposed that the *Achromatium* spp. obtained from Lake Fuchskuhle (sequence AFK 57) is a new species *Candidatus Achromatium minus*. T, type strains. Bar = 10% estimated sequence divergence.

affiliation with any of the recognized lineages within this group. *A. oxaliferum* represents a novel clade within the *Gammaproteobacteria* and is therefore unrelated to the gliding bacteria of the *Cytophaga-Flavobacterium* group, as had been inferred prior to recent phylogenetic studies (La Riviere and Schmidt, 1989). 16S rRNA-sequences recovered from *Achromatium* cells from several locations have demonstrated that within the clade, two distinct phylogenetic clusters, termed cluster A and cluster B, are present. This bifurcation is strongly supported by distance analysis and maximum likelihood analysis ($P < 0.01$). Support for cluster A by parsimony analysis is also high, while support for cluster B is less pronounced. The two clusters can also be distinguished based on secondary structure elements (helical motifs) in the V6 region of the 16S rRNA molecule. Sequences from cluster A have a characteristic deletion and lack helix 38 (1024–1036, *E. coli* numbering), as do sequences from a number of *Archaea* and *Eucarya*. In contrast, sequences in cluster B have a V6 region that is in accordance with that of the majority of the domain *Bacteria* (Gray et al., 1999; Glöckner et al., 1999). Cluster B comprises two genetically different subgroups.

Whole-cell hybridization with fluorescently-labeled oligonu-

cleotide probes specific for the different *Achromatium*-derived sequences has clearly demonstrated the co-existence of distinct subpopulations. In Lake Stechlin, Germany, a ratio of about 2:1:1 has been found for three different *Achromatium* populations identified by FISH, with the organism represented by 16S rDNA clone AST 01 accounting for nearly 50% of the total population.

In the habitats investigated using FISH, Lake Fuchskuhle (an acidic bog lake in Germany) was the only lake found to harbor a morphologically and genetically homogeneous population of *Achromatium*. Phylogenetic analysis of the 16S rDNA has assigned this population to cluster B of the *Achromatium* clade. Based on the unique features of the habitat, the homogeneous morphology and cell size, and the phylogenetic position of the population, a new species *Candidatus Achromatium minus* has been proposed (Figs. BXII.γ.52 and BXII.γ.54).

Unfortunately, until now there have been no probes available that cover the whole *Achromatium* clade or allow an exact separation of the two clusters by FISH. With the six probes shown in Fig. BXII.γ.52, it is possible to distinguish several subclusters and clones or, when probes are used in combination, to detect all currently available *Achromatium* sequences. Furthermore, with

probes ARY655a and AFK433, it should be possible to distinguish cluster B from cluster A, excluding clone 5 derived from Rydal Water, UK. In addition, the combination of probes AJD655a, ARY655c, AST192, and AHK655b is specific for cluster A, again with the exception of clone 5 from Rydal Water. All probes should work under the same conditions and must be applied with appropriate competitor probes as described in Glöckner et al. (1999) and Gray et al. (1999).

Morphological heterogeneity with extreme differences in cell size (Figs. BXII.γ.53 and BXII.γ.54) has been noted by all investigators, but evidence that differently sized cells represent genetically different populations has been obtained only recently. Different sub-populations coexist within the same habitat (Head et al., 2000b).

The cell volume can reach $50,000\text{--}80,000\text{ }\mu\text{m}^3$, and the abundance of *Achromatium* cells in sediments is in the range of $10^3\text{--}10^5\text{ cells/cm}^3$. Because of the large size of individual cells, *Achromatium* populations can be the major component of the bacterial community in some surficial sediments in terms of biomass. Cells divide by constriction in the middle, and before separation takes place, the daughter cells are connected by a hyaline “bridge” (Fig. BXII.γ.55). Often, *Achromatium* populations share the same habitat with other sulfide-oxidizing bacteria, like *Beggiatoa* spp. (Babenzien, 1991).

The most striking feature of the cells is the intracellular deposition of calcium carbonate crystals (calcite), a property unique in the *Bacteria*. These inclusions can reach $5\text{ }\mu\text{m}$ in size (Fig. BXII.γ.56) and the mean calcium content is approximately

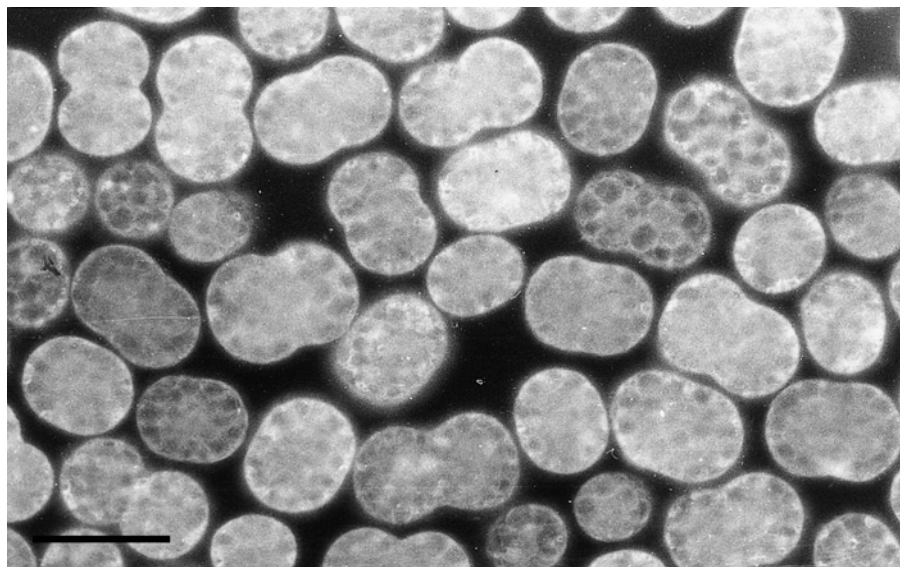


FIGURE BXII.γ.53. *Achromatium* cells collected from sediment of Lake Fuchskuhle (acidic bog lake, Brandenburg, Germany). In contrast to all other habitats this lake harbors a population that is homogeneous in morphology and size. Dark-field micrograph. Bar = $25\text{ }\mu\text{m}$.

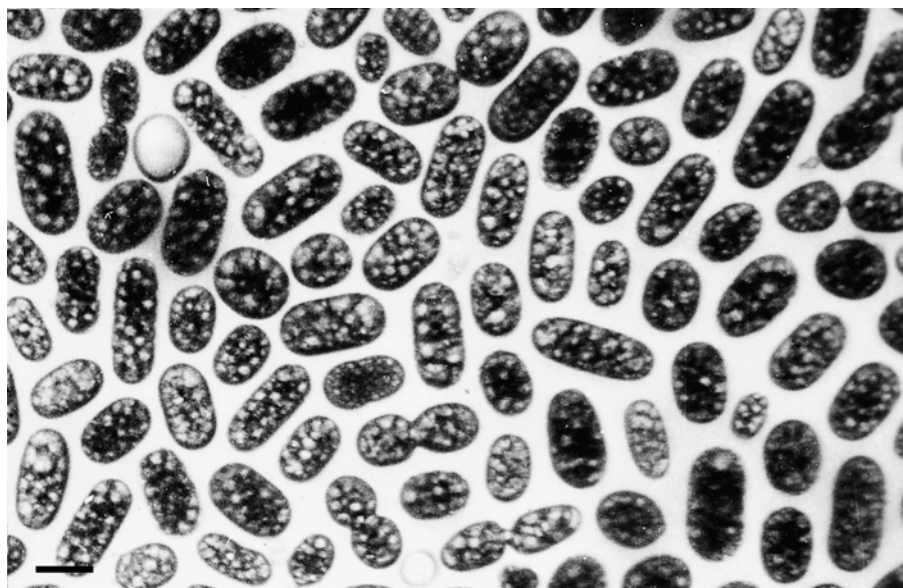


FIGURE BXII.γ.54. *Achromatium* cells collected from sediment of Lake Stechlin (Brandenburg, Germany). Considerable variation in size can be seen. Light micrograph. Bar = $50\text{ }\mu\text{m}$.

2 ng/cell. In SEM preparations, the inclusions below the cell surface become visible due to shrinkage of the cell envelope (Fig. BXII.γ.57).

The function of the calcite inclusions is unknown, but it seems likely that the intracellular precipitation fulfils a metabolic role. The hypothesis that *Achromatium* possesses an active mechanism for concentrating calcium from the environment is consistent with the observation that cells in an acidic pond with very low levels of calcium in the water column contain calcite inclusions. In starvation experiments with physical cell enrichments, the inclusions tend to disappear. They can be easily removed by treating the cells with dilute acetic acid.

The presence of sulfur inclusions within the cells and the predominance of *Achromatium* in the micro-oxic zone of sediments suggest that *Achromatium* spp. may be sulfur-oxidizing chemolithoautotrophs. Microautoradiographic studies of the uptake of ^{14}C -labeled substrates have shown that *Achromatium* cells are able to assimilate ^{14}C from bicarbonate, acetate, and protein hydrolysate; however, glucose is not assimilated. The autotrophic potential for at least a proportion of cell enrichments is evident not only from inorganic carbon fixation, but also from the presence of genes similar to that encoding the large subunit of ribulose-1,5-bisphosphate carboxylase/oxygenase (RubisCo) (Head et al., 2000b).

From experiments incubating *Achromatium*-bearing sediment cores with ^{35}S -labeled sulfate in the presence or absence of sodium molybdate, it was deduced that the population was capable of oxidizing sulfide to intracellular S^0 and ultimately to sulfate. Moreover, sequences homologous to APS reductase genes (*aprBA*), were detected in purified DNA extracted from physically enriched cell preparations (Head et al., 2000b; Gray et al., 1997).

However, the pattern of organic substrate uptake within a single population suggests that physiological diversity occurs in natural *Achromatium* communities (Head et al., 2000a). In contrast to the response of some sulfide-oxidizing bacteria, e.g. *Beggiatoa* spp., sulfur droplets do not diminish in *Achromatium* enrichments under starvation conditions. It is possible that putatively heterotrophic members of the population can oxidize sulfide and deposit S^0 intracellularly as a mechanism for the detoxification of metabolically produced hydrogen peroxide in the

manner previously suggested for heterotrophic *Beggiatoa* spp. (Burton and Morita, 1964).

The cells are surrounded by a specific glycocalyx. SEM-preparations of cells show a typical network that covers the whole cell surface (Fig. BXII.γ.58), which is consistent with results of TEM-preparations (Fig. BXII.γ.59). Presumably, these structures play an important role in the colonization of solid surfaces (adhesion) and possibly in the peculiar type of cell movement exhibited by *Achromatium* cells.

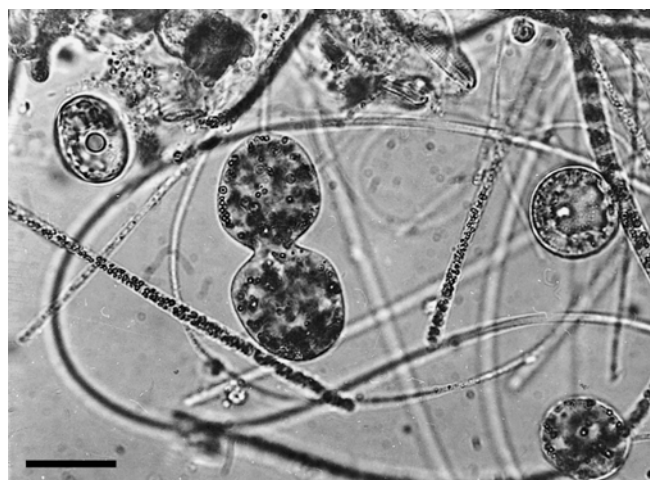


FIGURE BXII.γ.55. *Achromatium oxaliferum*, dividing stage of one cell. Sediment sample from Lake Dagow (Brandenburg, Germany). Filaments of *Beggiatoa* spp. and some diatoms are visible. Light micrograph. Bar = 25 μm .

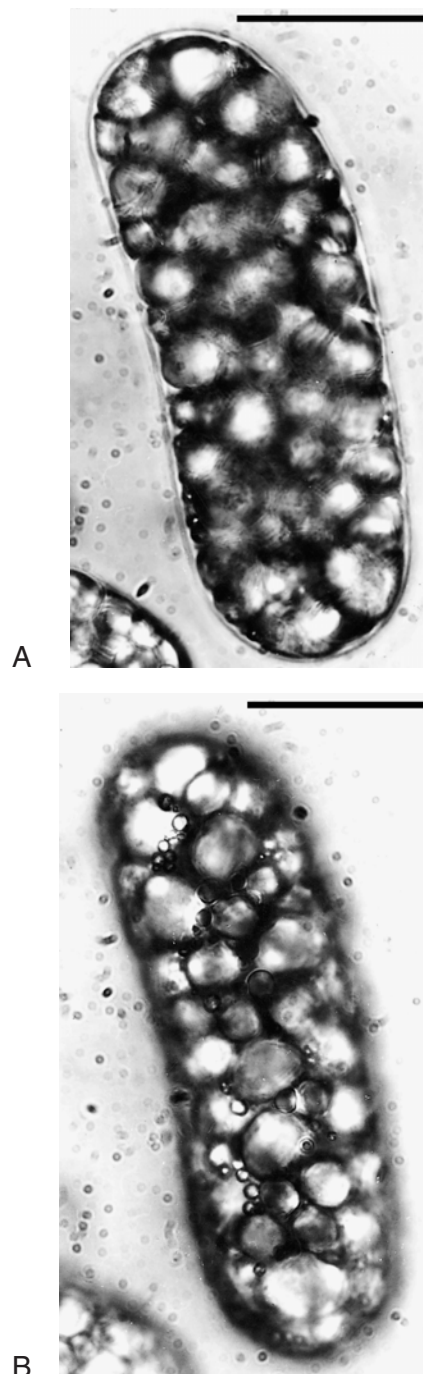


FIGURE BXII.γ.56. Light micrographs of an identical *Achromatium* cell, focused at various levels. The whole cell volume is filled with calcite crystals and small sulfur droplets. (A) surface of the cell, (B) center of the cell. Bars = 25 μm .

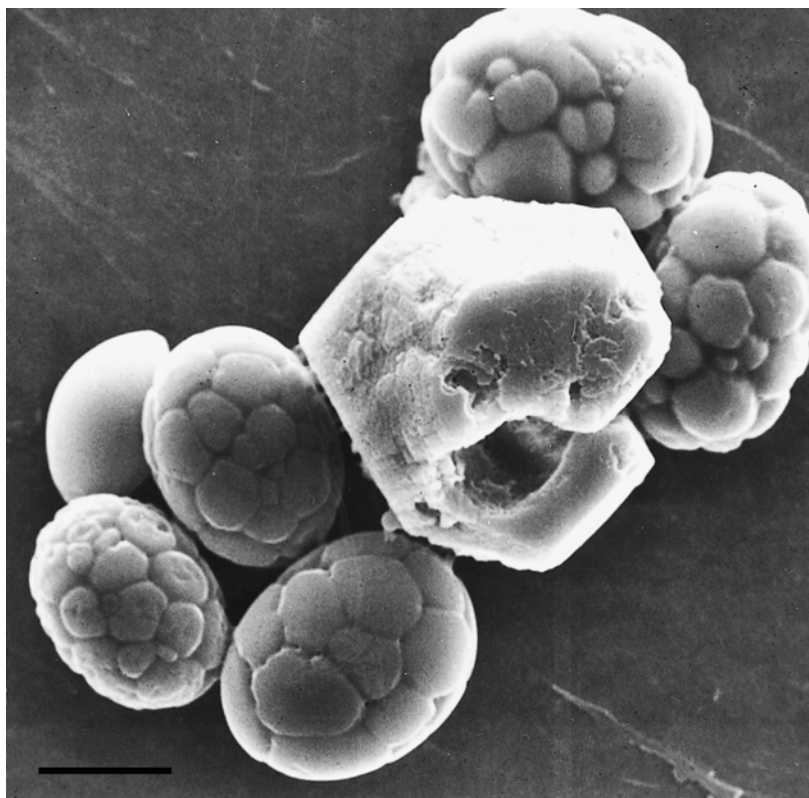


FIGURE BXII.γ.57. Scanning electron micrograph (SEM) showing some cells stuck to a calcite crystal from the sediment (Lake Stechlin, Brandenburg, Germany). The inclusions below the cell surface can be seen. Bar = 25 μm (Reproduced with permission from H.-D. Babenzien and H. Sass, *Ergebnisse der Limnologie*, 48: 247–251, 1996, ©E. Schweizerbart'sche, Stuttgart.)

ENRICHMENT AND ISOLATION PROCEDURES

No enrichment or isolation procedures are available. For experiments, cells must be collected physically from sediments by a swirling technique ("gold panning method"; De Boer et al., 1971). These cell enrichments may be washed carefully in sterile water and transferred with a Pasteur pipette several times to obtain highly purified preparations of cells.

TAXONOMIC COMMENTS

There are two remarkable misinterpretations in the history of the genus *Achromatium*. The most striking inclusions present within cells were interpreted by Schewiakoff (1893) to be Ca-oxalate (hence the name of the species originally described); however, they consist of calcium carbonate. This was correctly described by Bersa (1920).

From the electron micrographs shown from De Boer et al. (1971), the glycocalyx was interpreted as peritrichously arranged appendages (flagella), and it was noted that "gliding motility should be caused by peritrichous flagella beating in a slime layer that surrounds the cell". The assumption of "gliding movement" was the main reason for inclusion of the organism in Section 23—Nonphotosynthetic, nonfruiting gliding bacteria—of the first edition of *Bergey's Manual of Systematic Bacteriology*. Today the means of movement of *Achromatium* remains unclear, but it has no flagella.

Although only one species (*A. oxaliferum*) is included in the Approved List of Bacterial Names, a second, "*A. volutans*", was described earlier by Hinze (1903) under the proposed name

"*Thiophysa volutans*". The organism was found only in saline/marine waters, where it normally contains sulfur inclusions but lacks calcium carbonate crystals (La Riviere and Schmidt, 1992). More recently, Dando et al. (1998) described "*A. volutans*" at hydrothermal vents around the Greek island of Milos. The cells (diameter 5–90 μm) appeared with high density in the upper 1–15 mm of the sediment, forming "white sand" due to the reflective sulfur globules within the cells. Enzyme activity measurements of cell extracts (ribulosebiphosphate carboxylase, adenyphosphosulfate reductase, sulfate adenytransferase) were indicative of autotrophic sulfur oxidizing activity.

The phylogenetic position of these populations is unclear and until further investigations are available, the position of "*Achromatium volutans*" as a genuine member of the genus *Achromatium* remains doubtful. Although only one species is validly published, the occurrence of genetically distinct subpopulations in *Achromatium* communities with greater than 2.5% divergence in their 16S rRNA sequences provides evidence to support the delineation of *Achromatium* into several different species.

ACKNOWLEDGMENTS

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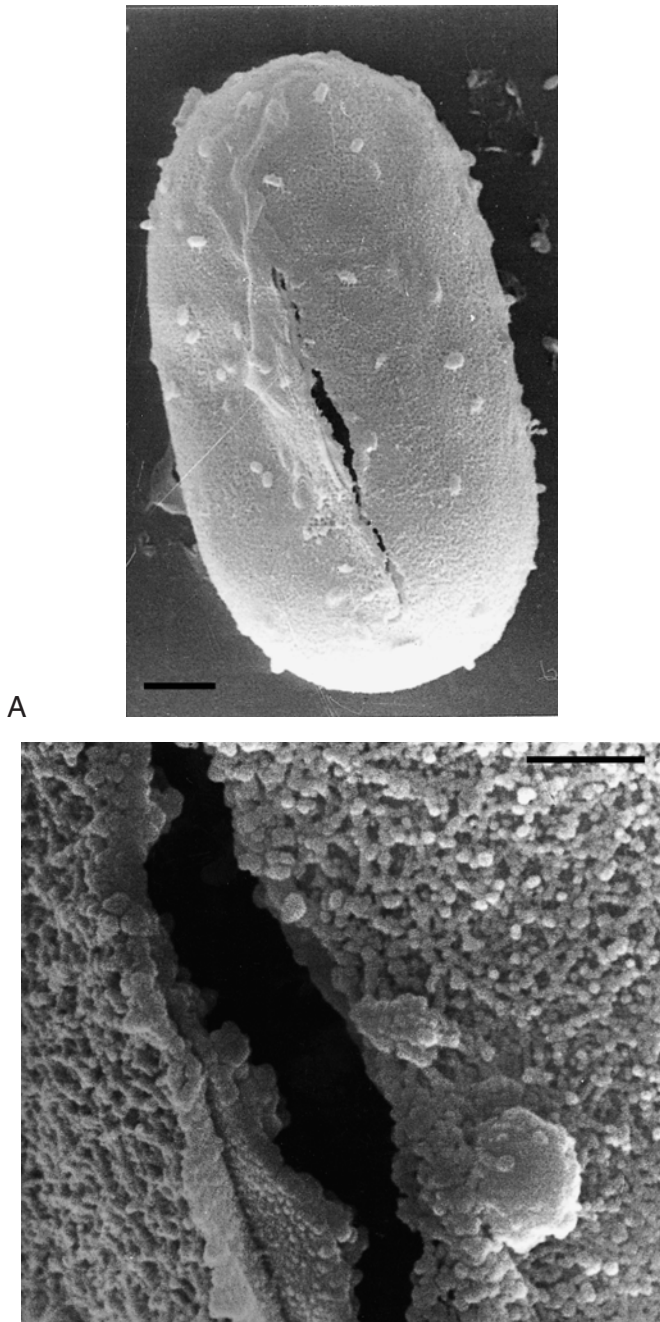


FIGURE BXII.γ.58. Scanning electron micrographs (SEM) from an *Achromatium* cell. The ruptured area of the cell is shown under a different magnification. A typical network can be seen. (A) whole cell. Bar = 5 μm. (B) Rupture of the cell, showing the thick glycocalyx. Bar = 1 μm.

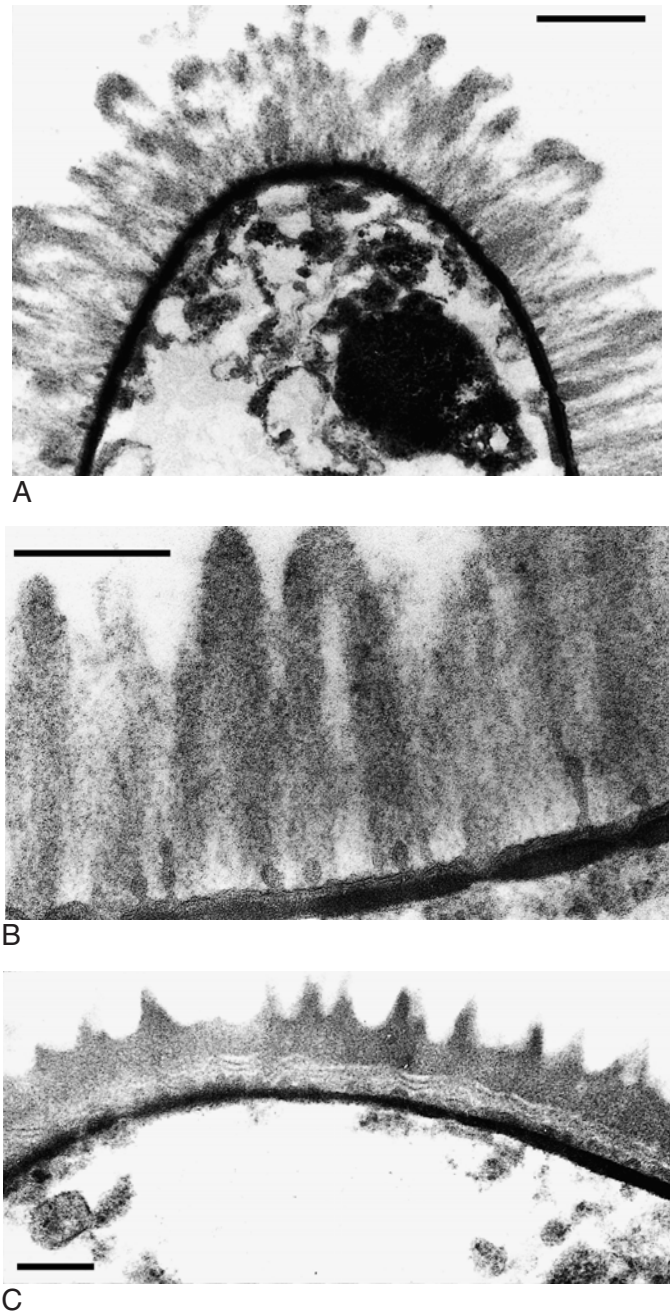


FIGURE BXII.γ.59. Transmission electron micrographs (TEM) of ultra-thin sections of *Achromatium* cells showing the extent of the glycocalyx covering the cell and its lobular structure. Calcite inclusions are solubilized. (A) Bar = 1 μm. (B), Bar = 0.5 μm. (C) Bar = 1.0 μm.

FURTHER READING

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List of species of the genus *Achromatium*

1. ***Achromatium oxaliferum*** Schewiakoff 1893, 1^{AL}
ox.al.i'fe.rum. M.L. n. *oxalatum* oxalate; Gr. n. *oxalis* sorrel, a sour plant; L. v. *fero* to carry; M.L. adj. *oxaliferum* oxalate-containing.

The characteristics are as the same as those described for the genus.

The mol% G + C of the DNA is: not known.

Type strain: No culture isolated.

Genus III. *Beggiatoa* Trevisan 1842, 56^{AL}

WILLIAM R. STROHL

Beg.gi.a.to'a. M.L. fem. n. *Beggiatoa* a genus of bacteria; named in remembrance of F.S. Beggiato, a physician of Vicenza, (g.b. 1807), who authored the *Delle Terme Euganea*.

Colorless cells, ~1.0–200 × 2.0–10.0 μm, occurring in filaments 5.0–10 cm in length. Filaments usually have a consistent width over the entire length. Freshwater strains mostly, if not universally, possess filament diameters less than 5.0 μm, whereas marine beggiatoas with filament diameters ranging from 1.0 to 200 μm have been observed microscopically. No strain wider than 5 μm has been isolated in axenic culture from any source. The filaments may contain up to 100 or more cells, and in rare cases, up to several thousand. Cells in filaments are cylindrical and are longer than they are wide in the thinner strains (i.e., those less than ~5 μm in diameter). In wider strains (i.e., those greater than ~7 μm in diameter), cells are usually disk-shaped and typically are wider than they are long. Filaments occur singly, in sheets of cottony masses on sediment surfaces, or in mats in which each filament retains its individuality. Reproduction is by transverse binary fission of cells within filaments; division occurs by septation, in which the peptidoglycan and cytoplasmic membranes close like the iris of a diaphragm. Filament dispersion is via sacrificial cell death (necridial cells), filament breakage, or simple disintegration. In some strains, the disintegration of filaments occurs until mostly single or double cell units (hormogonia) exist; a hormogonium then can grow to become a filament. **Cells contain inclusions of sulfur when grown in the presence of H₂S and, for some strains, thiosulfate.** Intracellular inclusions of poly-β-hydroxybutyrate or polyphosphate may be present, particularly in freshwater strains. **Marine or brackish water strains having a filament diameter of >10 μm contain a large (~80% of the cell volume), central liquid vacuole that contains concentrated nitrate.** Resting stages are not known. **Attachment holdfasts or sheaths are not present.** Capsules are not formed, but filaments usually produce a slime matrix. Gram negative. **Filaments and hormogonia have gliding motility.** No motility organelles are known. **Aerobic or microaerophilic, having a strictly respiratory type of metabolism,** with oxygen and, in some instance, nitrate as the terminal electron acceptor. Anaerobic growth is not proven with pure cultures, but may occur with nitrate as the terminal electron acceptor for some strains studied in nature. Internally stored sulfur may also serve as an electron acceptor by freshwater strains for short-term maintenance in the absence of oxygen. Sulfate does not substitute as the terminal electron acceptor for anaerobic growth in strains thus far studied. **Chemoorganotrophic (freshwater strains), and either facultatively or constitutively chemolithoautotrophic (marine strains).** H₂S or thiosulfate may be used as the electron donor for chemolithotrophic metabolism. Acetate is oxidized to CO₂ by all freshwater strains tested thus far. **Several C₂, C₃, and C₄ organic acids and, sometimes, their amino acid equivalents are utilized as sole carbon and energy sources for heterotrophic growth.** Growth factors are not required by most strains, although some strains may require vitamin B₁₂. Gelatin and starch are not hydrolyzed. **N₂ is fixed by all freshwater and marine strains tested thus far.** Nitrate, nitrite, ammonium, N₂, or certain amino acids may be used as the sole nitrogen source. **Oxidase positive. Catalase negative.** Freshwater, estuarine, and marine strains are known. **Beggiatoas are inherently gradient organisms, growing in horizontal layers in sediments at the interface between the un-**

derlying anoxic sulfide-liberating zone and the overlying oxic zone. Growth can occur in the range of 0–40°C. Obligately thermophilic strains have not been characterized, although some beggiatoas have been observed in high temperature runoffs associated with thermal springs, or in mats adjacent to thermal vents in the ocean floor.

The mol% G + C of the DNA is: 35–39.

Type species: *Beggiatoa alba* (Vaucher 1803) Trevisan 1842, 56.

FURTHER DESCRIPTIVE INFORMATION

Three subgroups of *Beggiatoa* can be discerned at present. The type and only named species is *Beggiatoa alba*, which is a member of the first subgroup. The three subgroups are: (i) thin, heterotrophic freshwater strains; (ii) thin, obligately or facultatively chemolithoautotrophic marine strains; and (iii) wide, autotrophic, vacuolated marine strains. Characteristics of the three subgroups *Beggiatoa* are listed in Table BXII.γ.39. Other features are described below.

Motility Gliding motility of beggiatoas is relatively rapid (1.0–8.0 μm/sec) and is often accompanied by flexing, bending, and/or rotation of the filaments. Gliding motility determines the nature of growth and colony formation; on agar media containing relatively few nutrients, spiral colonies are usually produced. The filaments are capable of gliding on agar surfaces to form various patterns of waves and concentric rings (Fig. BXII.γ.60).

Cell structure and life cycle of thin freshwater strains Cells within the filaments are separated by the membranes and the first cell wall layer (Fig. BXII.γ.61; Strohl and Larkin, 1978a). Cell division occurs by separation of those two inside layers (Fig. BXII.γ.61). Nearly one fourth of the cells in a trichome may be dividing simultaneously (Hinze, 1901). The outer cell wall layers do not take part in septation (Strohl et al., 1982) and are longitudinally continuous with the length of the filament. Beggiatoas do not contain a classical sheath but do exude large quantities of extracellular neutral polysaccharide slime (Strohl and Larkin, 1978b; Larkin and Strohl, 1983).

Filament dispersion in the thin freshwater strains occurs via sacrificial cell death (necridia formation) and filament breakage (Strohl and Larkin, 1978a) in a manner reminiscent to that observed in *Oscillatoria* spp. Filament dispersion may also occur via simple separation of an end cell to produce a hormogonium (Wiessner, 1981), or the filaments may divide by simple fragmentation (Pringsheim, 1949). Fragmentation, however, is often associated with filament death.

Although all beggiatoas stain Gram-negative (Strohl and Larkin, 1978b), they have a cell envelope which is much more complex than that of typical Gram-negative bacteria (Fig. BXII.γ.62; Maier and Murray, 1965; Strohl et al., 1982, 1986b). The A layer of the cell wall is probably a peptidoglycan layer, and the B layer is a unit membrane-like layer that is positionally similar to the lipopolysaccharide layer of Gram-negative bacteria (Fig. BXII.γ.62; Strohl et al., 1982). Depending on the strain and the method of fixation, two or three further external layers may be observed (Maier and Murray, 1965; Strohl et al., 1981a, 1982), the most external of which usually has a fibrillar pattern (Strohl et al., 1982).

TABLE BXII.γ.39. Characteristics of the three subgroups of the genus *Beggiatoea*^{a,b}

Characteristic	Thin, heterotrophic freshwater strains (including <i>B. alba</i>)	Thin, obligately or facultatively autotrophic marine strains	Wide, autotrophic, vacuolated marine strains
Typical filament diameters, mm	~1–5	~1–7	~10–200
Sulfide oxidized to sulfur	+	+	+
Sulfur oxidized to sulfate	–	+	+
Sulfur reduced to sulfide	+	?	?
Heterotrophic growth	+	–	–
Acetate assimilated	+	+	?
Acetate oxidized to CO ₂	+	?	?
Autotrophic growth	–	+	+
Mixotrophic growth	?	–	–
CO ₂ fixed by RubisCo	w	+	+
Large, central vacuoles	–	–	+
Nitrate accumulated	–	–	+
Nitrate reduced	+	+	+
N ₂ product	+	?	+
NH ₃ product	+	?	?
N ₂ fixed	+	+	?
Anaerobic growth suggested	–	–	+
Microaerophilic	+	+	+
c-type cytochromes	+	+	+
CO-binding c-type cytochromes	+	?	–
Growth in distinct mats	+	+	+
Photophobic response	+	+	+
O ₂ -phobic response	+	+	+
H ₂ S-phobic response	+	+	+
<i>Phylogenetic placement:</i>			
<i>Gamma</i> proteobacteria	+	+	+
Closest relatives	<i>Vitreoscilla beggiatoidea</i>	NA	wide, marine thioplocas
mol% G + C of the DNA	35–39 ^c	NA	NA
<i>Locations found:</i>			
Marine sediments		+	+
Sediments	+		
Warm mineral springs			+
Representative strains	B18LD, OH-75-2a	MS-81-1c, MS-81-6	Monterey Canyon ^d , Guaymas Basin ^d , Warm Mineral Springs ^d

^aSymbols: +, >90% positive; –, <90% negative; ?, not known; w, weak, insignificant levels; NA, not available; RubisCo, ribulose 1,5-bisphosphate carboxylase/oxygenase.

^bThe data presented in this table are not meant to be all-inclusive, but rather representative of the data, as they are known currently. There are likely to be exceptions to the given data based on individual strains or experimental conditions used.

^cThe mol% G + C content of the DNA of the freshwater *B. alba* strains B18LD (ATCC 33555), B15LD (ATCC 33556), and OH-75-2a are 37.1 mol%, 35.5 mol%, and 38.5 mol% (0.3 in duplicate and triplicate experiments), respectively, by *T_m* analysis (D.C. Nelson, personal communication).

^dFor the wide, marine example strains, there are no isolates. Thus, locations from which these organisms have been studied are given rather than strain names or numbers. It cannot be assumed, however, that the *beggiatoea* at these locations represent single species.

All *Beggiatoea* cells are known to contain at least three types of inclusions: poly-β-hydroxybutyrate (PHB; Pringsheim and Wiessner, 1963; Strohl and Larkin, 1978b; Strohl et al., 1982), polyphosphate (Maier and Murray, 1965; Strohl and Larkin, 1978b), and sulfur (Lawry et al., 1981; Strohl et al., 1981b). It is the latter inclusion that has traditionally separated organisms of the genus *Beggiatoea* from those in *Vitreoscilla* (Pringsheim, 1951; Leadbetter, 1974a; Strohl et al., 1986a). Several investigators have observed sulfur inclusions in thin sections of fixed and dehydrated *Beggiatoea* trichomes (Morita and Stave, 1963; Maier and Murray, 1965; Lawry et al., 1981; Strohl et al., 1981b; Larkin and Henk, 1996). The sulfur is dissolved away during dehydration with ethanol or acetone, leaving empty, electron-translucent spaces (Strohl et al., 1981b). Sulfur inclusions within these cytoplasmic membrane invaginations may be bound by envelopes typically composed of a single dense layer (Fig. BXII.γ.63; Strohl et al., 1981b). Unusual pentilaminar sulfur inclusion envelopes have

been observed in cells of *B. alba* strain B15LD (Strohl et al., 1981b, 1982). Rudimentary pentilaminar envelopes have been observed in cells of strain B15LD grown in the total absence of reduced sulfur, indicating that a “primer” envelope was always present (Strohl et al., 1982). This may partially explain the rapidity with which sulfur inclusions appear after exposure of *Beggiatoea* cells to sulfide (Winogradsky, 1887; Burton and Morita, 1964; Schmidt et al., 1986). It may be physiologically significant that the sulfur inclusions of freshwater *Beggiatoea* strains are found within invaginations of the cytoplasmic membrane and are therefore periplasmic (i.e., external to the cytoplasmic membrane) in nature (Strohl et al., 1982).

Under certain growth conditions, PHB comprises over 50% of the cell dry weight of thin freshwater, heterotrophic *B. alba* strains (Güde et al., 1981; Strohl et al., 1981a). Furthermore, in one experiment, ~50% of the [¹⁴C]acetate assimilated by heterotrophically grown cells was incorporated into PHB (Strohl et

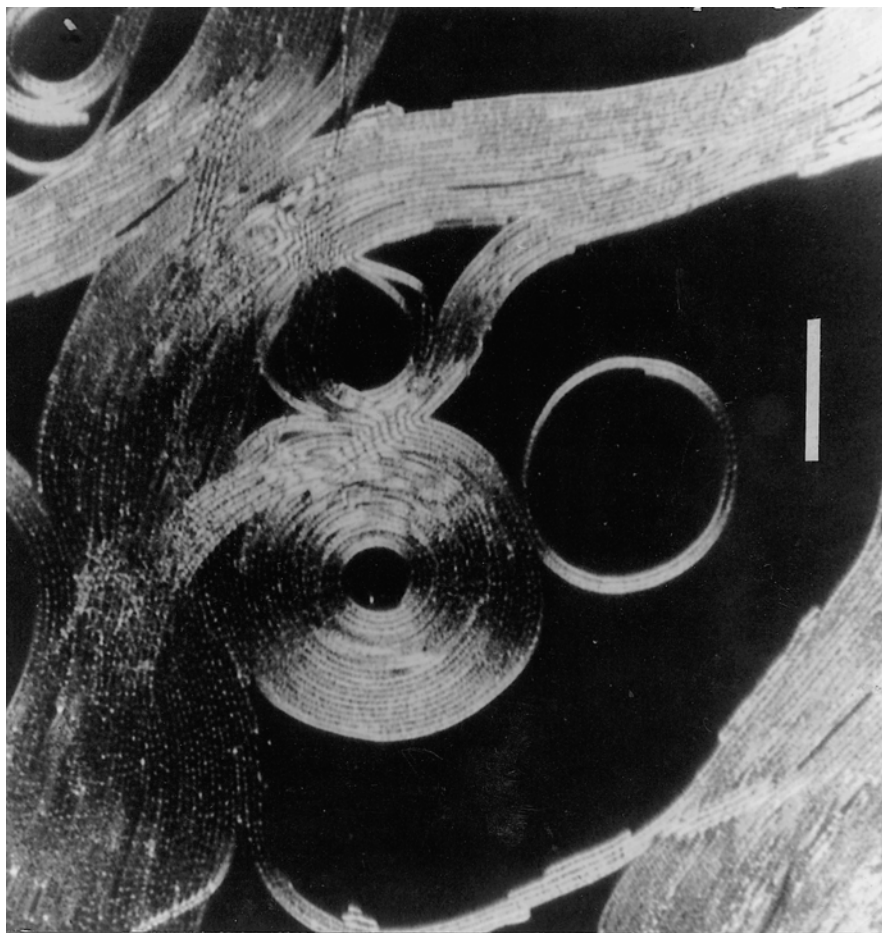


FIGURE BXII.γ.60. Light micrograph of a *Beggiatoa alba* colony on a 1.4% agar surface, taken through a dissecting microscope. Bar = 50 μ m.

al., 1981a). The amount of PHB in the cells is usually proportional to the amount of acetate in the growth medium (Kowallik and Pringsheim, 1966). Cultures containing PHB have been shown to survive in media containing sulfide but lacking organic nutrients (Kowallik and Pringsheim, 1966), suggesting a survival role for the endogenous metabolism of PHB during carbon starvation. Furthermore, a freshwater *B. alba* strain containing massive PHB inclusions has been shown to be able to survive up to 7 days of starvation in a mineral salts medium lacking a carbon and an energy source (W.R. Strohl, unpublished observation).

Cell structure of wide, vacuolated, marine strains Larkin and Henk (1989) and Nelson et al. (1989b) simultaneously recognized that the large marine beggiatoas with diameters greater than 10 μ m contain centrally located vacuoles (Figs. BXII.γ.64 and BXII.γ.65), similar to those previously observed in wide, marine *Thioploca* strains (Maier and Gallardo, 1984b). Larkin and Henk (1989) proposed that the vacuoles serve the purpose of generating a high surface to volume ratio, allowing for the large size of the trichomes. Later, Fossing et al. (1995) demonstrated that the vacuoles in *Thioploca* contain high concentrations (100–500 mM) of nitrate; similar analyses of the marine *Beggiatoa* central vacuoles also found high concentrations (130–150 mM) of nitrate (McHatton et al., 1996).

The cytoplasm, sulfur inclusions, and other cytoplasmic membrane intrusions are all found in a very narrow band at the outer

edge of these large, vacuolated cells (Fig. BXII.γ.64 and BXII.γ.65). Although no high-magnification data are yet available, lower-magnification photographs of large, vacuolated beggiatoas (Nelson et al., 1989b; Larkin and Henk, 1989; Larkin et al., 1994) suggest a less complex cell wall with perhaps fewer layers than observed in the heterotrophic freshwater strains. Larkin and his colleagues also observed what appear to be symbiotic bacteria ultrastructurally similar to type II methanotrophic bacteria living within the cytoplasm of the wide, vacuolated marine beggiatoas (Larkin and Henk, 1996). Larkin and Henk (1996) proposed that the large, marine beggiatoas possessing internal symbionts might in fact be “dual” organisms much like the tube-worms and mussels found in the same habitat. This finding and its potential implications (e.g., potential symbiotic relationship) may have important implications for the potential ability to culture large beggiatoas axenically.

Sulfur metabolism of beggiatoas Oxidation of sulfide to sulfur by the heterotrophic freshwater strain *B. alba* is constitutive (Schmidt et al., 1986) and does not promote an increase in growth rate (Güde et al., 1981) or in the rate of protein synthesis (Schmidt et al., 1986). Sulfide oxidation by *B. alba* B18LD is oxygen-dependent (Vargas and Strohl, 1985b; Schmidt et al., 1987) and is inhibited by several specific electron transport inhibitors (Strohl and Schmidt, 1984; Schmidt et al., 1987), suggesting that sulfide oxidation is coupled with oxygen reduction

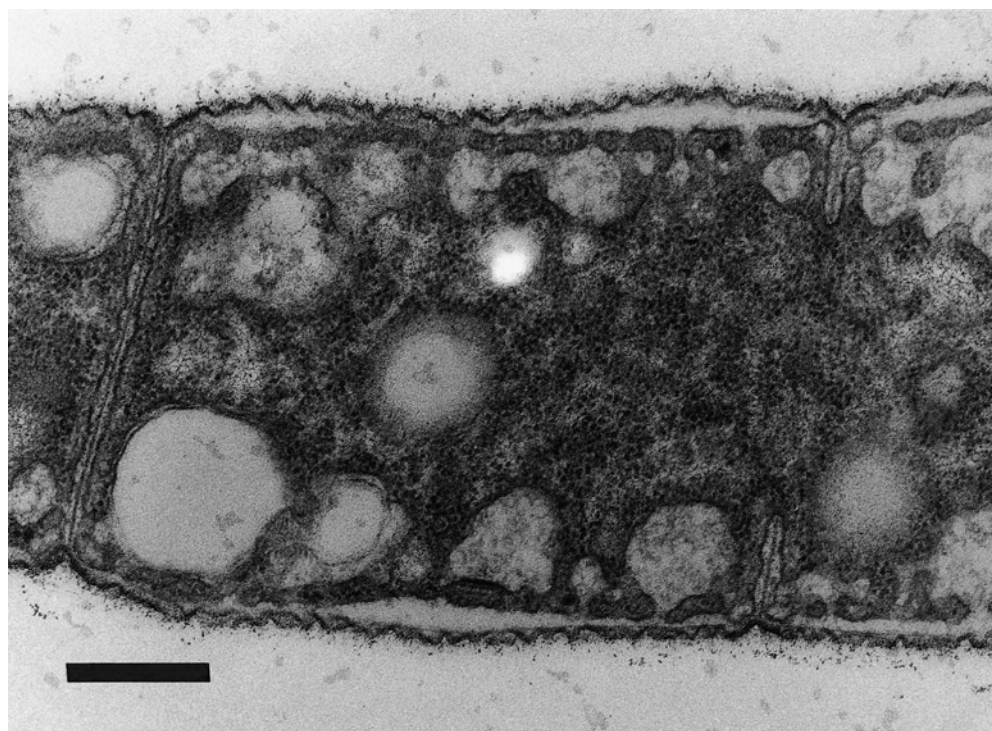


FIGURE BXII.γ.61. Thin section micrograph of *B. alba* B18LD showing the separation of cells in the filament by the cytoplasmic membranes and the peptidoglycan layer. A newly formed septation is also visible at the right. The outer walls, not well preserved by the modified Ryter-Kellenberger fixation technique (Strohl and Larkin, 1978b), are continuous with the length of the filament. Bar = 50 µm.

via the electron transport system. Nelson and Castenholz (1981b) have demonstrated that sulfur may comprise as much as 30% of the cell dry weight of the freshwater *Beggiatoa* strain OH-75-2a. Nelson et al. (1989b) have studied sulfide oxidation by the wide marine beggiatoas found near a hydrothermal vent at Guaymas Basin. They have shown that the sulfur, present mostly in the S_8 form, comprises ~14% of the cell dry weight (Nelson et al., 1989b). Although the form of sulfur in freshwater strains has not been determined, it is assumed to be in the S_8 form, as this is consistent with the sulfur found in other sulfur bacteria (Schmidt et al., 1971).

The significance of oxidation of the sulfur formed by freshwater strains such as *B. alba* to sulfate is not clear. Sulfur oxidation by strain OH-75-2a has been reported to be very slow (50 µmol sulfate produced/75 h/~30 mg cells; Nelson and Castenholz, 1981b); this rate—assuming 50% of cell dry weight is protein—calculates to a rate of ~0.4 nmol/min/mg protein. Hagen and Nelson (1997) have reported ATP sulfurylase activity in extracts of OH-75-2a to be about 1 nmol/min/mg of protein—slower than the rates of 23–30 nmol/min/mg protein and 1523–1710 nmol/min/mg protein for the facultatively autotrophic marine strain, MS-81-6, and the chemolithoautotrophic strain, MS-81-1c, respectively. In other experiments, Schmidt et al. (1987) did not observe significant *in vivo* oxidation of sulfur to sulfate (~5–10% maximum conversion) over periods of 22–72 hours, as evidenced by isotopic and chemical studies; a control strain, *Thiothrix nivea* strain JP3, produced considerable sulfate from sulfur inclusions in similar experiments (Schmidt et al., 1987). Similarly, another freshwater strain showed no (sulfuric) acid production from sulfur inclusions, whereas *Thiothrix* strains isolated from similar

sources did so (Williams and Unz, 1985a). On the other hand, Hagen and Nelson (1997) have recently demonstrated the presence, albeit low, in a heterotrophic, freshwater strain (OH-75-2a) of an enzyme activity (sulfite-dependent, AMP-independent ferricyanide reduction) that is normally considered to be involved in oxidation of sulfur to sulfate. This observation supports the contention that sulfur can be oxidized to sulfate by freshwater strains of *Beggiatoa*, albeit rather slowly as compared to the same process by marine strains (Hagen and Nelson, 1997).

Although sulfate appears to be a slowly produced product of sulfide oxidation by the heterotrophic, freshwater strains, the sulfur accumulated by these strains can be reduced back to sulfide under anaerobic conditions (Nelson and Castenholz, 1981b; Schmidt et al., 1987). For *B. alba* B18LD, this process occurs at a rate of ~6–7 nmol/min/mg protein (Schmidt et al., 1987), a 5–10-fold greater rate than the sulfur oxidation rates observed by Nelson and colleagues (Nelson and Castenholz, 1981a; Hagen and Nelson (1997)), and to be coupled with the oxidation of H_2 (Schmidt et al., 1987). Whether this process is ecologically relevant for these strains is as yet unknown, but it is possible that the heterotrophic beggiatoas utilize sulfur reduction as a temporary form of anaerobic respiration under anaerobic conditions, allowing them a form of energy to fuel the movement back to favorable conditions. Thus, sulfur reduction by the freshwater strains may provide an ecological role analogous to that proposed for anaerobic nitrate respiration in the wide, marine beggiatoas and thioplocas (Jørgensen and Gallardo, 1999).

Using cultures with vertical, convergent oxygen and sulfide gradients, Nelson et al. (1986a) have demonstrated in the thin, marine strain MS-81-6 a molar growth yield (Y_{H_2S}) on sulfide of

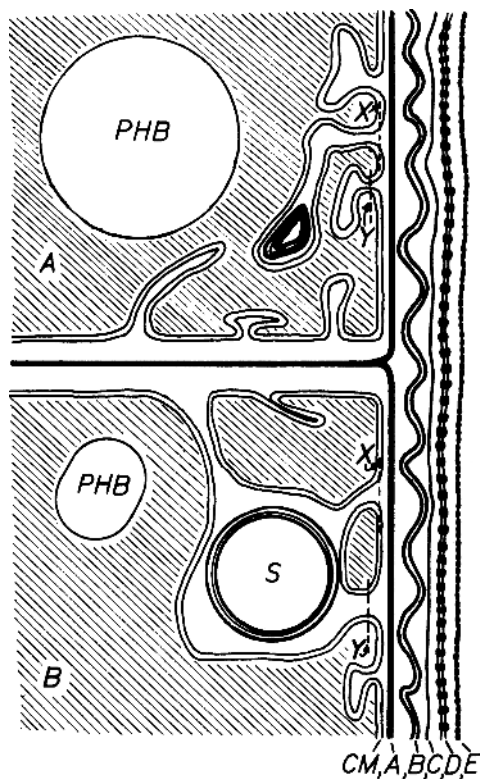


FIGURE BXII.7.62. Model of a freshwater *Beggiatoa* cell wall (strain B15LD; Strohl et al., 1982). Note the pentilaminar sulfur (S) inclusion (which is single-layered in other *Beggiatoa alba* strains; Strohl et al., 1981b) external to the cytoplasmic membrane (CM); poly-beta-hydroxybutyrate, (PHB); cell wall layers from inside to outside (A–E); plane through which freeze-fracture shows invaginations (X); and plane through which thin sections have been made that give a different view of the invagination of membranes around sulfur inclusions (Y).

~8.4 g/mol. The marine strain grew linearly in a horizontal plate with a doubling time of ~11 h at the position at which the oxygen and sulfide gradients converged. Hagen and Nelson (1997) have compared the sulfide oxidation of two marine strains, the obligately chemolithoautotrophic marine *Beggiatoa* strain MS-81-1c and the facultatively autotrophic strain MS-81-6. They demonstrated that MS-81-1c stores more elemental sulfur and has a higher yield for sulfide oxidation than does strain MS-81-6 (Y_{H_2S} for MS-81-1c on sulfide was 15.9 g/mol, compared with 8.3–8.4 g/mol for MS-81-6). Similarly, O_2 consumption rates by strain MS-81-1c were observed to be consistently 2–3-fold greater than those of the facultatively autotrophic strain, MS-81-6. Moreover, strain MS-81-1c oxidizes sulfur to sulfate by either the adenosine phosphosulfate pathway or a highly regulated sulfite: acceptor oxidoreductase, whereas strain MS-81-6 uses only the sulfite: acceptor oxidoreductase (Hagen and Nelson, 1997).

Electron transport in beggiatoas Although Burton and Morita (1964) stated that *Beggiatoa* strains lacked cytochrome systems, all beggiatoas thus far analyzed have been found to contain cytochromes (Cannon et al., 1979; Strohl et al., 1986b; Prince et al., 1988). Room temperature spectra have indicated that *B. alba* and other strains might contain several *b* type, *c* type, *a* type, and CO-binding cytochromes, as well as ubiquinone 8 and NAD(P)H dehydrogenases; however, the spectra obtained in those studies

were not of high quality (Strohl et al., 1986b). Schmidt and Dispirito (1990) have reinvestigated the cytochromes of *B. alba* B18LD and found four *c*-type cytochromes, three of which bind CO. One of the cytochromes was found to be a flavocytochrome *c*, somewhat analogous to cytochromes present in photosynthetic sulfide oxidizing bacteria (Schmidt and Dispirito, 1990).

Prince et al. (1988) determined that a sample of marine *Beggiatoa* filaments obtained from a marine sediment (suspension 1615B, which contained organisms of ~30 μ m diameter; Nelson et al., 1989b) contained only *c*-type cytochromes. These *c*-type cytochromes from marine *Beggiatoa* filaments did not bind CO, nor were flavocytochromes detected (Prince et al., 1988), thus making them substantially different from the *c*-type cytochromes isolated from the freshwater strain, *B. alba* B18LD (Schmidt and Dispirito, 1990).

Carbon metabolism of beggiatoas Freshwater strains of *Beggiatoa* grow heterotrophically (Pringsheim, 1964; Strohl and Larkin, 1978b; Nelson and Castenholz, 1981a; Strohl et al., 1981a; Larkin and Strohl, 1983). They can utilize acetate as the sole carbon and energy source (Larkin and Strohl, 1983; Strohl et al., 1986b). Moreover, several *Beggiatoa* strains have been shown to grow on C_2 , C_3 , and C_4 organic acids, although none of the beggiatoas described thus far has grown on C_5 or C_6 organic acids or on hexoses (Burton and Morita, 1964; Burton et al., 1966; Nelson and Castenholz, 1981a; Mezzino et al., 1984; Williams and Unz, 1985a). Although Burton et al. (1966) have suggested a type of reverse tricarboxylic acid cycle for CO_2 fixation and acetate utilization by a freshwater *Beggiatoa* strain, the accumulated data indicate that freshwater, heterotrophic *Beggiatoa* strains likely utilize the glyoxylate and tricarboxylic acid cycles during growth on acetate (Nelson and Castenholz, 1981a; Larkin and Strohl, 1983).

A small amount of CO_2 is fixed by freshwater strains of *Beggiatoa*, but most of it is assimilated primarily via heterotrophic metabolism mechanisms, rather than by the Calvin-Benson cycle (Nelson and Castenholz, 1981a; Strohl et al., 1981a). Low levels of ribulose 1,5 biphosphate carboxylase/oxygenase (RubisCo) activity have been observed in extracts of the freshwater strains OH-75-2a and *B. alba* B18LD (Nelson et al., 1989a). In addition, a probe derived from the gene encoding the large subunit of *Anacystis nidulans* RubisCo has been found to hybridize strongly to DNA isolated from strain OH-75-2a but not to that from strain B18LD (Nelson et al., 1989a). Thus, although the data strongly suggest that the freshwater beggiatoas possess RubisCo, the levels of activity demonstrated so far are not of physiological significance (Nelson et al., 1989a).

Thin, marine strains of *Beggiatoa* can grow autotrophically. During such growth, CO_2 is fixed by RubisCo at levels consistent with the support of autotrophic growth (Nelson and Jannasch, 1983; Hagen and Nelson, 1996). Strain MS-81-1c, an obligately chemolithoautotrophic strain, has been shown to lack 2-ketoglutarate dehydrogenase activity and does not regulate its RubisCo activity greatly. This strain accumulates acetate into biomass but does not oxidize acetate to CO_2 —similar to what has been observed with the chemolithoautotrophic thiobacilli (Hagen and Nelson, 1996). Strain MS-81-6, on the other hand, regulates both 2-ketoglutarate dehydrogenase and RubisCo activities and appears to possess facultative chemolithoautotrophic metabolism (Hagen and Nelson, 1996).

Suspensions of wide, marine beggiatoas have been shown by Nelson et al. (1989b) to fix CO_2 by RubisCo at levels consistent with chemolithoautotrophic growth, and the CO_2 fixation can

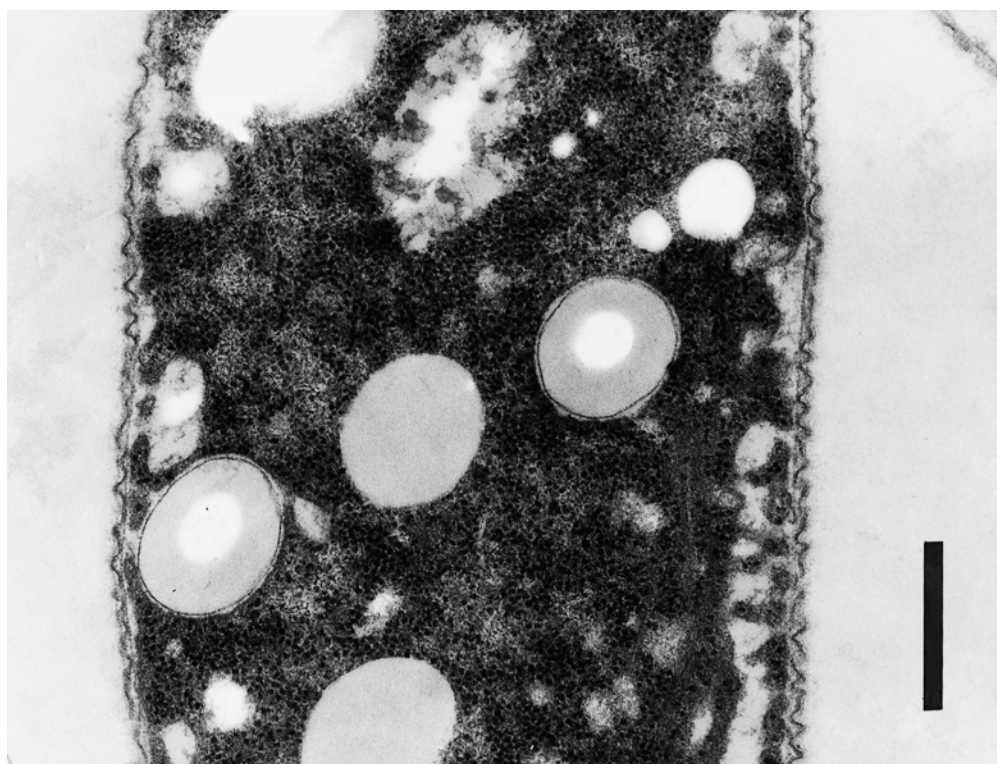


FIGURE BXII.γ.63. Thin section micrograph of *B. alba* B18LD showing the monolayered (4 nm thick envelope) sulfur inclusions external to the cytoplasmic membrane. Bar = 0.5 μm.

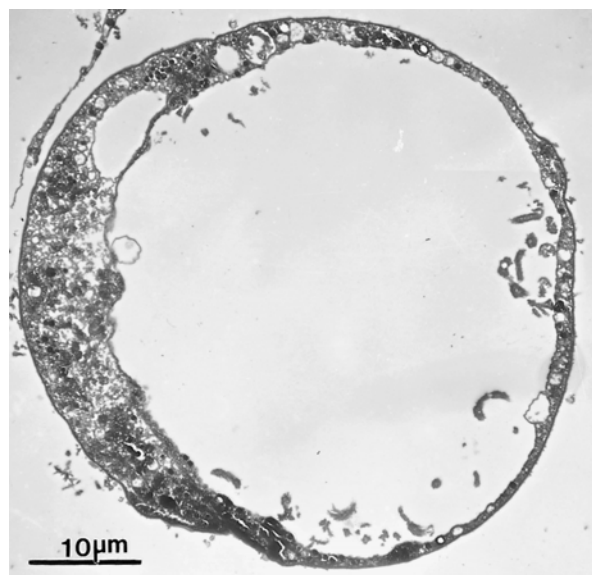


FIGURE BXII.γ.64. Thin section, cross-section of a wide, vacuolated, marine *Beggiatoa* strain showing an enormous central vacuole, with the cytoplasm and sulfur inclusions all squeezed into the thin outer "shell" surrounding the vacuole (Reproduced with permission from J.M. Larkin and M.C. Henk, *Microscopy Research Technique* 33: 23–31, 1996, ©John Wiley & Sons, Inc.).

be stimulated approximately 7-fold by the addition of sulfide. Both RubisCo and phosphoribulokinase activities have been observed at significant levels in the suspensions of these marine

beggiatoas (Nelson et al., 1989b), whereas 2-ketoglutarate dehydrogenase activities are below the limit of detection (McHatton et al., 1996). These data are supported by autoradiography, which has demonstrated extensive $^{14}\text{CO}_2$ fixation by intact strains (Nelson et al., 1989b). Interestingly, the vent beggiatoas from the Guaymas Basin exhibit 1.6–4.8-fold greater RubisCo activity at 50°C than at 30°C (whereas the thin, marine strain has approximately 4-fold more RubisCo activity at 30°C), suggesting a relatively high optimal temperature for their enzyme activity. Whole cell autoradiography of these organisms, however, demonstrates that CO_2 is not fixed above 40°C (Nelson et al., 1989b).

Steele et al. (1995) have reported that suspensions of the wide beggiatoas found in Warm Mineral Springs, FL, assimilate CO_2 at a level that would predict a 27-hr doubling time, suggesting that this organism also grows chemolithoautotrophically. This organism grows in an environment that is at approximately half the salinity of seawater (Lackey et al., 1965); thus, it is expected that it is more similar to the wide marine beggiatoas than to the freshwater strains.

Nitrogen metabolism Ammonia is a nitrogen source used by every *Beggiatoa* tested thus far (Larkin and Strohl, 1983). In studies by Vargas and Strohl (1985a), the glutamine synthetase (GS)–glutamate synthase (GOGAT) pathway was found to be the primary route for ammonia assimilation by *B. alba* B18LD. Glutamate dehydrogenase activity has not been observed and only low alanine dehydrogenase activity is observed under "high" ammonia conditions (Vargas and Strohl, 1985a). Other nitrogen sources that support growth of *B. alba* B18LD are nitrate, nitrite, urea, aspartate, and, to a lesser degree, asparagine, alanine, and thiourea (Vargas and Strohl, 1985a). Glutamate, aspartate, or

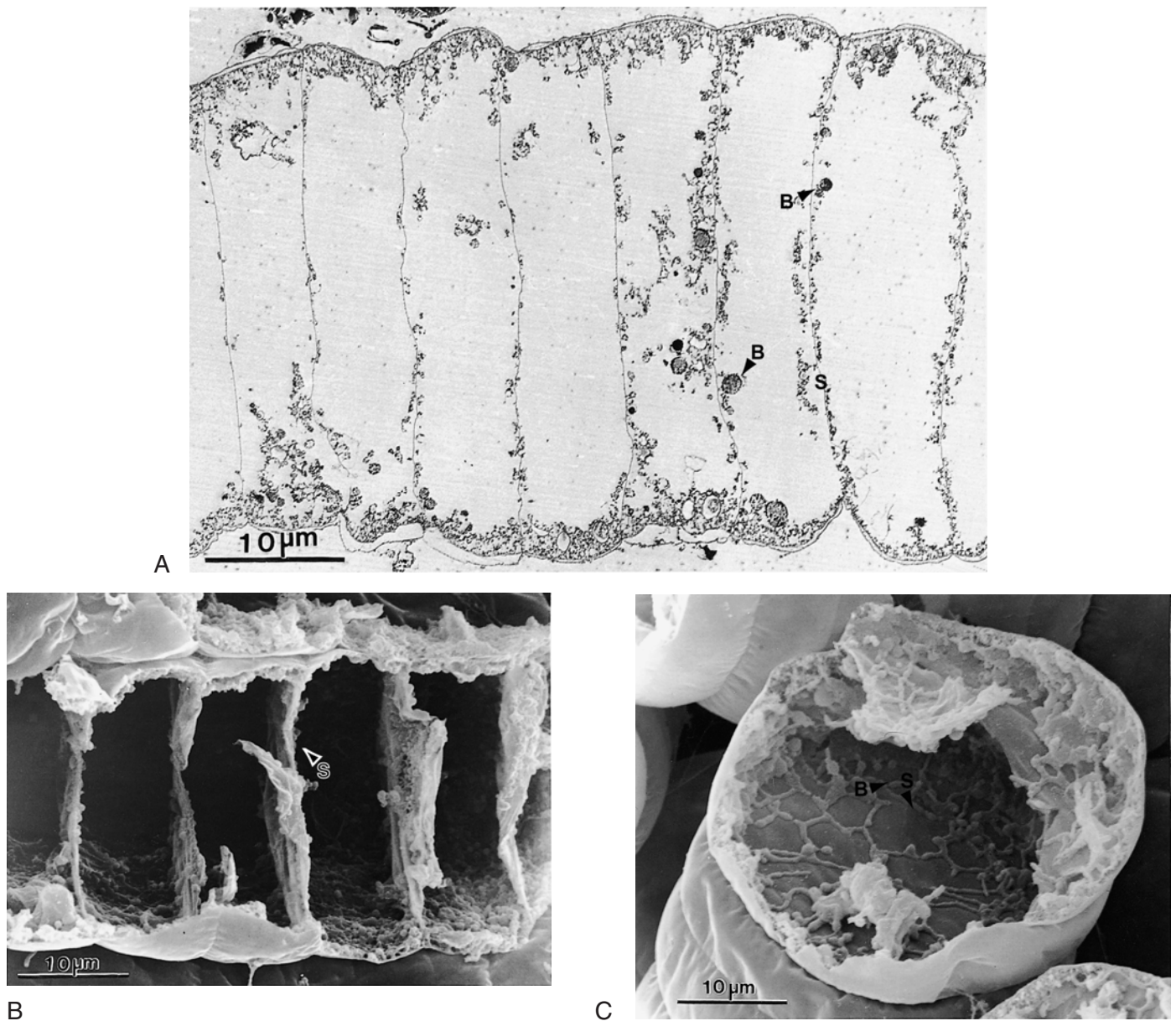


FIGURE BXII.7.65. A typical thin section (A) of a *Beggiatoa* filament with a diameter of greater than 10 µm. Small blebs (B) of cytoplasm can be seen on the septa (S) and the remainder of the cytoplasm is restricted to a thin layer beneath the cell membrane. Scanning electron micrographs (B, C) of wide, marine, vacuolated *Beggiatoa* spp. The hollowed-out spaces where the nitrate vacuoles are located are easily observed. (Reproduced with permission from J.M. Larkin and M.C. Henk, *Microscopy Research and Technique* 33: 23–31, 1996, ©John Wiley & Sons, Inc.)

asparagine support the growth of certain other beggiatoas, indicating that they can be used as sole nitrogen, carbon, and energy sources for those strains (Pringsheim, 1964).

Beggiatoas can utilize nitrate (Nelson et al., 1982; Vargas and Strohl, 1985b; Williams and Unz, 1985a) or nitrite (Vargas and Strohl, 1985b) as a sole nitrogen source. The product of nitrate reduction by *B. alba* is not N_2 or N_2O but ammonia, indicating that it has an assimilatory type nitrate reduction mechanism (Vargas and Strohl, 1985b). *B. alba* B18LD is unable to grow anaerobically by using nitrate as an electron acceptor, and nitrate reduction is not coupled with sulfide oxidation (Vargas and Strohl, 1985b). The nitrate reductase of *B. alba* B18LD has also been shown to be primarily in the soluble fraction (Vargas and Strohl, 1985b). Later experiments by Sweerts et al. (1990) showed that

a freshwater mat of *Beggiatoa* spp. denitrified nitrate to N_2 *in situ*. They also suggested, based on minimal experimental results, that under certain conditions *B. alba* B18LD may be able to denitrify nitrate to N_2 (Sweerts et al., 1990).

Perhaps the most significant recent discovery is that the wide, vacuolated marine strains of *Beggiatoa* (i.e., those strains greater than 10 µm in diameter) have the ability to assimilate and accumulate nitrate in their large, central liquid vacuoles (McHatton et al., 1996), in a manner similar to that of large, marine *Thioploca* strains (Fossing et al., 1995). These large, marine organisms concentrate nitrate to 130–160 mM, or ~3000–4000-fold greater than the concentration of ~40 µM measured in the sea water in which they are found (Fossing et al., 1995; McHatton et al., 1996; Otte et al., 1999). On the other hand, the nonvacuolated thin,

marine strains MS-81-1c and MS-81-6 do not accumulate nitrate, indicating that the ability to accumulate and concentrate nitrate is correlated with the presence of the large central vacuoles (McHatton et al., 1996). Because the nitrate reductase activities of vacuolated, wide, marine beggiatoas are largely membrane bound—thus differing from the soluble nitrate reductase found in *B. alba* B18LD (Vargas and Strohl, 1985b) and the two thin, marine strains (McHatton et al., 1996)—the nitrate stored in these large vacuoles was originally thought to be denitrified to N_2 (McHatton et al., 1996). More recent evidence, however, suggests that ammonia is the major end product of nitrate reduction (D.C. Nelson, personal communication). Based on the thickness of the mats found at Guaymas Basin, the low level of oxygen present in the sea water, and the inadequacy of oxygen diffusion to support the sulfide oxidation present in such biomass, McHatton et al. (1996) made a strong case for the coupling of sulfide oxidation to nitrate reduction in those marine *Beggiatoa* mats. They calculated that enough nitrate was stored to support the organism for approximately 4.8 h of anaerobic growth without input of fresh nitrate (McHatton et al., 1996). Otte et al. (1999) have recently demonstrated that wide, marine *Thioploca* enrichments reduce most of their stored nitrate to NH_4^+ , with a minor fraction ($\sim 15\%$ of the NH_4^+ produced) going to N_2 . Based on the rate of nitrate reduction to ammonia and the rate of sulfide oxidation, Otte et al. (1999) have suggested that the nitrate present in the vacuoles of marine *Thioploca* spp. might be sufficient for 200 hours of maintenance without additional input of nitrate. It has been calculated that the nitrate accumulation and reduction by these large marine beggiatoas and thioplocas should have a major impact on the nitrogen cycling in the oceanic sediments in which these organisms are found (Fossing et al., 1995; McHatton et al., 1996; Otte et al., 1999). Clearly, the fate of the nitrate reduction products (N_2 versus NH_4^+) by these organisms ought to be significant in the balance of nitrogen in the marine niches they occupy.

Nine strains of *Beggiatoa*—freshwater and marine, all with diameters less than $5\ \mu m$ —have been shown to fix N_2 at rates of 3–12 nmol/min/mg protein when grown in tubes of semisolid agar in which the redox, sulfide, and oxygen gradients had been established (Nelson et al., 1982). The gradient cultures are required to provide proper redox conditions for N_2 fixation by both the freshwater and marine strains (Nelson et al., 1982). Nitrate or ammonia inhibits the fixation of nitrogen, and sulfide is required by some of the beggiatoas to fix nitrogen (Nelson et al., 1982). Polman and Larkin (1988) later found that ammonia and urea inhibit the activity of *B. alba* B18LD nitrogenase *in vitro*, whereas nitrate, nitrite, glutamine, and asparagine do not. Nitrite and nitrate, however, were found to inhibit *in vivo* nitrogenase induction. On the other hand, glutamine stimulates nitrogenase activity of *B. alba* B18LD (Polman and Larkin, 1988).

Autotrophy and sulfide oxidation Winogradsky (1887) developed his theory of chemoautotrophy based on his observations of sulfide oxidation by beggiatoas in slide cultures. The beggiatoas were considered autotrophs until the mid-1900s when Cataldi (1940), Faust and Wolfe (1961), Scotten and Stokes (1962), Burton and Morita (1964), Pringsheim (1964), and Burton et al. (1966) described heterotrophic growth for various freshwater strains of *Beggiatoa*. At that point, controversy began between various investigators about the role of sulfide oxidation in *Beggiatoa* metabolism. Pringsheim and his colleagues claimed that *Beggiatoa* was a facultative autotroph with the ability to grow autotrophically with sulfide used as the sole energy source or

heterotrophically with acetate used as the sole carbon and energy source (Kowallik and Pringsheim, 1966). Burton and Morita (1964) claimed that sulfide oxidation was responsible for the detoxification of peroxides produced during *Beggiatoa* growth. Because sulfide was oxidized in the presence of organic carbon, Pringsheim (1967, 1970) later rescinded his claim of autotrophy and suggested that the beggiatoas were capable of “mixotrophic” growth, obtaining their carbon from acetate and CO_2 and their energy from sulfide oxidation (and perhaps from acetate oxidation). *B. alba* B18LD was postulated to grow mixotrophically, with sulfide or thiosulfate serving as a supplemental energy source (Güde et al., 1981; Strohl and Schmidt, 1984).

On the other hand, very careful experiments carried out by Nelson and Castenholz (1981b) demonstrated that the increased increments in dry weights ($\sim 30\%$ increases) brought about by the oxidation of sulfide could be entirely accounted for by the weight of the sulfur itself, indicating that no yield could be accounted for by energy conserved by the sulfide oxidation. Moreover, the continuous culture experiments carried out by Güde et al. (1981) have been brought into question by others (c.f., Kuenen and Beudeker, 1982; Nelson, 1989, 1992), leading most investigators in the field to conclude that mixotrophic growth could not be inferred by the outcome of those experiments. Finally, the oxidation of sulfur to sulfate by freshwater strains is, at best, very inefficient (at least under the laboratory conditions used in the various experiments; Nelson and Castenholz, 1981b; Schmidt et al., 1987; Hagen and Nelson, 1997), indicating that the most favorable energy conservation steps, i.e., oxidation of sulfur to sulfate, were not being utilized efficiently by these freshwater *Beggiatoa* strains. Thus, the overwhelming bulk of data available currently suggest that the thin, freshwater beggiatoas are solely heterotrophic and that the oxidation of sulfide must serve a physiological purpose other than energy conservation.

What purpose does sulfide oxidation serve for these strains, if not for the conservation of energy? As stated above, Burton and Morita (1964) suggested that this process might be coupled to the detoxification of peroxides. Experiments by Nelson and Castenholz (1981b) support that contention. On the other hand, both Nelson and Castenholz (1981b) and Schmidt et al. (1987) have demonstrated that sulfur can be reduced back to sulfide anaerobically. Schmidt et al. (1987) showed that this reduction can be coupled with the oxidation of H_2 , an energetically favorable reaction; thus, perhaps sulfur is utilized as a short-term electron acceptor for maintenance metabolism under anaerobic growth conditions; this is somewhat analogous to the role suggested for nitrate metabolism by the wide, vacuolated marine beggiatoas (McHatton et al., 1996).

Although all of the experiments to date have fallen short in proving mixotrophic growth for freshwater beggiatoas, there are three bodies of evidence that are intriguingly suggestive as to its occurrence. First, experiments by Strohl and Schmidt (1984) and Schmidt et al. (1987) have demonstrated clearly that specific electron transport inhibitors inhibit not only the oxidation of acetate to CO_2 , but also the oxidation of H_2S to internally stored sulfur, suggesting a linkage of sulfide oxidation with electron transport. *B. alba* B18LD has also been shown to possess a flavocytochrome *c*, not unlike those linking sulfide oxidation and electron transport in photosynthetic sulfur bacteria (Schmidt and Dispirito, 1990). Second, both strains OH-75-2a and B18LD were shown by Nelson et al. (1989a) to possess RubisCo activity, albeit at insignificant levels with respect to growth. Nevertheless, it is possible that conditions have not yet been found in the labo-

ratory to optimize this activity. Finally, Hagen and Nelson (1997) have recently shown that the freshwater *B. alba* strain OH-75-2a possesses a regulated, sulfide-inducible sulfite:acceptor oxidoreductase, suggesting that perhaps under certain conditions, these strains might be capable of oxidizing sulfur to sulfate, a distinctly "lithotrophic-like" metabolism. While the rate of sulfur oxidation is very slow under the conditions tested thus far, oxidation does appear to occur. These three lines of data are still the most critical evidence to date for the potential conservation of energy from H_2S oxidation by heterotrophic freshwater strains. Nevertheless, without convincing orthogonal data to substantiate the mixotrophy claims, the freshwater strains should still be considered to be (only) heterotrophic. Perhaps the evolutionary progenitors of these freshwater strains possessed the ability to couple sulfide and sulfur oxidation with energy conservation mechanisms, but lost it, retaining the vestigial biochemical remains that we see today.

There is no controversy about the autotrophic nature of the marine strains. Careful experiments carried out by Nelson and colleagues (Nelson and Jannasch, 1983; Nelson et al., 1986b; Hagen and Nelson, 1996, 1997) have shown definitively that two thin, marine *Beggiatoa* strains, MS-81-1c and MS-81-6, can grow autotrophically in gradient cultures. Further experiments have in fact shown that strain MS-81-1c is an obligately chemolithoautotrophic marine *Beggiatoa* strain and that MS-81-6 is a facultatively autotrophic strain, capable of increasing growth yields by the assimilation of organic carbon. The data accumulated on the wide, marine *Beggiatoa* all support chemolithoautotrophic growth with either oxygen or nitrate as the terminal electron acceptor (Nelson et al., 1989b; McHatton et al., 1996).

Ecological characteristics *Beggiatoa* exhibit four physiological characteristics that describe their ecological niche. First, they appear to be oligotrophic organisms that generally do not tolerate or utilize high concentrations of organic (W.R. Strohl, unpublished observations) or nitrogenous (Vargas and Strohl, 1985b) nutrients. Low concentrations (e.g., ≤ 20 mM) of organic acids and, in some cases, amino acids are apparently optimal for the growth and metabolism of freshwater, heterotrophic *Beggiatoa*. Autotrophic, marine *Beggiatoa* are even more fastidious and may or may not utilize organic nutrients.

Second, *Beggiatoa* are respiratory organisms that couple the oxidation of organic acids (heterotrophic strains; Scotten and Stokes, 1962; Nelson and Castenholz, 1981a; Strohl et al., 1986b) and/or sulfide (autotrophic strains; Hagen and Nelson, 1997), with the reduction of oxygen (Schmidt et al., 1987; Hagen and Nelson, 1996, 1997) or nitrate (McHatton et al., 1996) as the terminal electron-accepting step. It appears that stored S^0 can be utilized, at least by the heterotrophic freshwater strains, as an alternative electron acceptor under strictly anaerobic conditions (Nelson and Castenholz, 1981b; Schmidt et al., 1987). Because the pool of internally stored sulfur would be limiting, however, the sulfur is probably used only as an electron acceptor *in situ* for maintenance purposes while the filaments glide to a new sulfide-oxygen interface. It is not yet known whether autotrophic strains can also utilize sulfur as an electron acceptor.

Third, *Beggiatoa* are microaerophilic bacteria (Larkin and Strohl, 1983) that exhibit a phobic response to oxygen (Møller et al., 1985). Microaerophilic bacteria have been described as "aerobic or facultatively anaerobic bacteria, which under aerobic conditions use O_2 as hydrogen acceptor in a respiratory way of energy production and show optimal growth at low oxygen ten-

sions" (Stouthamer et al., 1979). Heterotrophic *Beggiatoa* apparently produce hydrogen peroxide in potentially autolytic quantities (Burton and Morita, 1964); however, peroxide accumulation has not been quantified. High levels of nutrients usually induce post-growth autolysis of *Beggiatoa* (Faust and Wolfe, 1961; Scotten and Stokes, 1962; Strohl and Larkin, 1978b), which may be directly related to the amount of peroxide produced. Catalase is not produced to degrade the potentially toxic peroxides (Burton and Morita, 1964; Joshi and Hollis, 1976; Strohl and Larkin, 1978b; Nelson and Castenholz, 1981b). Addition of catalase to growth medium may enhance growth yields (Burton and Morita, 1964) or survival of certain *Beggiatoa* strains (Strohl and Larkin, 1978b). In a carefully performed set of experiments, however, Nelson and Castenholz (1981a) showed that catalase does not increase growth rates or yields significantly over cultures lacking catalase. Moreover, their experiments indicated that peroxides are not formed during exponential growth phase (Nelson and Castenholz, 1981b).

Finally, and most importantly, *Beggiatoa* are gradient organisms that live in sediments, in which they occupy the horizontal interface between sulfide emanating upward from below and O_2 diffusing downward from above (Jørgensen and Revsbech, 1983; Nelson et al. 1986b). Jørgensen and his colleagues (Jørgensen and Revsbech, 1983; Møller et al., 1985; Nelson et al., 1986b) have described the sulfide/oxygen interface location of *Beggiatoa* mats and plates both *in situ* and *in vivo*. In marine *Beggiatoa* mats, gradients of O_2 and sulfide are created by the *Beggiatoa* above and below the horizontal mats, respectively (Jørgensen and Revsbech, 1983). In creating such concentration gradients, the *Beggiatoa* apparently do not produce high concentrations of most nutrients at their surface, a phenomenon probably preferential to their oligotrophic nature. O_2 in high concentrations, especially if coinciding with concentrations of carbon nutrients, is apparently toxic to the *Beggiatoa*, probably due to formation of peroxide or other toxic product. Thus, the vertical gradient effect, which permits only minute amounts of oxygen to the organisms, is seemingly ideal for them (Nelson et al., 1986b). Stimulation of certain enzymes, such as nitrogenase (Nelson et al., 1982) and RubisCo (Nelson and Jannasch, 1983), and even the ability to grow (Nelson and Jannasch, 1983) may depend on the attributes of the gradient effect (i.e., gradients of oxygen and sulfide in which the organism selects its best position).

Beggiatoa possess unique physiological mechanisms to maintain their proper position in nutrient gradients. Nelson and his colleagues have elaborated a series of phobic responses by *Beggiatoa* that guide their mat-forming behavior. First, the *Beggiatoa* were found to possess a photophobic response that results in a daily migration of *Beggiatoa* to the surface at night and down into the sediments during the sunlight hours (Nelson and Castenholz, 1982). It was later discovered that *Beggiatoa* also possess phobic responses to both O_2 (Møller et al., 1985) and sulfide (Jørgensen and Postgate, 1982), resulting in the movement to the proper position in sediments at which the two substrates overlap (Jørgensen and Revsbech, 1983; Nelson et al., 1986b). A significant difference is observed between the freshwater, heterotrophic strains and the marine, chemolithoautotrophic strains in gradient cultures. The marine strains utilize sulfide much more efficiently, with the sulfide and oxygen gradients overlapping by only ~ 50 – 100 μm (D. Nelson, personal communication), whereas sulfide and oxygen overlap significantly in gradients of the less efficiently sulfide-oxidizing freshwater strains (Nelson et al., 1986b). The rapid gliding motility exhibited by

beggiatoas allows them to position themselves at the optimal nutrient concentrations and to move quickly as environmental conditions change.

The interwoven, horizontal mats formed by wide, marine beggiatoas found near the hydrothermal vents in the Guaymas Basin can be up to 30 cm in thickness and are virtual monocultures of *Beggiatoa* filaments (Nelson et al., 1989b; McHatton et al., 1996). This is in striking contrast to mats formed by freshwater filaments, which typically reach only 1–2 mm in thickness at most, as well as to mats of ~1–2 cm formed by marine beggiatoas in other locations (Larkin et al., 1994). Marine *Thioploca* spp., on the other hand, form vertical columns of filaments reaching up out of the sediments, indicating that they possess an ecology that is somewhat unique when compared to the beggiatoas found in similar environments (McHatton et al., 1996; Jørgensen and Galardo, 1999).

Beggiatoas of various morphological and physiological types have been found in environments as diverse as hydrothermal vents at 2000 m of depth (McHatton et al., 1996), desert hypersaline pools (Lackey et al., 1965), freshwater and marine marsh sediments, mineral and sulfur springs (Lackey et al., 1965), run-offs from thermal springs and geysers, activated sludge in sewage treatment plants (particularly those utilizing rotating biological contractors; Chung and Strom, 1997), microbial mats, and sediments ranging from sandy and pristine to polluted and silty. In marine microbial mats, a succession of colorless sulfur bacteria occur, starting with *Thiospira* and *Macromonas*, followed by *Thiovulum*, and then finally *Beggiatoa* spp. (Bernard and Fenchel, 1995).

An environment in which *Beggiatoa* strains appear to play a significant "economic" role is the rice rhizosphere, another gradient environment, set up by oxygen transport by the rice plants to their roots and the surrounding submerged, anoxic sediments (Pitts et al., 1972; Joshi and Hollis, 1977). A symbiotic relationship between the rice plants and *Beggiatoa* has been postulated in which the rice plants provide not only the gradient environment, but also catalase to the beggiatoas to protect them from toxic oxygen, while the beggiatoas protect the plants from sulfide by oxidizing it to harmless elemental sulfur (Joshi and Hollis, 1977).

Finally, a marine *Beggiatoa* spp., along with the cyanobacterium, *Phormidium corallyticum*, and sulfate-reducing bacteria, has been observed to be associated with black band disease of coral (Carlton and Richardson, 1995; Richardson, 1996). While not directly implicated in causation of the disease itself, the *Beggiatoa* is hypothesized to be a stable and contributing member to the consortium of bacteria associated with the presence of the disease (which is most likely caused by sulfide generated by the sulfate reducers) (Carlton and Richardson, 1995; Richardson, 1996).

Genetic characteristics Little has been done to characterize the beggiatoas genetically. The molecular mass of the *B. alba* genome, as characterized by C_{ot} analysis, is 2.02×10^9 (3.03×10^3 kilobase pairs; Genthner et al., 1985). Thus, the *B. alba* genome is approximately the same size as that of *Escherichia coli* (Genthner et al., 1985). Three *B. alba* strains contain plasmids with molecular masses of $12.3\text{--}12.8 \times 10^6$ (18.9–19.7 kb; Minges et al., 1983); however, no function has been ascribed to these plasmids. No phages specific for infection of *Beggiatoa* species have been found, either in sediment samples or endogenously in the genome of four strains tested (W.R. Strohl, unpublished data). No mutants or strains containing auxotrophic markers have yet been described for any *Beggiatoa* species.

ENRICHMENT AND ISOLATION PROCEDURES

Freshwater strains Keil (1912) isolated *Beggiatoa* by washing the filaments repeatedly with sterile stream water. The washed filaments were placed in Petri dishes containing stream water and were incubated under a bell jar containing the gases H_2S (7.8 Pa), O_2 (147 Pa), CO_2 (245 Pa), and H_2 . It has been suggested that Keil's methods have been the only isolation procedures to be specifically designed to enrich for and isolate autotrophic strains.

Cataldi first observed *Beggiatoa* during her studies with iron bacteria (Cataldi, 1940), and she used the extracted hay technique that Winogradsky had developed for enrichment of iron bacteria. Recent investigators (Joshi and Hollis, 1976; Strohl and Larkin, 1978b) have successfully used various modifications of Cataldi's boiled hay procedure. Burton and Lee (1978) used an alternative procedure in which gallon jars were filled with raw sewage and aerated for 3 days, after which bundles of *Beggiatoa* filaments were observed on the surface.

For enrichment of salt marsh or estuarine beggiatoas, a marine salt mix can be added to a typical hay enrichment to approximate 50% seawater (Strohl and Larkin, 1978b). An excellent enrichment procedure for marine beggiatoas uses the sulfuretum concept of Baas-Becking (1925). A 1–2-inch layer of sea sand is covered by a 1-inch layer of sulfide-emanating mud and a few inches of 50–80% seawater. Decaying leaves (or boiled hay) and $CaSO_4$ (~1 g/l) are then added. Two thirds of the tank is darkened by wrapping in aluminum foil, and a light is shined on the remaining third so that a natural sulfur cycling takes place between the photosynthetic and the dark zones, as well as between the aerobic and reduced zones. Combined cyanobacterial, photosynthetic bacterial, and *Beggiatoa* mats develop in about 2 weeks and remain stable for months.

Depending on the nutritive conditions of the hay enrichments and the agar plates upon which the beggiatoas are isolated, it is likely that a wide variety of different nutritional types of beggiatoas can be obtained (Strohl and Larkin, 1978b). Cataldi (1940) used the gliding motility of *Beggiatoa* strains to isolate them. She placed washed filaments from enrichments into the center of a Petri dish containing 0.1% yeast extract, 0.6% ethanol, and 1.2% agar. After 48 h, the contaminant-free trichomes at the edge of the plates were picked up by a Pasteur pipette and were placed into the liquid medium (Cataldi, 1940). Most investigators have used various modifications of Cataldi's isolation procedures (Faust and Wolfe, 1961; Scotten and Stokes, 1962; Pringsheim, 1964; Burton and Lee, 1978; Strohl and Larkin, 1978b). The following is a general isolation procedure for *Beggiatoa* strains. Filaments from enrichments are washed 5 times with sterile basal salts containing 5 mM neutralized Na_2S . Tufts of filaments are transferred from one wash bath to the next using microforceps. The tufts of filaments are teased and prodded to remove as much macroscopic contaminating material as possible. The washed tufts are placed onto a dry 1.6% agar plate for about 1 min to absorb away as much of the excess fluid as possible. They can be touched with a corner of filter paper to further remove excess fluid (Burton and Lee, 1978). The tufts of filaments are then removed from the "drying plate" and placed onto pre-scored (Burton and Lee, 1978), freshly prepared 1.4% agar plates containing 2 mM Na_2S , 0.01% each of yeast extract and sodium acetate, and basal salts (Vargas and Strohl, 1985a). After 5–48 h, the plates may be observed with a dissecting microscope to detect any contaminant-free filaments, which, along with a small

amount of the agar beneath them, are removed using flame-sterilized 23–26-gauge needles. It is usually best to wait for at least 24 h, so that the contaminated filaments are overgrown, and to make sure that the pure filaments are far away from the contaminants, making them easy to isolate.

For growth of most freshwater beggiatoas, a defined medium can be constructed that contains the following nutrients: 0.2–10.0 mM sodium acetate, succinate, or malate as carbon source, ~2.0 mM ammonium or nitrate as nitrogen source (Vargas and Strohl, 1985b), and standard basal salts (Vargas and Strohl, 1985a). An apparent requirement for a high concentration of calcium and a low concentration of phosphate is notable. A reduced sulfur source, such as sulfide or thiosulfate, can be added at concentrations of 1–2 mM to the heterotrophic medium above. Increasing carbon nutrients shortens the survival time of the organism, perhaps due to higher levels of the peroxides produced. A good general purpose medium for growth, isolation, and maintenance of most freshwater beggiatoas is one modified from the MY medium described by Larkin (1980). It contains 0.01% each of sodium acetate, yeast extract, and nutrient broth (Difco, Detroit, Michigan), 2 mM neutralized Na₂S·9H₂O, and basal salts (Vargas and Strohl, 1985a). For routine maintenance, Na₂S can be added prior to autoclaving, after which the medium should appear gray due to production of FeS. About 90% of the sulfide is lost upon autoclaving (Vargas and Strohl, 1985a). The medium should be inoculated immediately after cooling so that some of the sulfide can be used before it is chemically oxidized or volatilized. For critical experiments, the sulfide should be neutralized and then autoclaved separately under a nitrogen atmosphere in sealed Wheaton serum bottles. It may then be added to the medium just prior to inoculation.

Marine strains To date, the only marine strains that have been successfully cultured in the laboratory are two relatively thin, marine, chemolithoautotrophic strains—one obligately autotrophic and the other facultatively autotrophic (Nelson and Jannasch, 1983; Hagen and Nelson, 1996, 1997). One of the keys to the study of these thin, marine beggiatoas was the development and use of gradient cultures using gel-stabilized sulfide/oxygen gradients (Nelson et al., 1982, 1986a; Nelson and Jannasch, 1983). Wide, marine strains have been studied, however, by carefully washing the trichomes from thick marine mats in order to reduce the contamination to virtually background noise levels. These types of studies have generated significant physiological information about these strains, allowing investigators to determine their overall physiological and phylogenetic parameters.

Additionally, Lackey et al. (1965) reported that they were able to culture wide, brackish water beggiatoas in mixed-culture environmental chambers, such as the sulfuretum described above, for several months, although those of intermediate size ranges (described as “*Beggiatoa arachnoidea*” [5–14 μm diameter] and “*Beggiatoa mirabilis*” [15–21 μm diameter]) did not survive under those conditions.

MAINTENANCE PROCEDURES

The best media for general maintenance of freshwater beggiatoas are usually low nutrient media with adequate sulfide or thiosulfate concentrations and relatively low oxygen concentrations. The modified MY medium described above is a good general medium for the maintenance of freshwater beggiatoas. Biphase cultures can be constructed by overlaying a 1.6% agar plug of MY medium, modified by increasing the sulfide concentration to 4 mM, with either MY liquid medium or MY and 0.2% semi-solid agar medium. Normally, cultures will survive about 1 month under any of these conditions. For growth and maintenance of marine cultures, refer to methods described by Nelson et al. (1982) and Nelson and Jannasch (1983).

Beggiatoa strains have been preserved for long term viability by freezing them in the presence of 20–30% glycerol at –70°C or –196°C. Survival at –196°C is at least several months for the strains tested.

DIFFERENTIATION OF THE GENUS *BEGGIATO*A FROM OTHER GENERA

Although the beggiatoas make up a relatively heterogeneous group, considering the major differences observed between the thin, freshwater strains and the wide, vacuolated, marine strains, they can still be differentiated as a group from members of other genera (Table BXII.γ.40). The primary distinguishing characteristics include their abilities to glide and deposit sulfur when grown on sulfide, their inability to produce sheaths or holdfasts, and their absence of pigmentation.

TAXONOMIC COMMENTS

Much has been learned about the physiology, ecology, and phylogeny of beggiatoas since the last edition of *Bergey’s Manual of Determinative Bacteriology* (Strohl, 1989)—particularly in reference to the large, marine strains—that bear on *Beggiatoa* taxonomy. The three general groupings listed in Table BXII.γ.39 are heterogeneous and should not be construed as “taxonomic” designations; nevertheless, they do represent the three major physiological, morphological, ecological, and phylogenetic groups of

TABLE BXII.γ.40. Characteristics differentiating the genus *Beggiatoa* from other phylogenetically or phenotypically similar genera^a

Characteristic	<i>Beggiatoa</i>	<i>Thioploca</i>	<i>Thiothrix</i>	<i>Vitreoscilla</i>
Colorless	+	+	+	+
Strains deposit sulfur	+	+	+	–
Ensheathed	–	+	+	–
Holdfast	–	–	–	+
Gliding motility	+	+	+	+
Filament formation	+	+	+	± b
Individual cells observable within filaments	–	–	–	±
Class:				
<i>Betaproteobacteria</i>				+
<i>Gamma</i> proteobacteria	+	+	+	–

^aSymbols: +, >90% positive; –, >90% negative.

^bDifferent strains of this species give different results, and some individual strains do not give consistent results.

beggiatoas studied to date. As exemplified by the notable exceptions below, discovery and characterization of additional strains in various niches may significantly alter this profile in the future.

Currently, there are three possible exceptions to the generalized groupings given above. Both the marine strain L1401-14 ("strain 14") and a Pringsheim marine strain isolated from a marine estuary near Plymouth exhibit a requirement for salt-water and grow heterotrophically (Pringsheim, 1964). Also, Uphof (1927) reported the presence of wide (i.e., >10–20 μm diameter) beggiatoas in Florida freshwater sulfur springs; Lackey and Lackey (1961) later showed that many of the Florida sulfur springs are brackish, and suggested that Uphof may have observed large beggiatoas similar to those seen at Warm Mineral Springs, Florida, which apparently require salt for growth (Lackey et al., 1965; Larkin et al., 1994). The third possible exception is the most intriguing—the finding of a natural population of freshwater *Beggiatoa* filaments from Hunters Hot Springs that shows activities of RubisCo comparable to those of autotrophic marine strains (D.C. Nelson, personal communication). Further physiological and phylogenetic investigations of this group of organisms could clarify the taxonomy of the members of the genus *Beggiatoa*.

Phylogenetic analyses have greatly clarified the taxonomy of these organisms, but they have also opened new questions about taxonomic relationships, particularly between the wide, marine beggiatoas and the thioplocas (see the chapter on *Beggiatoa* and the chapter on *Thioploca* in this volume). All of the beggiatoas thus far analyzed fall into the *Gammaproteobacteria*, which is most closely aligned with the thioplocas (see the chapter on *Thioploca* in this volume and Teske et al., 1995, 1999; Ahmad et al., 1999; Jørgensen and Gallardo, 1999). Teske et al. (1995) have pointed out that positioning of the beggiatoas and thioplocas in the *Gammaproteobacteria* places them as developing in the Proterozoic Period, suggesting that they are relatively "young" organisms phylogenetically.

While the resemblance of *Beggiatoa* filaments to those of cyanobacteria caused Pringsheim to label them as colorless forms of cyanobacteria (Pringsheim, 1949), none of the *Beggiatoa* species is phylogenetically related to any cyanobacterium (Reichenbach et al., 1986; Stahl et al., 1987). In addition, as pointed out by Stahl et al. (1987), phylogenetic analysis of beggiatoas and similar organisms has demonstrated that gliding motility is not a particularly useful taxonomic or phylogenetic marker. Additionally, the separation of *Beggiatoa* spp. from *Thioploca* spp. has traditionally been dependent on the formation of sheaths by the thioplocas. Several phylogenetic studies have now shown that wide, marine, autotrophic beggiatoas are more closely related to both the wide, marine and narrow, freshwater thioplocas than they are to the narrow, freshwater heterotrophic *Beggiatoa* strains. Although *Thiothrix* spp. are members of the *Gammaproteobacteria* (Polz et al., 1996), they are not closely related to either *Beggiatoa* or *Thioploca* spp. (Stahl et al., 1987; Lane et al., 1992; Polz et al., 1996; Teske et al., 1999; see chapter on *Thioploca* in this volume).

Recent 16S rRNA sequence analyses have shown that the freshwater *Beggiatoa* strains (OH-75-2a, B15LD, B18LD) cluster very tightly (see again chapter on *Thioploca*), making up the type species (which now can be more narrowly defined as freshwater only). Two other freshwater strains, L1401-13 (Pringsheim, 1964) and AA5A (unpublished strain from Azeem Ahmad; D.C. Nelson, personal communication; also see *Beggiatoa* and *Thioploca* in this volume) which cluster near the three *B. alba* strains but are clearly different from them.

Sequence analysis of the 5S rRNA placed *Beggiatoa alba* B18LD (neotype strain) and *Vitreoscilla beggiatoides* strain B23SS (neotype strain) in the *Gammaproteobacteria* (Stahl et al., 1987; Woese, 1987; also see chapter on *Beggiatoa* in this volume). Stahl et al. (1987) stated, however, that *V. beggiatoides* was "specifically, but distantly, related to *B. alba*" strain B18LD. Based on the 5S rRNA similarity, it is likely that *Vitreoscilla beggiatoides*—a nonsulfur-depositing species that is similar in other features to the *B. alba* species—would fit within this cluster. However, the 16S rRNA of *V. beggiatoides* has not yet been sequenced. Subsequent analyses may show that deposition of sulfur—the key physiological criterion separating *V. beggiatoides* from freshwater, heterotrophic *Beggiatoa* spp.—may not be a valid criterion for separation of the two genera (just as gliding motility [see again chapter on *Beggiatoa* in this volume] and sheath formation by wide, marine thioplocas [see chapter on *Thioploca* in this volume] have proven to be invalid as taxonomic separators).

The two thin, chemolithoautotrophic marine strains of *Beggiatoa* (i.e., MS-81-1c and MS-81-6) cluster with the *Beggiatoa-Thioploca* group, but clearly are more closely related to the wide, marine beggiatoas than to the thin, freshwater, heterotrophic *B. alba* types. These two strains, however, are clearly separated from each other, both physiologically (Hagen and Nelson, 1996, 1997) and phylogenetically (Jørgensen and Gallardo, 1999) (also see again chapter on *Thioploca* in this volume).

Finally, based on 16S rRNA sequence analyses, the wide marine "strains" of *Beggiatoa* from Bay of Concepción and Monterey Canyon form a discrete cluster with the wide, marine, autotrophic thioplocas (*Thioploca araucae* and *T. chileae*), and the freshwater, thin *Thioploca* strain, *T. ingrica* (Teske et al., 1995, 1999; Ahmad et al., 1999; Jørgensen and Gallardo, 1999; see chapter on *Thioploca* in this volume), as well as the newly described marine sulfur microorganism, *Thiomargarita namibiensis* (Schulz et al., 1999a). These phylogenetic data suggest that taxonomic changes are due for the genera *Beggiatoa* and *Thioploca*. While no specific changes can be made at this time, it is clear that rearrangements within at least these two genera, as well as within *Vitreoscilla* (see chapter on *Vitreoscilla* in this volume), will need to be made based on the phylogenetic data. Additional 16S rRNA sequence analyses and strain-strain nucleic acid hybridizations with a greater variety of beggiatoas, thioplocas, and similar non-sulfur-depositing strains are clearly needed to place these organisms into their proper taxonomic locations with respect to related organisms.

Historical information *Beggiatoa alba* was the only species of *Beggiatoa* recognized in the ninth edition of *Bergey's Manual of Determinative Bacteriology* (Strohl, 1989). This remains true for the present edition, although evidence is accumulating that should allow additional species descriptions by the next version. In the seventh (Buchanan, 1957) and eighth (Leadbetter, 1974a) editions of *Bergey's Manual of Determinative Bacteriology*, six species were recognized, each differentiated on the basis of filament diameter alone: *B. alba* (type species; 2.5–5 μm diameter), "*Beggiatoa arachnoidea*" (5–14 μm diameter), "*Beggiatoa gigantea*" (26–55 μm diameter), "*Beggiatoa leptomitiformis*" (1–2.5 μm diameter), "*Beggiatoa minima*" (<1 μm diameter), and "*Beggiatoa mirabilis*" (15–21 μm diameter). In the eighth version, an additional 18 species names were listed as *species incertae sedis*, differentiated based on filament diameter, habitats and other general characteristics. Until strain B18LD was proposed as the neotype strain of the *B. alba* (Mezzino et al., 1984), species designations in *Beggiatoa* had been made solely based on filament diameter, general morphology, or habitat.

B. alba was originally described in 1803 as "*Oscillatoria alba*", an organism characterized by "white filaments, of diameter of $1/800^{\text{th}}$ of the length, distance between septa equal to diameter, extremities rounded and not pointed" (Vaucher, 1803). The length of *Beggiatoa* filaments is variable (Strohl and Larkin, 1978a, b), however; thus, the diameter of the first observed "*O. alba*" cannot be calculated. Trevisan (1842) described the genus *Beggiatoa*, in which "*O. alba*" (Vaucher) was included as a species. The two original species of *Beggiatoa*, *B. alba* and "*Beggiatoa leptomitiformis*", were differentiated by their filament tips being rounded and pointed, respectively (Trevisan, 1842, 1845), not by filament diameter as in more recent years. Strohl (1989) has described in detail how the use of filament diameters has been both arbitrary and inconsistent (i.e., different authors using different filament diameters for the same species) over the years, especially for the thinner strains, making that criterion virtually useless in defining species. For example, historically, "*B. mirabilis*" has been described by different investigators as having a filament diameter (in μm) of 16 (original description), 20–40, 20–38, 20–45, 30–42, 16–45, and 15–45 (Klas, 1936). This interesting, but confusing, state of affairs in *Beggiatoa* taxonomy prompted the deletion of all but *B. alba*, the type species, from acceptance (Skerman et al., 1980), with the hope that the move would induce future descriptions of new *Beggiatoa* species to include physiological and phylogenetic data as well as morphological data. In subsequent descriptions, filament width should be considered as a single characteristic among several that differentiate the various species.

On the other hand, good historical logic and experimental evidence exist that support using filament diameter as a differentiation tool for certain *Beggiatoa*s. In various field investigations, *Beggiatoa*s have been observed ranging from <1 to >50 μm in diameter. Klas (1936) found that two size ranges of marine *Beggiatoa*s, 14.8–21.4 μm (18.4 μm average) and 26.4–49.2 μm (34.4 μm average), existed among trichomes from environmental samples. Likewise, Jørgensen (1978) found clusters in the size

classes of 3–5 μm , 8–17 μm , and >23 μm among the marine *Beggiatoa*s he observed. McHatton et al. (1996) reported size clusters of 65–85 μm diameter from Clam Field Seep and 24–32 μm , 40–42 μm , and 116–122 μm diameter from Guaymas Basin. Some of these filaments were reported to reach 10 cm in length. Based on the maximum cell length of 36 μm reported in that study (McHatton et al., 1996), this would suggest that some filaments may contain as many as 270,000 cells, which would clearly qualify them as the most "multicellular" procaryotic organisms ever observed. Larkin and Henk (1996) similarly observed clusters of marine *Beggiatoa*s with diameters of 20–30 μm , 40–50 μm , and ~ 110 –130 μm . They also observed marine *Beggiatoa* filaments with diameters of nearly 200 μm (Fig. BXII.γ.66) and lengths of over 2 cm (Larkin and Henk, 1996), making these among the largest procaryotes in existence, along with the recently discovered *Thiomargarita namibiensis* (Schulz et al., 1999a). Whereas a continuum of sizes is not observed from individual sites in nature, a compilation of all wide *Beggiatoa*s observed by nine different groups of investigators results in a broad continuum of reported sizes of ~ 7 –50 μm , with much larger filament diameters also observed (Fig. BXII.γ.67). Clearly, more phylogenetic data on filaments of different sizes will be required to determine how to differentiate species among the larger marine *Beggiatoa*s.

ACKNOWLEDGMENTS

I sincerely thank Bo B. Jørgensen for sharing a preliminary version of the *Thioploca* chapter, which helped to stimulate thoughts about the taxonomy of these organisms, and Douglas C. Nelson for sharing his thoughts and expertise on *Beggiatoa* taxonomy and physiology, as well as for unpublished data and updates on his current research efforts in the area.

FURTHER READING

Ahmad, A.A., J.P. Barry and D.C. Nelson. 1999. Phylogenetic affinity of a wide, vacuolate, nitrate-accumulating *Beggiatoa* sp. from Monterey Canyon, California, with *Thioploca* spp. Appl. Environ. Microbiol. 65: 270–277.

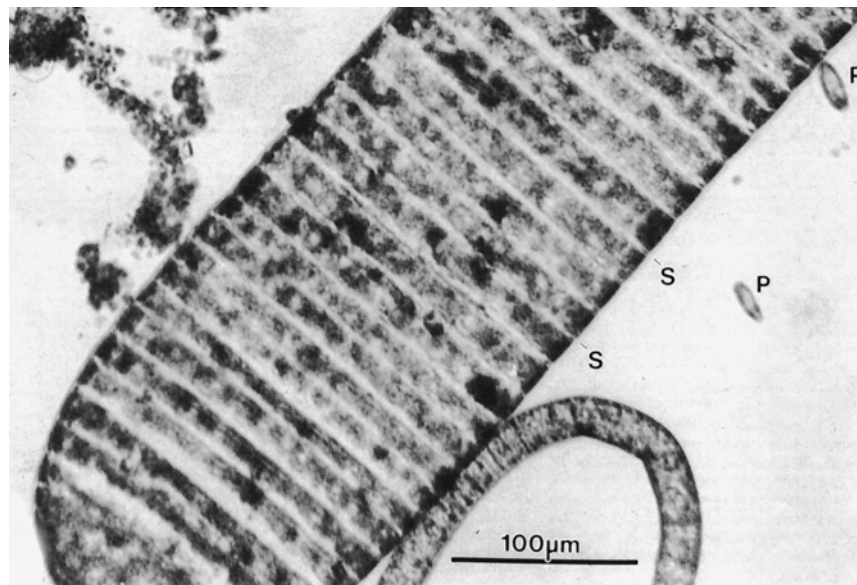


FIGURE BXII.γ.66. A light micrograph of a wide, marine *Beggiatoa* filament with a diameter of ~ 100 μm . Note how the cells, divided by visible septa (S) are the width of the filament but are only ~ 20 μm in length. Ciliated protozoa (P) are present, giving a comparison in size. (Reproduced with permission from J.M. Larkin and M.C. Henk, Microscopy Research and Technique 33: 23–31, 1996, ©John Wiley & Sons, Inc.)

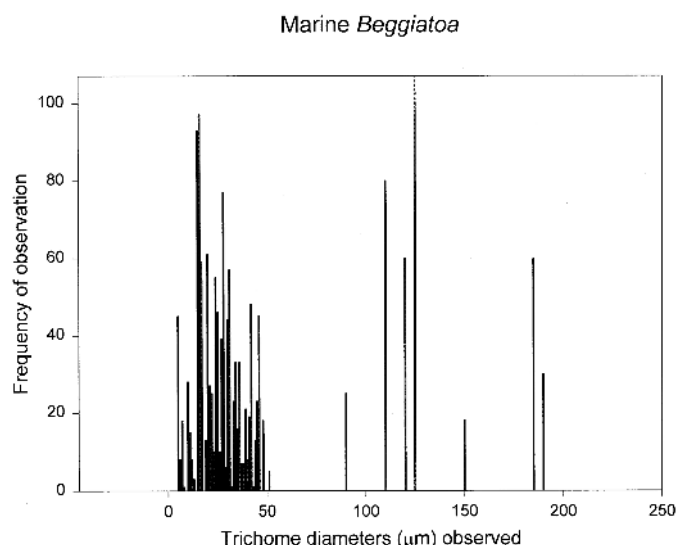


FIGURE BXII.γ.67. A bar graph depicting observations of wide, marine beggiatoas in nature at nine different locations. The Y-axis gives an arbitrary frequency of observation figure, as mentioned by several investigators, as a function of filament diameter (in μm).

List of species of the genus *Beggiatoa*

1. *Beggiatoa alba* (Vaucher 1803) Trevisan 1842, 56.^{AL}
al'ba. L. adj. *albus* white.

The characteristics are as described for the genus, with the following additional features. The filaments measure about 1.5–4.0 μm in diameter and are of uniform width; the filament diameter may vary with growth conditions. The cells are usually 3.0–9.0 μm long, with filament lengths averaging 60–120 μm . Ends of the filaments are rounded. Necridia and hormogonia may be produced. Circuitans type colonies are usually formed. Sulfur is deposited as inclusions external to the cytoplasmic membrane, but within the cell wall, when the organism is grown in the presence of sulfide or thiosulfate. PHB and polyphosphate deposits are often present.

Chemoorganotrophic growth is obtained by using acetate, fumarate, lactate, malate, pyruvate, succinate, or ethanol as sole carbon and energy sources. The oxidation of sulfide, thiosulfate, or H_2 has been proposed to be coupled with energy conservation for chemolithotrophic metabolism, although this has yet to be proven. Ammonia, nitrate, nitrite, or urea is used as the sole source of nitrogen.

Among the thin, heterotrophic, freshwater strains, a strain not presently assigned to the species *Beggiatoa alba* is L1401-13 ("strain 13" as described by Pringsheim, 1964; Kowallik and Pringsheim, 1966). The thin, marine strains include the obligately chemolithoautotrophic strain MS-81-1c and the facultatively chemolithoautotrophic strain MS-81-6 (Nelson and Jannasch, 1983; Nelson et al., 1986a; Hagen and Nelson, 1996, 1997)

- Hagen, K.D. and D.C. Nelson. 1996. Organic carbon utilization by obligately and facultatively autotrophic *Beggiatoa* strains in homogeneous and gradient cultures. *Appl. Environ. Microbiol.* 62: 947–953.
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Organisms grow by a respiratory metabolism, with molecular oxygen used as the terminal electron acceptor. Microaerophilic to aerobic. Maintenance under anaerobic conditions, with sulfur used as the terminal electron acceptor, may occur.

Gelatin, starch, and casein are not hydrolyzed. Filaments are catalase-negative, and *N,N,N,N*-tetramethyl-*p*-phenylenediamine (TMPD) cytochrome oxidase-positive. No growth is observed in the presence of 0.05% sodium dodecyl sulfate or 1.5% NaCl. Strains thus far have been isolated only from freshwater sediments.

The neotype strain, B18LD, was isolated from an enrichment obtained from a rice paddy in Lacassine, Louisiana, U.S.A. This strain, along with other *B. alba* strains, was described in detail by Mezzino et al. (1984).

The mol% G + C of the DNA is: 35–39 (T_m).

Type strain: LSU B18LD, ATCC 33555.

Additional Remarks: Based on phylogenetic and physiological data, other well-characterized strains that now should be considered as *B. alba* strains are OH-75-2a (Nelson and Castenholz, 1981a, b) and B15LD (ATCC 33556; Strohl et al., 1982; Ahmad et al., 1999).

Other Organisms

The wide, autotrophic, vacuolated, marine beggiatoas include, e.g., Monterey Canyon, Guaymas Basin, and Bay of Concepcion suspensions, which are phylogenetically related to marine thioplocas (Table BXII.γ.39) (Jannasch et al., 1989; Nelson et al., 1989b; Ahmad et al., 1999; Jørgensen and Gallardo, 1999; Teske et al., 1999).

Genus IV. *Leucothrix* Oersted 1844, 44^{AL}

JUDITH A. BLAND AND THOMAS D. BROCK

Leu'co.thrix. Gr. adj. *leucus* clear, light; Gr. n. *thrix*, *trichis* hair; M.L. fem. n. *Leucothrix* colorless hair.

Long filaments composed of short, cylindrical or ovoid cells (**gonidia**), cross-walls clearly visible, **colorless, unbranched**; typically uniform filaments may taper from base to apex, showing under some conditions an apical beady chain of gonidia connected from end to end. Filaments may attain a length of over 100 μm , and lengths of several hundred micrometers are not unusual (Harold and Stanier, 1955; Pringsheim, 1957). The filaments, particularly when attached to a substrate, may be somewhat tapered, with the apex being 2–3 μm and the base 5–6 μm in diameter (Harold and Stanier, 1955; Pringsheim, 1957; Brock, 1969). In nature, **filaments are usually attached to solid substrates by means of inconspicuous holdfasts; stalks and sheaths absent**. **Filaments do not glide**, but may wave sporadically from side to side. **Dispersal by means of gonidia** (single cells arising from cells of the filaments by rounding up, often released primarily from apices, but they may also be formed in an intercalary fashion); **gonidia often**, but not **always, show jerky gliding motion** on solid substrates. **Rosette formation** is a key diagnostic characteristic of the genus that is seen frequently in laboratory culture but less commonly in nature. The rosettes may be formed of gonidia or, after gonidial growth, of several or more filaments attached at their bases. Filaments in a laboratory culture often form true knots. These structures also occur in nature, but rarely. The organism morphologically resembles some filamentous cyanobacteria but does not form photosynthetic pigments. **Gram negative**. **Strictly aerobic, heterotrophic**. **Aquatic; typically marine**, although one freshwater strain has been reported. Strains require NaCl for growth; optimum concentration: about 1.5% NaCl; grow at concentrations of 0.3–6.0% NaCl. Most strains do not require growth factors. Optimum temperature: 25°C; maximum: 30–35°C; minimum: 0°C, forming visible colonies within 1–2 weeks. Strains from tropical waters are more stenothermal, not growing below 15°C. Catalase positive. Oxidase positive.

According to Woese et al. (1985), the “definition” of the *Leucothrix* group (two strains tested—*L. mucor* ATCC 25707 and a “*Thiomicrospira*-like isolate”) is the set of 16S rRNA signature sequences consisting of AACCUUAUCCAUCCCUUG, CACU-UUCAAUUG, UUAACUUUAG, and AUCUAUAG. The most specific 16S rRNA probe sequence published is CCCCTCTCCAA-ACTCTA, positions 652–669 (a non-ATCC culture; Wagner et al., 1994a). Sequences of 16S rRNA sequences are also available for two other isolates (Reichenbach et al., 1986; Nielsen et al., 1998). Also defined is a 23S rRNA sequence (Ludwig et al., 1995).

The mol% G + C of the DNA is: 46.9–51 (Brock and Mandel, 1966; Kelly and Brock, 1969a, b).

Type species: *Leucothrix mucor* Oersted 1844, 44.

FURTHER DESCRIPTIVE INFORMATION

Phylogenetic treatment *Leucothrix* was first isolated in pure culture from a marine source by Harold and Stanier (1955), who characterized the organism “as a chemoheterotrophic counterpart of the colorless sulfur-oxidizing organism *Thiothrix*.” The two organisms have many features in common, and the life cycles of both organisms can be superimposed. Both genera were placed in the family *Leucotrichaceae* (Buchanan and Gibbons, 1974). *Leucothrix* and *Thiothrix* were further described as colorless

evolutionary descendants of cyanobacteria (Harold and Stanier, 1955; Pringsheim, 1957; Raj, 1977), since each have a gliding form, the gonidium, and are loosely associated with other gliding, heterotrophic bacteria. Focus on the presence of a sheath in *Thiothrix*, its chemoautotrophy, and its sulfur metabolism eventually distanced this species from *Leucothrix*. Currently, as a result of 16S rRNA studies, *Leucothrix* is classified as a eubacterium belonging to the *Gammaproteobacteria* (Woese et al., 1985). Studies of the 23S rRNA support this classification (Ludwig et al. 1995). The findings of both Woese et al. (1985) and Ludwig et al. (1995) place *Leucothrix* by itself with its relationship to the other *Gammaproteobacteria*.

Cell morphology *Leucothrix* is a complex marine bacterium with diverse morphologies. A commonly seen form, especially in nature, is the filament (Bland and Brock, 1973; Harper and Talbot, 1984; Hansen and Olafsen, 1989), which is composed of individual cells that appear slightly beaded under the light microscope (Fig. BXII.γ.68). Filaments in nature extend perpendicularly outward from the surface to which they are attached via holdfast (Fig. BXII.γ.68). Cell division occurs throughout the filament (Brock, 1967). Filaments are of variable length, often much greater than 100 μm , with a diameter of 2–3 μm or more. They are colorless, unbranched, nonmotile (although they occasionally wave back and forth), and lack a sheath. Storage granules are often visible (Fig. BXII.γ.69). Filaments frequently grow intertwined or in dense tangles. One of the intriguing characteristics of *Leucothrix* is the ability of most strains to form true knots (Brock, 1964). The function of knot formation is unknown, and it may actually be an accidental process, but it seems to be characteristic of most *Leucothrix* strains as part of the growth process. Swollen cells often form apparently at random along filaments. Larger structures (bulbs) usually form in knotty cultures, possibly as the result of fusion of cells in the region of knots.

Gonidia (Figs. BXII.γ.69 and BXII.γ.70) are formed by rounding of cells in the filaments. At a low concentration of an organic energy source, filaments do not grow very long, and gonidia are formed more readily. Gonidial formation is also increased if growth rate is slowed by anaerobiosis, starvation, or reduction of the temperature of incubation (Pringsheim, 1957; Bland and Brock, 1973). Gonidia move by a jerky sort of gliding motility, which can only be seen well when observations are made in slide cultures. Single gonidia may move for extended periods before attaching and dividing to form filaments. If the gonidial density is high enough, gonidia appear to be attracted to each other by homotaxis, and they form aggregates, a process that ultimately results in rosette formation (Fig. BXII.γ.69). Rosettes are also seen in nature (Fig. BXII.γ.70; Solangi et al., 1979; Hansen and Olafsen, 1989). Free gonidia appear to lack holdfast material; however, once a gonidium has attached to a surface or to another gonidium (after elaborating holdfast), it remains attached indefinitely. Gonidia grow into filaments if conditions are favorable. Holdfasts can also be visualized by fluorescence microscopy. The fluorochrome dye primulin, when viewed in blue light, causes holdfast material to fluoresce red and the cell wall to fluoresce yellow.

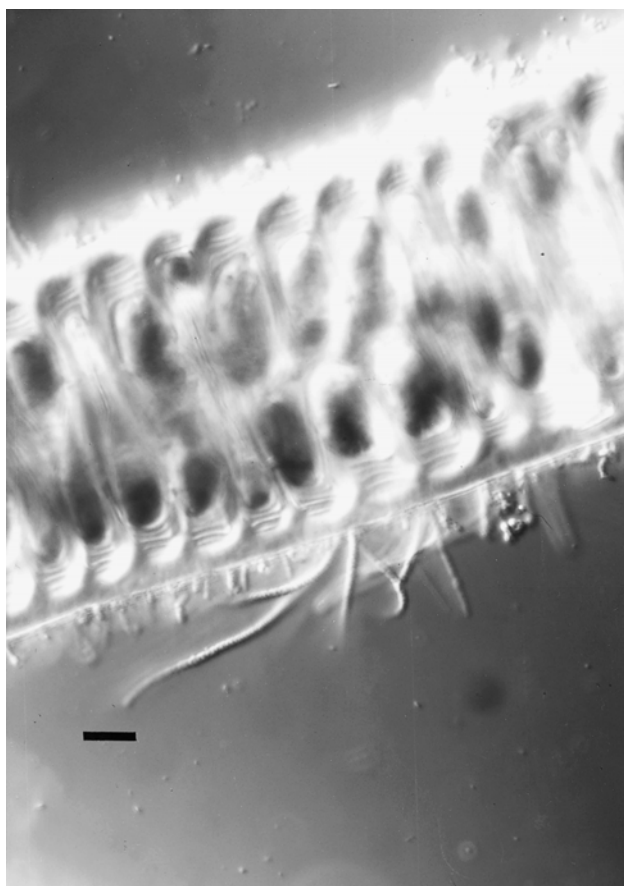


FIGURE BXII.γ.68. Filaments of *Leucothrix* attached to the red alga *Bangia fuscopurpurea*. Bar = 10 μm (Nomarsky interference optics).

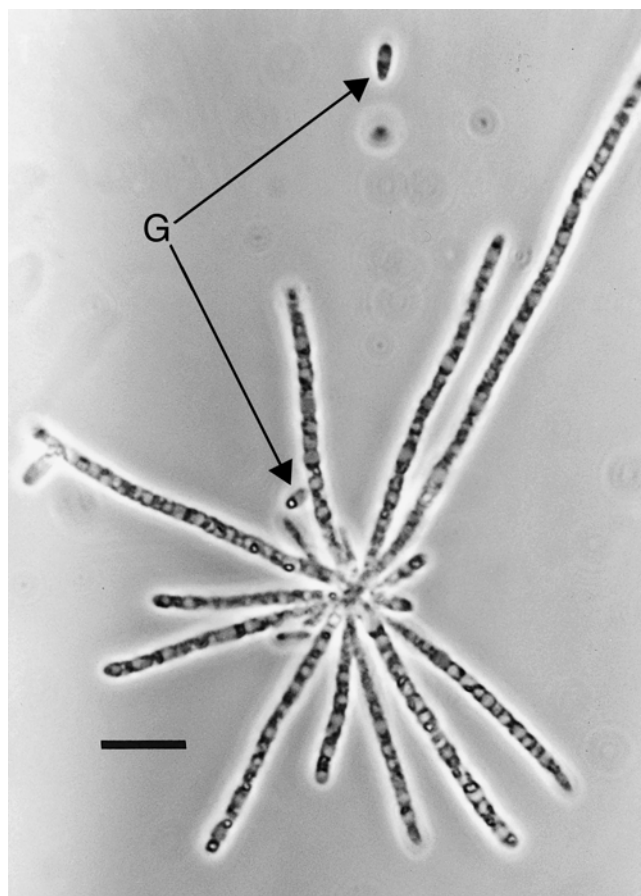


FIGURE BXII.γ.69. Rosette of *L. mucor* in laboratory culture, with storage granules visible in the filaments. Free gonidia (G) are visible. Bar = 15 μm (Phase microscopy).

Cell wall composition To date, no chemistry for the cell walls of *Leucothrix* has been published.

Fine structure Electron microscopic studies of *Leucothrix* have been presented by Brock and Conti (1969), Snellen and Raj (1970), Raj (1977), Couch (1978), Solangi et al. (1979), and Hansen and Olafsen (1989). The filaments of *Leucothrix* are multicellular, with the individual, somewhat square cells being separated by well-defined cross-walls (Fig. BXII.γ.71). Membranous and storage vesicles are obvious in both filament cells and gonidia (Figs. BXII.γ.71 and BXII.γ.72). The cell wall shows structural features characteristic of the Gram-negative bacterial envelope. No sheath is apparent (Figs. BXII.γ.71 and BXII.γ.72). In gonidia formation, the vegetative cells in the filament become rounded with no apparent internal fine-structural changes (Brock and Conti, 1969) and disassociate from each other (Harold and Stanier, 1955). Released gonidia may either attach singly to a surface via holdfast or aggregate into rosettes and elaborate holdfast at the aggregated cell ends. Figure BXII.γ.72 depicts aggregated gonidia with the associated holdfast material easily seen in the electron microscope as an electron-dense material, possibly a polysaccharide, peripheral to the outer wall layer. A micrograph by Solangi et al. (1979) shows holdfast material external to a brine shrimp structure. A similar finding is demonstrated by Couch (1978). His pictures reveal *Leucothrix* attached via holdfast to the gill cuticle of pink shrimp. The holdfast adheres to the

cuticle (e.g., Fig. BXII.γ.73) and does not penetrate into the organism. Snellen and Raj (1970) have demonstrated that many bizarre forms of *Leucothrix*, including knots and bulbs, can be induced by calcium deficiency, and they have shown the resultant fine structure.

Life cycle The life cycle of *Leucothrix* is shown in Fig. BXII.γ.74. Rosette and gonidia formation are not obligatory; the organism can grow indefinitely in the filamentous form. Gonidia formation occurs in nature, ensuring the dispersal of the organism.

Nutrition and growth conditions *Leucothrix* is a versatile heterotroph that is able to grow on a variety of organic energy sources, including sugars, amino acids, sugar alcohols, alcohols, lipids, and organic acids (Raj, 1977; Williams and Unz, 1985a, 1989; Brock, 1992). Walker et al. (1975) reported utilization of a mixed hydrocarbon substrate by the type strain, ATCC 25107. Strains exhibit variable use of both inorganic and organic nitrogen sources (Williams and Unz, 1989), but *Leucothrix* is unable to fix N₂ (Mague and Lewin, 1974). Many strains do not need growth factors; however, a stenothermal strain required vitamin B₁₂ for growth (Kelly and Brock, 1969b).

Metabolism and metabolic pathways The physiology and the metabolism of *Leucothrix* have been extensively reviewed by Raj (1977). *Leucothrix* is a strict aerobe, with an optimum temperature